UNITED STATES
SECURITIES AND EXCHANGE COMMISSION

Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to

Commission file number: 001-36146

CommScope Holding Company, Inc.

(Exact name of registrant as specified in its charter)

Delaware	**27-4332098**
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)

3642 E. US Highway 70	**28610**	**(828) 459-5000**
Claremont, North Carolina	(Zip Code)	(Registrant's telephone number, including area code)
(Address of principal executive offices)		

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	**Trading symbol**	**Name of each exchange on which registered**
Common Stock, par value $.01 per share	COMM	Nasdaq

Securities registered pursuant to Section 12(g) of the Act: **NONE**

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer" "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐		

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

The aggregate market value of shares of Common Stock held by non-affiliates of the registrant was approximately $1,179.2 million as of June 30, 2023. For purposes of this computation, shares held by affiliates and by directors and officers of the registrant have been excluded.

As of February 14, 2024 there were 212,270,576 shares of the registrant's Common Stock outstanding.

Documents Incorporated by Reference

Portions of the registrant's Proxy Statement for the 2023 Annual Meeting of Stockholders are incorporated by reference in Part III hereof.

CommScope Holding Company, Inc.
Form 10-K
December 31, 2023
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PART I

Unless the context otherwise requires, references to "CommScope Holding Company, Inc.," "CommScope," "the Company," "Registrant," "we," "us," or "our" are to CommScope Holding Company, Inc. and its direct and indirect subsidiaries on a consolidated basis.

This Annual Report on Form 10-K includes certain statements that constitute "forward-looking statements" within the meaning of Section 27A of the Securities Act of 1933, as amended, and Section 21E of the Securities Exchange Act of 1934, as amended, which reflect our current views with respect to future events and financial performance. These forward-looking statements are generally identified by their use of such terms and phrases as "intend," "goal," "estimate," "expect," "project," "projections," "plans," "potential," "anticipate," "should," "could," "designed to," "foreseeable future," "believe," "think," "scheduled," "outlook," "target," "guidance" and similar expressions, although not all forward-looking statements contain such terms. This list of indicative terms and phrases is not intended to be all-inclusive.

These statements are subject to various risks and uncertainties, many of which are outside of our control. Item 1A, "Risk Factors," of this Annual Report on Form 10-K sets forth more detailed information about the factors that may cause our actual results to differ, perhaps materially, from the views stated in such forward-looking statements. Although the information contained in this Annual Report on Form 10-K represents our best judgment as of the date of this report based on information currently available and reasonable assumptions, we can give no assurance that the expectations will be attained or that any deviation will not be material. Given these uncertainties, we caution you not to place undue reliance on these forward-looking statements, which speak only as of the date made. We are not undertaking any duty or obligation to update any forward-looking statements to reflect developments or information obtained after the date of this Annual Report on Form 10-K, except to the extent required by law.

ITEM 1. BUSINESS

Company Overview

CommScope Holding Company, Inc. was incorporated in Delaware on October 22, 2010, and our initial public offering for our common stock was on October 25, 2013. Since our founding as an independent company in 1976, we have consistently played a significant role in many of the world's leading communication networks. Our evolution has been driven by technological innovation and strategic acquisitions that expanded our product offerings and complemented our existing solutions. We are a global provider of infrastructure solutions for communication, data center and entertainment networks. Our solutions for wired and wireless networks enable service providers, including cable, telephone and digital broadcast satellite operators and media programmers, to deliver media, voice, Internet Protocol (IP) data services and Wi-Fi to their subscribers and allow enterprises to experience constant wireless and wired connectivity across complex and varied networking environments. Our solutions are complemented by services including technical support, systems design and integration. We are a leader in digital video and IP television (IPTV) distribution systems, broadband access infrastructure platforms and equipment that delivers data and voice networks to homes including fiber to the home technologies (FTTH). Our global leadership position is built upon innovative technology, broad solution offerings, high-quality and cost-effective customer solutions, and global manufacturing and distribution scale.

We have a team of over 20,000 people who serve our customers in over 150 countries through a network of world-class manufacturing and distribution facilities strategically located around the globe. Our customers include substantially all the leading global telecommunications operators, data center managers, cable television providers or multi-system operators (MSOs) and thousands of enterprise customers, including many Fortune 500 companies. We have long-standing, direct relationships with our customers and serve them through a direct sales force and a global network of channel partners.

Since 2021, we have been engaged in a transformation initiative referred to as CommScope NEXT, which is designed to drive shareholder value through three pillars: profitable growth, operational efficiency and portfolio optimization. We believe these efforts are critical to making us more competitive and allowing us to invest in growth, de-leverage our indebtedness and maximize stockholder and other stakeholder value in the future. In 2022, CommScope NEXT generated positive impacts on net sales, profitability and cash flow from our execution on pricing initiatives, capacity expansion and operational efficiencies. In 2023, we experienced headwinds related to a slow-down in spending by our customers, but we continued to execute under CommScope NEXT by driving operational efficiencies and focusing on portfolio optimization that should enable us to take advantage of the expected recovery in demand in the second half of 2024. To that end, we incurred $29.7 million, $63.0 million and $85.1 million of restructuring costs and $27.1 million, $35.1 million and $50.6 million of transaction, transformation and integration costs during the years ended December 31, 2023, 2022 and 2021, respectively, primarily related to CommScope NEXT. We expect to continue to incur such costs in 2024 as we continue executing on CommScope NEXT initiatives, and the resulting charges and cash requirements could be material.

In 2021, as a step in our CommScope NEXT transformation plan, we announced a plan to separate the Home Networks (Home) segment and began analyzing the financial results of our "Core" business separately from Home. On October 2, 2023, we entered into a Call Option Agreement with Vantiva SA, a société anonyme organized under the Laws of France (Vantiva), pursuant to which we granted Vantiva a binding call option to acquire our Home segment and substantially all of the associated segment assets and liabilities (Home business), which was subsequently exercised and a Purchase Agreement signed on December 7, 2023. The transaction closed on January 9, 2024.

We determined the anticipated divestiture of our Home business met the "held for sale" criteria and the "discontinued operations" criteria in accordance with accounting guidance. All prior period amounts in these consolidated financial statements have been recast to reflect the discontinuation of the Home business. The discussions in these consolidated financial statements relate solely to our continuing operations, unless otherwise noted. For further discussion of the discontinued operations related to the Home business, see Note 3 in the Notes to Consolidated Financial Statements.

For the year ended December 31, 2023, our revenues were $5.79 billion and our loss from continuing operations was $851.3 million. For further discussion of our current and prior year financial results, see Part II, Item 7, "Management's Discussion and Analysis of Financial Condition and Results of Operations" and the Consolidated Financial Statements included in Part II, Item 8 of this Annual Report on Form 10-K.

Operating Segments

As a result of the divestiture of the Home business, we are now reporting financial performance based on the following remaining four operating segments, which excludes our Home segment: Connectivity and Cable Solutions (CCS), Outdoor Wireless Networks (OWN), Networking, Intelligent Cellular and Security Solutions (NICS) and Access Network Solutions (ANS). For all periods presented, amounts have been recast to reflect the discontinuation of our Home business. The distribution of net revenues among our four segments was as follows:

	Year Ended December 31,		
	2023	2022	2021
CCS	46.8%	50.4%	45.4%
OWN	15.2	19.5	21.0
NICS	19.3	12.5	12.8
ANS	18.7	17.6	20.8
Total	100.0%	100.0%	100.0%

CCS Segment (2023 Net Sales of $2.7 billion)

Our CCS segment provides fiber optic and copper connectivity and cable solutions for use in telecommunications, cable television, residential broadband networks, data centers and business enterprises. The CCS portfolio includes network solutions for indoor and outdoor network applications. Indoor network solutions include optical fiber and twisted pair structured cable solutions, intelligent infrastructure management hardware and software and network rack and cabinet enclosures. Outdoor network solutions are used in both local-area and wide-area networks and "last mile" fiber-to-the-home installations, including deployments of fiber-to-the-node, fiber-to-the-premises and fiber-to-the-distribution-point to homes, businesses and cell sites.

OWN Segment (2023 Net Sales of $0.9 billion)

Our OWN segment focuses on the macro and metro cell markets. The segment includes base station antennas, radio frequency (RF) filters, tower connectivity, microwave antennas, metro cell products, cabinets, steel, accessories and our wireless spectrum management business, Comsearch.

NICS Segment (2023 Net Sales of $1.1 billion)

Our NICS segment provides wireless networks for enterprises and service providers. Product offerings include indoor cellular solutions such as distributed antenna systems, public key infrastructure solutions, indoor and outdoor Wi-Fi and long-term evolution (LTE) access points, access and aggregation switches; an Internet of Things (IoT) suite, on-premises and cloud-based control and management systems; and software and software-as-a-service applications addressing security, location, reporting and analytics.

ANS Segment (2023 Net Sales of $1.1 billion)

Our ANS segment's product solutions include cable modem termination systems (CMTS), video infrastructure, distribution and transmission equipment and cloud solutions that enable facility-based service providers to construct a state-of-the-art residential and metro distribution network.

Industry Background

We participate in the large and growing global market for connectivity and essential communications infrastructure. This market is being driven by the growth in bandwidth demand associated with the continued demand of smartphones, tablets and machine-to-machine (M2M) communication as well as the proliferation of data centers, Big Data, cloud-based services, streaming media content and IoT. In addition, video distribution over the broadband IP network is transforming how content is managed and consumed. IP facilitates new forms of video such as Over-the-Top (OTT) and interactive television. We continue to see a mix of connectivity needs in homes, in offices and while on the move. We also continue to see higher upstream network usage than downstream usage. Additionally, the network has become more bi-directional and interactive, driving the need for lower and more consistent latency.

There are several major trends that we expect to continue to drive network deployments and investment, including the following:

Evolution of Network Architecture and Technology

The pace of change in networking continues to increase as consumers and data-driven businesses utilize more bandwidth and cloud and mobile applications. Exponential growth in video and mobile data consumption is continuing to revolutionize how we connect to each other and changing the network architecture needed to support consumer demand. This trend requires better network coverage, greater broadband access, and increased capacity and data storage.

Our customers are continuing to develop networks that are faster, more responsive, more efficient and more reliable. We believe the following key network trends will continue to impact CommScope and the industry during 2024:

1)***Network Convergence*:** Operators are continuing to move toward converged or multi-use network architectures. Rather than building upon independent wireline and wireless networks, operators are utilizing networks that combine voice, video and data communications into a single converged data network for wired and wireless services.

2)***Continued Disruption by OTT TV:*** Although content consumption continues to increase, subscriptions to pay TV continue declining. As a result, cable operators are continuing to invest in upgrades to their networks for broadband but continue to have mixed feelings about investments in their video and voice services. While past data trends have been defined by rapid growth in the downlink, more interactive experiences and IoT are driving the need for major network change in the uplink.

3)***Densification*****:** As wireless operators work to meet consumer demand, cell splitting, in the form of densification, is a key driver for continuing to fulfill the promise of 5G networks. Increased sectorization at macro cell sites and establishment of better inbuilding coverage play significant roles in the 5G network. We expect that densification will require significant fiber cable and connectivity between wireless cell sites.

4)***Virtualization, Centralization and Disaggregation:*** Operators are continuing to virtualize and centralize their networks to make them more flexible and efficient. Wireless operators have deployed centralized radio access networks (CRAN) as a first step in the evolution to a virtualized radio access network. This enables servers and switches to replace some of the hardware-specific equipment that exists today and allows much of the processing to be performed on general purpose processors wherever and whenever it is needed throughout the network. Cable operators have also virtualized their networks by moving from a traditional converged cable access platform (CCAP) architecture to a distributed access architecture (DAA). This moves some of the processing from the head end to the node and virtualizes the rest on traditional switches and servers.

5)***Low Latency Services:*** To support the increased demands of a growing game-playing subscriber base, all operators are continuing to seek new ways to reduce the latency and jitter of the gaming packet streams. As an example, Data Over Cable Service Interface Specification (DOCSIS) deployments are being added to the new low latency DOCSIS technologies to their CMTS and customer premises equipment (CPE) gear. Node-splits are also used to reduce congestion. Over time, we expect these low latency services to allow support of Web3.0 and the metaverse, as access networks are increasing and used in a more interactive way. The densification of 5G networks also reduces congestion and decreases latency.

6)***Capacity Expansion*****:** Wired and wireless network providers continue to be cognizant of the need to stay ahead of the traffic growth that occurs every year. This traffic growth results from increases in average subscriber consumption levels and in maximum service level agreement (SLA) levels. Cable providers are beginning to see many increases in spectrum, including moves to DOCSIS 3.1 upstream mid-splits (85 MHz) and DOCSIS 3.1 upstream high-splits (204 MHz) and downstream DOCSIS 3.1 transitions to 1.2 GHz. Capacity is also increasing via the increased use of the spectrally-efficient DOCSIS 3.1 orthogonal frequency-division multiplexing (OFDM) and orthogonal frequency-division multiple access (OFDMA) channels within the cable spectrum. Additionally, some operators are beginning their upgrade path to DOCSIS 4.0 in certain regions of their networks. New DOCSIS equipment is needed for this expansion. Wireless providers are seeing continued deployment of 5G that will include ongoing additions to their wireless spectrum in the Citizens Broadband Radio Service (CBRS) band, the C-band and the millimeter wave (mmWave) bands. New antenna equipment is needed for this expansion.

7)***Government-Sponsored Broadband Improvements*****:** Several government-sponsored programs aimed at improving the Broadband infrastructure connecting to rural and other under-served areas launched in the second half of 2022. The funds from these programs and initiatives to build more equitable access—in particular, in the United States (U.S.), the Broadband Equity, Access, and Deployment (BEAD) Program, American Rescue Plan Act (ARPA) and Rural Digital Opportunity Fund (RDOF)—are expected to drive technology and device sales across the board. The U.S. initiatives aim to ensure that every person in the U.S. has access to reliable and affordable broadband by 2030. While we did not see money flowing under the BEAD Program in 2023, we expect a minimal amount of funds to begin being distributed in the second half of 2024. We do expect a more significant impact from ARPA in 2024. The RDOF is a ten year program that will only be less than halfway through the anticipated build timeline by the end of 2024 for those service providers that won awards.

8)***Open Radio Access Network (O-RAN) Standardization:*** We ae a leader in the O-RAN ecosystem through our internal research and development activities and our external participation in standardization efforts through alliances and advocacy organizations. There are various government initiatives and legislative proposals to provide federal investment in O-RAN research and development that would encourage federal agencies to foster the deployment of open and interoperable 5G networks in their various pilot programs and procurement guidelines. We anticipate many commercial deployments of our O-RAN compliant products in large public venues in 2024.

Transition to 5G

5G wireless is continuing to evolve from an industry vision toward a tangible next generation wireless technology. Operators continue to transition to 5G networks with countries in North America, Northeast Asia, Gulf Cooperation Council and Western Europe continuing to lead the way in 5G subscription penetration. The number of 5G-enabled devices is expected to continue to increase during 2024. The primary benefits of 5G include the following:

- Enhanced mobile broadband—to support significant improvement in data rates and user experience in both the uplink and downlink;
- IoT communications—to support the expected billions of connections between machines, as well as short bursts of information to other systems;
- Low latency, high reliability—to support applications that are critical or are needed in real time, like factory machines, virtual reality and augmentation; and
- Underlying capacity—to support fixed broadband services in underserved areas.

As described above, wireless operators are having to both acquire and launch new spectrum for 5G, as well as continue their strategy of re-allocation of spectrum from one generation to another. Some of this spectrum is at much higher frequencies and uses new technologies to deliver exceptional amounts of bandwidth to subscribers. 5G also requires significant fiber infrastructure to connect wireless access points to each other to improve the response time of the network. As wireless operators continue to transition toward 5G, they are also managing the fundamental network deployment issues of site acquisition, power, backhaul and in-building wireless proliferation.

In addition to investment required by wireless operators, the continued transition to 5G also creates an investment cycle by cable operators as they upgrade their networks to compete with fixed wireless broadband, which is a viable alternative to traditional broadband internet access. Many cable operators continue to offer or plan to offer 5G wireless services on top of their wired cable services, and one approach employs convergence techniques that utilize wired networks, such as DOCSIS or passive optical networks (PON), to support Crosshaul (xHaul) to the more heavily-densified wireless access points and radio units of 5G. All of these transitions are expected to continue driving increased investment.

Fiber Deep Deployments

Residential and business bandwidth consumption continues to grow substantially. The proliferation of OTT video, multiscreen viewing, cloud services and social media are continuing to prompt operators to accelerate fiber deployment. Operators are increasing network capacity by installing fiber deeper into their networks. Although consumer devices are increasingly connected to the network via a wireless connection such as LTE or Wi-Fi, these wireless access points must have abundant optical backhaul capacity available to provide consumers the experience they expect. Operators around the globe are deploying fiber deep to build next generation networks. These networks use the capabilities of fiber to enable consumers access to content at higher speeds with improved network response time.

As networks improve and deliver higher speed and greater reliability, many operators have chosen to provide both residential and business services over a common physical layer infrastructure, saving them time and money. In addition, with the deployments of metro cells, outdoor small cells and fixed wireless broadband to the home, these same service providers are utilizing this common physical layer infrastructure to provide connectivity to these wireless access points.

Ethernet passive optical networks (EPON) and XGS-PON are both being included in the plans of network operators, and CommScope has developed optical line terminal (OLT) and optical network terminal (ONT) equipment for both technologies.

Shift in Enterprise Spending

Several trends in the enterprise market are expected to continue creating opportunities and challenges for us. First, the shift toward mobility in business enterprises is expected to continue impacting the amount and type of structured copper connectivity needed over the longer term. As the bandwidth requirements for Wi-Fi, indoor cellular networks (private and public), and IoT devices continue to increase, more access points will be needed throughout commercial buildings. As a result, enterprises continue to adjust in-building cabling designs to deliver both power and high-speed data to those devices. Power-over-ethernet is becoming increasingly important as the number of devices used for Wi-Fi and indoor cellular networks multiplies. While enterprises continue to need copper connectivity to power edge devices, enterprises are deploying fiber more extensively in data centers. Over the next several years, we expect the growing demand for fiber and Wi-Fi solutions to result in decelerating demand for copper solutions in networks.

Due to huge increases in data traffic and migration of applications to the cloud, enterprises continue to shift spending toward multi-tenant (co-located) data centers and hyperscale cloud service providers, which offer cloud data center services as a replacement for in-house corporate data centers. Multi-tenant and hyperscale data center managers are focused on ultra-low loss, high density, scalable fiber connectivity solutions.

Enterprises are also using LTE and 5G for their own, private uses. Private networks are becoming more important to an enterprise's information technology plans and provide a level of reliable connection that enterprises have not been able to get from their Wi-Fi networks, further moving the demand of enterprise communications into the wireless domain.

Metro Cell, Distributed Antenna System (DAS) and Small Cell Investment to Enhance and Expand Wireless Coverage and Capacity

As demand growth continues to outpace macro cell capacity growth, new solutions are required for densely populated areas. Metro cells and indoor networks have emerged as important layers of the network. Metro cells are smaller outdoor cell sites, located closer to the ground, having a lower power level than traditional macro cell sites. Metro cells blend into their environment and are often found integrated with traditional street furniture, which helps alleviate zoning restrictions that have made traditional deployments difficult.

Small cell and DAS solutions address the capacity and speed requirements from an indoor perspective. These systems provide coverage and capacity to the indoor environment and reduce the load from the macro and metro layers, which improves overall network performance. Small cell and DAS systems may range from small single-operator, single-band, low-capacity systems for use in enterprise buildings to large multi-carrier, multi-technology, multi-band systems for use in high-capacity public venues.

Transition to Wi-Fi 7

Wi-Fi 7, the latest generation of tri-band (2.4/5/6GHz) Wi-Fi, is designed to deliver unparalleled performance. It aims to provide extremely high throughput, near-wired levels of latency and increased capacity, enabling immersive and content-rich wireless experiences in cutting-edge applications such as XR, gaming, 4K/8K streaming and real-time collaboration. Wi-Fi 7 is expected to deliver exceptional user experiences and empower an entirely new class of advanced connected devices and demanding applications through the most efficient use of spectrum.

Strategy

Since 2021, we have been engaged in a transformation initiative referred to as CommScope NEXT, which is designed to drive shareholder value through three pillars: profitable growth, operational efficiency and portfolio optimization. We believe these efforts are critical to making us more competitive and allowing us to invest in growth, de-leverage our indebtedness and maximize stockholder and other stakeholder value in the future. In 2022, CommScope NEXT generated positive impacts on net sales, profitability and cash flow from our execution on pricing initiatives, capacity expansion and operational efficiencies. In 2023, we experienced headwinds related to a slow-down in spending by our customers, but we continue to execute under CommScope NEXT by driving operational efficiencies and focusing on portfolio optimization so that we are in a better position to take advantage of the recovery later in 2024.

Profitable Growth

Organic growth is fundamental to achieving the financial returns that investors expect from us. While acquisitions and inorganic growth can change the structure of a business and reset financial expectations resulting in short-term financial returns, the only reliable means for consistently producing long-term positive financial performance is strong organic growth. Our plan to achieve our growth opportunities are driven by five themes:

- Become more market and customer centric – work to truly understand the needs of our customers and applications for data and video networking solutions.
- Expand to service providers outside of North America – expand market share with service providers in the rest of the world.
- Expand enterprise sales coverage – enhance sales coverage in historically underpenetrated top metropolitan statistical areas and verticals in North America, as well as targeted country/vertical combinations around the world.
- Introduce new products and scale software solutions – build and scale our differentiated products, software and technology.
- Invest in capacity – expand capacity for products with high backlog, fast growth and long-term demand visibility.

The underpinning of our growth opportunities is also optimizing pricing across our products and solutions. We are revamping our pricing processes, policies, tools and governance structure to simplify and create more ownership and accountability so that we can better react to changes in the market and maintain acceptable margins.

Operational Efficiency

We are pursuing strategic initiatives aimed at optimizing our utilization of resources by improving direct procurement processes, increasing transparency and control over indirect procurement spend, driving operational improvements to lower manufacturing costs and streamlining and optimizing our period overhead cost structure. Our management team has a strong track record of improving operational efficiency and successfully executing on formalized annual profit improvement plans, cost-savings initiatives and working capital improvements to drive future profitability and cash flow.

Portfolio Optimization

We utilize a general management model in our segments. This enables us to manage our portfolio more granularly, assign responsibilities and build a culture of accountability and ownership. We continuously review our portfolio and look for ways to better manage and optimize our product offerings. We also regularly review our product offerings, business lines and segments and holistically assess opportunities to deliver the most value to stakeholders.

For the past two years, as a step in optimizing our portfolio, we have been committed to finding the right strategic opportunity for our non-Core Home segment. As discussed above, we believe we found the right opportunity for its future success and closed on a transaction to divest of our Home business on January 9, 2024.

The Future of CommScope

We are positioned as a leader in most of our segments already and will work to defend our leadership in the more mature parts of these markets, while also shifting resources towards our targeted growth choices within them. With CommScope NEXT, we are striving to achieve the following:

- Deliver organic growth
- Create a well-positioned comprehensive portfolio of products and services
- Stimulate market leading innovation, delivering powerful software and services
- Maintain world class operational efficiency and cost structures
- Architect a simplified organization, with more accountability, responsibility and visibility

With CommScope NEXT, we are transforming our organization into one that has better operational efficiency, speed and resilience and one that can better service our existing customers, as well as new ones. As demand in our industry recovers, we expect CommScope NEXT to drive adjusted EBITDA expansion that will enable us to increase our cash flow to de-leverage our indebtedness and further invest in our growth.

Customers

Our customers include substantially all the leading global telecommunications operators, data center managers, cable television providers or MSOs and thousands of enterprise customers, including many Fortune 500 companies. Major customers and distributors include companies such as AT&T Inc.; Charter Communications, Inc.; Comcast Corporation (Comcast); Graybar Electric Company, Inc.; KGP Companies, Inc.; Power & Telephone Supply Company; Talley Inc.; TD SYNNEX Corporation; Verizon Communications Inc.; and Wesco International, Inc. (including Anixter International Inc.). For the year ended December 31, 2023, we derived approximately 17% of our consolidated net sales from our top two direct customers, however, for the years ended December 31, 2023, 2022 and 2021, no single direct customer accounted for 10% or more of our net sales.

Products from our CCS segment are primarily sold directly to cable television system operators, broadband operators and other service providers that deploy broadband networks. CCS segment products are also sold through independent distributors or system integrators for large telecommunications operators.

Products from our OWN segment are primarily sold directly to wireless operators, original equipment manufacturers (OEMs) that sell equipment to wireless operators and other service providers that deploy elements of wireless networks at the direction of wireless operators. Our customer service and engineering groups maintain close working relationships with these customers due to the significant amount of customization associated with some of these products. Although we sell to most wireless operators globally, we are dependent on a small number of large operators.

Products from our NICS segment are primarily sold through independent distributors or system integrators for large telecommunications operators and to customers in a broad range of enterprise vertical markets, including hospitality, education, smart cities, government, venues and service providers, indirectly through channel partners. We also sell directly to cable television system operators, broadband operators and service providers that deploy broadband networks. In certain circumstances, we sell NICS segment products directly to end customers, but it is a relatively small part of the overall business.

Products from our ANS segment are primarily sold directly to wireline network service providers, such as telephone companies and cable television network providers, to be deployed into their service delivery networks. In some cases, we sell through specialized resellers and distributors who primarily provide logistics support, and in certain circumstances, post-sale service and support. Our customer service and engineering groups maintain close working relationships with these customers due to the significant amount of customization associated with some of these products. We sell these products to most of the wireline and satellite operators globally.

We generally have no minimum purchase commitments from any of our distributors, system integrators, channel partners, value-added resellers, wireless operators or OEM customers, and our contracts with these parties generally do not prohibit them from purchasing from our competitors or offering products or services that compete with ours. Although we maintain long-term relationships with these parties and have not historically lost key customers, we have experienced significant variability in the level of purchases by our key customers. Any significant reduction in sales to these customers, including as a result of the inability or unwillingness of these customers to continue purchasing our products, could materially and adversely affect our business, financial condition, results of operations and cash flows. See Part 1, Item 1A, "Risk Factors."

Competition

The markets in which we participate are dynamic and highly competitive, requiring companies to react quickly to capitalize on opportunity. We retain skilled and experienced personnel and deploy substantial resources to meet the changing demands of the industry and to capitalize on change. The market for our products is also subject to rapid technological change.

We encounter significant domestic and international competition across all segments of our business. Our competitors include large, diversified companies some of whom have substantially more assets and greater financial resources than we do. We also face competition from small to medium-sized companies and less diversified companies that have concentrated efforts in one or more areas of the markets we serve. Major competitors by segment include the following: CCS segment – Amphenol Corporation, Belden Inc., Clearfield, Inc., Corning Incorporated and Sterlite Technologies Limited; OWN segment – Comba Telecom Systems Holdings Limited, Huawei Technologies Co., Ltd., Rosenberger North America and Telefonaktiebolaget LM Ericsson; NICS segment – Cisco Systems, Inc., Comba Telecom Systems Holdings Limited, Corning Incorporated, Extreme Networks, Inc., Hewlett Packard Enterprise Development LP, Huawei Technologies Co., Ltd., JMA Wireless, Juniper Networks, Inc., SOLiD, Inc. and Ubiquiti Inc.; and ANS segment – ATX Networks Corp., Casa Systems, Inc., Cisco Systems, Inc., Harmonic Inc., Technetix Group Limited, Teleste Oyj and Vecima Networks Inc.

We compete primarily on the basis of delivering solutions, product specifications, quality, price, customer service and delivery time. We believe that we differentiate ourselves in many of our markets based on our market leadership, global sales channels, intellectual property, strong reputation with our customer base, the scope of our product offering, the quality and performance of our solutions, and our service and technical support.

Competitive Strengths

We are a global leader in connectivity and essential infrastructure solutions for communications and entertainment networks, and we believe we hold leading market positions in most of our segments. Since our founding in 1976, CommScope has been a leading brand in connectivity solutions for communications networks. In the cable television and video network equipment industry, CommScope and ARRIS are longstanding market leaders, along with other brands we own such as RUCKUS, ADC and many smaller brands. In the wireless industry, ANDREW is one of the world's most recognized brands and a global leader in RF solutions for wireless networks. In the enterprise market, SYSTIMAX, NETCONNECT and UNIPRISE are recognized as global market leaders in enterprise connectivity solutions for business enterprise and data center applications.

We believe the following competitive strengths have been instrumental to our success and position us well for future growth and strong financial performance:

Differentiated Solutions Supported by Ongoing Innovation and Significant Proprietary Intellectual Property

Our integrated solutions for building better networks are differentiated in the marketplace and are a significant global competitive advantage. We invested $459.7 million in research and development (R&D) during 2023 to advance product innovation and decrease total cost of deployment and ownership. Our ongoing innovation, supported by proprietary intellectual property and technology know-how, has allowed us to build and sustain a competitive advantage.

Established Sales Channels and Customer Relationships

We serve customers in over 150 countries and have become a trusted advisor to many of them through our industry expertise, quality products, leading technology and long-term relationships. These factors enable us to provide mission-critical connectivity solutions that our customers need to build and maintain high-performing communication networks. Our customers include substantially all the leading global telecommunications operators, data center managers, cable television providers or MSOs and thousands of enterprise customers, including many Fortune 500 companies. We are a key supplier within the wireless infrastructure market and enjoy established sales channels across all geographies and technologies. Our long-standing relationships with telecommunications operators enable us to work closely with them in providing highly customized solutions aligned with their technology roadmaps. We have a global sales force with sales representatives based in North America, Europe, Latin America, Asia and other regions, and an extensive global network of channel partners, including independent distributors, system integrators and value-added resellers. Our sales force has direct relationships with our customers and end users which generates demand for our products, with a significant portion of our sales fulfilled through channel partners. Our direct sales force and channel partner relationships give us extensive reach and distribution capabilities to customers globally. Given our understanding of their existing networks, when it comes to deploying networks at scale, these customers trust CommScope and hold high regards for our ability to help them achieve their goals.

Global Scale, Manufacturing Footprint and Quality

Our global manufacturing and distribution footprint and worldwide sales force give us significant scale within our addressable markets. We believe our scale, stability and quality make us an attractive strategic partner to our large global customers, and we have been repeatedly recognized by key customers for these attributes.

Our manufacturing and distribution facilities are strategically located to optimize service levels and product delivery times. We also utilize lower-cost geographies for high labor content products and largely automated plants in higher cost regions. Most of our manufacturing employees are in lower-cost geographies such as Mexico, China, India and the Czech Republic. The combination of our dynamic manufacturing organization, our global network of third-party manufacturers and our distribution organization allows us to:

- Flex our capacity to meet market demand and expand our market position;
- Deliver high-quality customer solutions;
- Provide high customer service levels due to proximity to the customer; and
- Effectively integrate acquisitions and capitalize on related synergies.

Manufacturing and Distribution

We maintain a balance of internal and external manufacturing providers to continue offering our customers a competitive combination of quality, cost and flexibility in meeting their needs. We develop, design, fabricate, manufacture and assemble many of our products and solutions in-house at our facilities located around the world. We have strategically located our manufacturing and distribution facilities to provide superior service levels to customers. We utilize lower-cost geographies for high labor content products while investing in largely automated plants in higher-cost regions close to customers. Most of our manufacturing employees are located in lower-cost geographies such as Mexico, China, India and the Czech Republic.

In addition, we utilize contract manufacturers located throughout the world for many of our product groups, including certain products in our CCS, OWN and ANS segments and all of our Ruckus products. There can be no guarantee that the Company will be able to extend or renew agreements with contract manufacturers on similar terms, or at all.

Our global footprint allows us to mitigate macroeconomic headwinds in an everchanging environment. We continuously evaluate and adjust operations to improve service, lower cost and improve the return on our capital investments, and we expect to continue modifying our global operations to adapt to changing product demand and business conditions.

Raw Materials and Components

Our products are manufactured or assembled from both standard components and parts that are unique to our specifications. Our internal manufacturing operations are largely process oriented and we use significant quantities of various raw materials, including aluminum, copper, steel, bimetals, optical fiber and plastics and other polymers, among others. Portions of the requirements for these materials are purchased under supply arrangements where some portion of the unit pricing may be indexed to commodity market prices for these metals. We may occasionally enter forward purchase commitments or otherwise secure availability for specific commodities to mitigate our exposure to price changes for a portion of our anticipated purchases. Certain of the raw materials utilized in our products may only be available from a few suppliers, and we may enter into longer term agreements to secure access to certain key inputs. We may, therefore, encounter significant price increases and/or availability issues for the materials we obtain from these suppliers as we have seen in recent years. These supply chain constraints have limited our ability to manufacture and deliver products to our customers in the past and could have similar impacts in the future.

Our profitability has been and may continue to be materially affected by changes in the market price of our raw materials and components, most of which are linked to the commodity markets. Prices for aluminum, copper, plastics, silicon and certain other polymers derived from oil and natural gas have fluctuated substantially during the past several years. We have adjusted our prices for certain products and may have to adjust prices again. Delays in implementing price increases, failure to achieve market acceptance of price increases, or price reductions in response to a rapid decline in raw material costs, could have a material adverse impact on the results of our operations.

In addition, some of our products are assembled from specialized components and subassemblies manufactured by third-party suppliers. We depend upon sole suppliers for certain of these components, including capacitors, memory devices and silicon chips. Our results of operations have been and may continue to be materially affected if these suppliers cannot provide these components in sufficient quantity and quality on a timely and cost-efficient basis. We believe that our supply contracts and our supplier contingency plans mitigate some of this risk. Our supply agreements include technology licensing and component purchase contracts, and several of our competitors have similar supply agreements for these components. There can be no guarantee that the Company will be able to extend or renew these supply agreements on similar terms, or at all. In addition, we license software for operating network and security systems or sub-systems and a variety of routing protocols from different suppliers.

Research and Development

We operate in an industry that is subject to rapid changes in technology, and our success is largely contingent upon anticipating and reacting to such changes. Accordingly, R&D is important to preserve and expand our position as a market leader and to provide the most technologically advanced solutions in the marketplace. We invested $459.7 million in R&D during 2023, and we expect to continue with substantial investments in future years. We intend to focus our major R&D activities on high-growth opportunities such as fiber optic connectivity for fiber-to-the-x (FTTX) and data centers, Wi-Fi 7 and 6GHz, CCAP, DAA, DOCSIS 4.0, gigabit passive optical network (GPON), active and passive base-station antennas and metro cell and small cell wireless solutions. We are also developing solutions that support the convergence of wireline and wireless networks in connection with the rollout of 5G. Several of our professionals are leaders and active contributors in standards-setting organizations, which helps ensure that our products can be formulated to achieve broad market acceptance.

Backlog and Seasonality

At December 31, 2023 and 2022, we had an order backlog of $1,151.5 million and $2,925.4 million, respectively. Orders typically fluctuate from quarter to quarter based on customer demand and general business conditions. Our backlog includes only orders that are believed to be firm. Sometimes, unfilled orders may be canceled prior to shipment of goods, but cancellations historically have not been material. However, our current order backlog may not guarantee future demand. We expect a majority of our backlog as of December 31, 2023 to be recognized as revenue during 2024.

Due to the variability of shipments under large contracts, customers' seasonal installation considerations and variations in product mix and in profitability of individual orders, we can experience significant fluctuations in quarterly sales and operating income. Our operating performance is typically the weakest during the first quarter, and this pattern is expected to continue in the future. It may be more meaningful to focus on our annual rather than interim results.

Patents and Trademarks

We pursue an active policy of seeking intellectual property protection, including patents and registered trademarks, for new products and designs. For technology that is not owned by us, we have a program for obtaining appropriate licenses to help ensure that we have the necessary license coverage for our products. In addition, we have formed strategic relationships with leading technology companies to provide us with early access to technology that we believe will help keep us at the forefront of our industry.

As of December 31, 2023, we held over 13,000 patents and patent applications and over 2,700 registered trademarks and trademark applications worldwide. Over the next five years, approximately 1,650, or about 18%, of our issued patents will expire, while at the same time we intend to seek patents protecting new innovations. We consider our patents and trademarks to be valuable assets, and although no single patent is material to our overall operations, we believe the COMMSCOPE, ARRIS, RUCKUS, SYSTIMAX, NETCONNECT, NOVUX, ERA, ONECELL and HELIAX trade names and related trademarks are critical assets to our business. We intend to rely on our intellectual property rights, including our proprietary knowledge, trade secrets and continuing technological innovation, to develop and maintain our competitive position. From time to time there are disputes with respect to the ownership of the technology used in our industry and accusations of patent infringements. We will continue to protect our key intellectual property rights.

Government Regulation

We are subject to various domestic and international government regulations. For example, our international operations expose us to increased challenges in complying with anti-corruption laws and regulations of the U.S. government and various other international jurisdictions. We are also subject to governmental export and import regulations and sanctions programs that could subject us to liability or impair our ability to compete in international markets. In addition, because of the nature of information that may pass through or is stored on our solutions or networks, we and our end customers may be subject to complex and evolving U.S. and foreign laws and regulations regarding information privacy, data protection, cybersecurity and other matters. Further, we are subject to various federal, state, local and foreign environmental laws and regulations governing, among other things, substances used in our products, discharges to air and water, management of regulated materials, handling and disposal of solid and hazardous waste, and investigation and remediation of contaminated sites. These descriptions are not exhaustive, and these laws, regulations and rules frequently change and are increasing in number. See Part I, Item 1A, “Risk Factors” for additional discussion of our risks related to government laws and regulations.

Corporate Responsibility and Sustainability

We believe that corporate responsibility and sustainability means making decisions that have a positive impact on our people, planet and bottom line. Our company-wide sustainability mission is to enable faster, smarter and more sustainable solutions while demonstrating the utmost respect for our human and natural resources. Innovative technology, intelligent engineering and energy efficient design help us to build more sustainable networks that make our customers more agile, while at the same time allowing us to preserve the natural ecosystems from which we source our raw materials.

While we may provide technological solutions, it is our people who make the real difference in our communities. Their commitment to our customers, fellow employees and the communities in which they live and work drives them to provide creative solutions, services and practices that are safe and sustainable for our environment and future generations.

We understand how important it is to consider the larger impact of our actions beyond the balance sheet. We are proud of CommScope’s significant standing in one of the world’s most vital and dynamic industries. We are making great progress in delivering our sustainability actions while advancing the industry and creating a better and sustainable tomorrow. For the sake of our current and future generations, we will continue to grow as a sustainable, environmentally conscious business that benefits the whole planet.

For additional information, which is not incorporated by reference in this Annual Report on Form 10-K, see our Corporate Responsibility & Sustainability pages on the CommScope website: https://www.commscope.com/corporate-responsibility-and-sustainability/.

Human Capital Management

CommScope believes that human capital management, including attracting, developing and retaining a high-quality workforce, is critical to our long-term success. Our Board of Directors and its Committees provide oversight on a broad range of human capital management topics, including corporate culture, diversity, equity and inclusion, pay equity, health and safety, training and development and compensation and benefits.

We employed approximately 20,000 people worldwide as of December 31, 2023, with approximately 64% classified as manufacturing employees. The majority of these manufacturing employees are located in low-cost labor countries such as Mexico, China, India and the Czech Republic. Our U.S. workforce includes approximately 4,400 employees, comprised of a mix of manufacturing and non-manufacturing employees. We are party to numerous works' councils or similar statutory arrangements outside of the U.S. (none in the U.S.) and believe that relations with our employees are generally good.

Protecting the safety, health, and well-being of our associates around the world is CommScope's top priority. We strive to achieve an injury-free work environment. We believe that our culture of diversity, equity and inclusion is a competitive advantage that fuels innovation, enhances our ability to attract and retain talent and strengthens our reputation. We are also committed to providing competitive rewards and the continued growth and development of our employees through a variety of global training and development opportunities that build and strengthen employees' leadership and professional skills. Lastly, we aim to provide a positive employee experience and workplace environment that engages all employees.

We are proud that employees all over the world continue to unite behind our common purpose to "Create Lasting Connections." We collaborate and innovate to create the world's most advanced networks driven by our passionate employees who deliver on this vision every day.

Employee Health, Safety and Well-being

At CommScope, our employees' health, safety and well-being are our top priority. We are always seeking opportunities to protect the well-being of our employees, customers, suppliers, environment and communities.

A commitment to business practices that are innovative, safe and sustainable is key to our success. To achieve this, we maintain a robust Environment, Health & Safety (EHS) management system, set objectives and targets and measure them accordingly, provide necessary resources and create a comprehensive well-being and benefits program that supports a culture of safety and health first. Our EHS management system was designed and implemented based on the requirements of the International Standards of ISO45001 and ISO14001.

CommScope seeks to inspire a culture of proactive and productive health management so employees make lifestyle decisions that lead to rewarding careers and healthy, balanced lives. To realize this goal, we provide tools, services and programs that help employees achieve and maintain optimal personal health (physical, emotional and financial). The results of our efforts make us stronger and pave a path for innovation, which drives business differentiation, talent engagement and retention.

Employee Diversity, Equity and Inclusion

CommScope strives to create an equitable and inclusive environment that draws upon the strength of our diverse workforce to deliver exceptional results for our investors and all key stakeholders. CommScope's global workforce is comprised of individuals of many races, cultures, backgrounds, geographies and experiences. We focus on ensuring equity in the workplace and take pride in our diverse workforce and inclusive culture. CommScope has continued to strengthen its global Diversity & Inclusion Business Network that was established in 2020, providing over 1,600 employees with targeted opportunities to network, learn and lead, grow their careers and support their communities. CommScope's CEO also signed the pledge for the CEO Action for Diversity & Inclusion and committed to a set of actions several years ago. The results of our focused efforts make us stronger and pave a path for innovation, which drives business differentiation, talent engagement and retention.

Total Rewards

We compensate employees equitably, relative to experience and performance, regardless of gender, nationality or disability. Globally, we embrace a pay-for-performance compensation philosophy, conducting pay equity assessments to determine the results of our pay practices. CommScope's compensation plans and programs strive to: attract and retain skilled, high-performing individuals; pay base salaries and provide benefits that are competitive in our industry and the local markets in each country where we operate; and provide short- and long-term incentives (when appropriate) that are tied to exceptional employee and Company performance.

Employee Education, Training and Development

We are committed to developing the careers and capabilities of our current and future employees. We manage employees' performance and goals throughout the year, providing both classroom and virtual training and development opportunities for both our manufacturing and non-manufacturing employees. We believe employees learn best through a combination of work experience, coaching, feedback, training and education.

We augment in-person communications with technology to align and manage employees' performance and goals throughout the year, providing continuous development opportunities through coaching and feedback. We also offer classroom training as well as an online learning platform that offers a wealth of work-related employee development content (e.g., for managerial, technical and personal development). We focus heavily on interacting with our employees how, when and where it matters most.

Employee Engagement

CommScope prides itself on creating a collaborative, engaged and enabled workforce. We define engagement as the strength of the emotional connection employees feel toward their work, their teams and the Company. We believe communication and feedback are integral to building engaged employees and driving a high-performance culture. In support of this, we periodically "take the pulse" of our organization through a global employee engagement survey, one of the ways our global workforce can voice their opinions and provide ongoing feedback. Among other things, the survey seeks to understand how employees experience our Company values in their day-to-day work in order to measure our cultural health. We also ask questions to determine if employees feel a strong sense of inclusion and belonging, as well as measure overall engagement. With this valuable feedback, we can identify strengths and potential areas for focused improvement. CommScope continues to enhance employee engagement by leveraging technology, enabling managers, emphasizing communication, striving to become a destination for the best talent and providing competitive rewards, flexible work approaches and career development opportunities.

Available Information

Our Annual Report on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K and amendments to reports filed or furnished pursuant to Sections 13(a) and 15(d) of the Securities Exchange Act of 1934, as amended, are available free of charge on our website at www.commscope.com under Company — Investor Relations as soon as reasonably practicable after we electronically file such material with, or furnish it to, the Securities and Exchange Commission. The information posted to our website is not incorporated elsewhere in this Annual Report on Form 10-K.

ITEM 1A. RISK FACTORS

The following is a cautionary discussion of risks, uncertainties and assumptions that we believe are significant to our business. In addition to the factors discussed elsewhere in this Annual Report on Form 10-K, the following are some of the important factors that, individually or in the aggregate, we believe could make our results differ materially from those described in any forward-looking statements. It is impossible to predict or identify all such factors and, as a result, you should not consider the following factors to be a complete discussion of risks, uncertainties and assumptions related to us or our business.

Summary of Risk Factors

The following is a summary of some of the risks, uncertainties and assumptions that could materially adversely affect our business, financial position, results of operations and cash flows. You should read this summary together with the more detailed description of each risk factor contained below.

Competitive Risks

- •Our business is dependent upon third-party capital spending for data, communication and entertainment equipment, and reductions in such capital spending could adversely affect our business.
- •A substantial portion of our business is derived from a limited number of key customers and channel partners.
- •We face competitive pressures with respect to all our major product groups.
- •Our ability to sell our products is highly dependent on the quality of our support services after the sale, and our inability to provide adequate support after the sale would have a material adverse effect on business.
- •Changes to the regulatory environment in which our customers operate and changes in or uncertainty about government funded programs may negatively impact our business.

Financial Risks

- •We may be required to obtain additional financing in the future to address our liquidity needs, and subject to market conditions, we may seek to amend, refinance, restructure or repurchase our outstanding indebtedness and/or raise additional equity financing.
- •To service our indebtedness and pay dividends on our preferred stock, we will require a significant amount of cash, and our ability to generate sufficient cash depends on many factors beyond our control.
- •Our substantial indebtedness could adversely affect our ability to raise additional capital to fund our operations, limit our ability to react to changes in the economy or our industry, expose us to interest rate risk to the extent of our variable rate debt and prevent us from meeting our financial obligations.
- •Despite current indebtedness levels and restrictive covenants, we may still incur additional indebtedness that could further exacerbate the risks associated with our substantial financial leverage.
- •We may need to recognize additional impairment charges related to goodwill, identified intangible assets, fixed assets and right of use assets.
- •The Internal Revenue Service (IRS) may not agree that ARRIS International plc (ARRIS) was a foreign corporation for United States (U.S.) federal income tax purposes.

Supply Chain Risks

- •We are dependent on certain raw materials and components linked to the commodity markets and utilize a limited number of key suppliers for logistics support of certain of these raw materials and components, subjecting us to cost volatility and supply shortages or delays that could limit our ability to manufacture products.
- •If our integrated global manufacturing operations, including our contract manufacturers, suffer capacity constraints or production or shipping delays, we may have difficulty meeting customer demands.

Strategic Risks

- •The successful execution of our CommScope NEXT transformation plan is key to the long-term success of our business.
- •Difficulties may be encountered in the realignment of manufacturing capacity and capabilities among our global manufacturing facilities and our contract manufacturers that could adversely affect our ability to meet customer demand for our products.
- •The separation, discontinuance or divestiture of a business or product line is subject to various risks and uncertainties that could disrupt or adversely affect our business.
- •Our business strategy has historically relied, in part, on acquisitions to create growth. We may not fully realize anticipated benefits from past or future acquisitions or investments in other companies.
- •We may need to undertake additional restructuring actions in the future.
- •The Carlyle Group (Carlyle) owns a substantial portion of our equity, and its interests may not be aligned with yours.

Business and Operational Risks

- Our future success depends on our ability to anticipate and adapt to changes in technology and customer preferences and develop, implement and market innovative solutions.
- If we do not stay current with product life cycle developments, our business may suffer.
- If our products do not effectively interoperate with cellular networks and mobile devices, future sales of our products could be negatively affected.
- If our product or service offerings, including material purchased from our suppliers, have quality or performance issues, our business may suffer.
- We depend on cloud computing infrastructure operated by third parties, and any disruption in these operations could adversely affect our business.
- Our business depends on effective management information systems.
- Cybersecurity incidents, including data security breaches, ransomware or computer viruses, could harm our business by exposing us to various liabilities, disrupting our delivery of products and services and damaging our reputation.
- Climate change may have a long-term impact on our business.

Labor-Related Risks

- We may not be able to attract and retain key employees.
- Labor unrest could have a material adverse effect on our business, results of operations and financial condition.

International Risks

- Our significant international operations expose us to economic, political, foreign exchange rate and other risks.
- Additional or new tariffs or a global trade war could increase the cost of our products, which could adversely impact the competitiveness of our products.
- Our significant international operations expose us to increased challenges in complying with anti-corruption laws and regulations of the U.S. government and various other international jurisdictions.
- We are subject to governmental export and import controls and sanctions programs that could subject us to liability or impair our ability to compete in international markets.

Litigation and Regulatory Risks

- We may not be successful in protecting our intellectual property and in defending against claims that we are infringing on the intellectual property of others, and any such actions may be costly.
- Because of the nature of information that may pass through or be stored on certain of our solutions or networks, we, our vendors and our end customers are subject to complex and evolving U.S. and foreign laws and regulations regarding information privacy, data protection, cybersecurity, and other related matters.
- Compliance with current and future social and environmental laws, regulations, policies and provisions, customer and investor pressures, other efforts to mitigate climate change and potential environmental liabilities may have a material adverse impact on our business, financial condition and results of operations.

General Risks

- Any future public health crisis could materially adversely affect our business, financial condition, results of operations and cash flows.
- We do not intend to pay dividends on our common stock and, consequently, the ability of investors to achieve a return on their investment will depend on appreciation in the price of our common stock.

Competitive Risks

Our business is dependent upon third-party capital spending for data, communication and entertainment equipment, and reductions in such capital spending could adversely affect our business.

Our performance is dependent on third parties' capital spending for constructing, rebuilding, maintaining or upgrading data, communication and entertainment networks, which can be volatile and difficult to forecast. Capital spending in the communications industry is cyclical and can be curtailed or deferred on short notice. We experienced a significant decrease in customer capital spending in 2023, which negatively impacted our results of operations, and we may continue to experience significant fluctuations in sales and operating income due to the volatility in our industry. A variety of factors affect the timing and amount of capital spending in the communications industry, including:

- general economic and market conditions, including increased costs due to rising inflation or interest rates;
- customer-specific financial conditions or budget allocation decisions;
- competitive pressures, including pricing pressures;
- competing technologies;
- timing and adoption of the global rollout of new technologies;
- customer acceptance of new technologies and services offered;
- foreign currency fluctuations;
- seasonality of outdoor deployments;
- rollout of government funding for certain initiatives;
- changes in customer preferences or requirements;
- availability and cost of capital;
- governmental regulation;
- demand for network services;
- consumer demand for video content and pay TV services;
- variability of shipments under large contracts;
- industry consolidation; and
- real or perceived trends or uncertainties in these factors.

As a result of these factors, we may not be able to maintain or increase our sales in the future, and our business, financial condition, results of operations and cash flows could be materially and adversely affected.

A substantial portion of our business is derived from a limited number of key customers and channel partners.

Our customer base includes direct customers, original equipment manufacturers (OEMs) and channel partners, which include distributors, system integrators, value-added resellers and sales representatives. For the year ended December 31, 2023, we derived approximately 17% of our consolidated net sales from our top two direct customers. The concentration of our net sales with these key customers subjects us to a variety of risks, including:

- lower sales volumes that could result from the loss of one or more of our key customers;
- dependency on customers with substantial purchasing power and leverage in negotiating contractual obligations as well as the operational structure of the relationship, resulting in potential reductions in profit;
- less efficient operations that could result in higher costs from an inability to accurately forecast and plan for volatile spending patterns of key customers;
- financial difficulties experienced by one or more of our key customers that could result in reduced purchases of our products and/or delays or difficulties in collecting accounts receivable balances;

- election by our key customers to purchase products from our competitors in order to diversify their supplier base and dual-source key products, resulting in reduced purchases of our products; and
- reductions in inventory levels held by channel partners and OEMs, which may be unrelated to purchasing trends by end customers.

We are also exposed to similar risks to the extent that we have significant indirect sales to one or more end-users of our products, who may also be a direct customer.

A material portion of our sales is derived through our channel partners, including distributors, systems integrators and value-added resellers. Our channel partners have experienced financial difficulties in the past that have adversely affected our collection of accounts receivable. Our exposure to credit risks of our channel partners may increase if our channel partners and their end customers are adversely affected by global or regional economic conditions. One or more of these channel partners could delay payments or default on credit extended to them, either of which could materially and adversely affect our business, financial condition, results of operations and cash flows.

We generally have no minimum purchase commitments with any of our distributors, value-added resellers, operators, OEMs or other customers, and our contracts with these parties generally do not prohibit them from purchasing or offering products or services that compete with ours. We have historically experienced variability in the level of purchases by our key customers and expect that similar variability could affect future sales. Any significant reduction in sales to these customers, including as a result of the inability or unwillingness of these customers to continue purchasing our products, could materially and adversely affect our business, financial condition, results of operations, cash flows and stock price.

We face competitive pressures with respect to all our major product groups.

Competition in our industry depends on a number of factors, including: innovative product and service solution offerings; the ability to adapt to changing markets and customer preferences; product and service quality; timing of the introduction of new products and services; speed of delivery; pricing; and customer service, including the total customer experience. In each of our major product groups, we compete with a substantial number of foreign and domestic companies, some of which have greater financial, technical, marketing and other resources or lower operating costs. They may also have broader product offerings and market focus. This gives many of these enterprises a competitive advantage to withstand any significant reduction in capital spending by customers in our markets over the long term. Further, our industry continues to consolidate, and the combination of any of our competitors could further increase these advantages and result in competitors with broader market presence.

Some competitors may be able to bundle their products and services together and may be capable of delivering more complete solutions that better meet customer preferences than we are able to provide, which may cause us to lose sales opportunities and revenue. Competitors' actions, such as price reductions, acceptance of high-risk contractual terms or the introduction of new, innovative products and services, and the use of exclusively price-driven auctions by customers, have caused lost sales opportunities in the past and may cause us to lose sales opportunities in the future.

The rapid technological changes occurring in the communications industry could also lead to the entry of new competitors against whom we may not be able to compete successfully. For example, as networks become more virtualized, the functionality of our products is at risk of being subsumed by competitors who utilize software to provide the same functions as our products. A related trend that could affect us is the emerging interest in distributed access architecture (DAA), which disaggregates some of the functions of the converged cable access platform (CCAP) and the access and transport platforms to enable deployment of these functions in ways that could reduce traditional operator capital expenditures in hybrid fiber-coaxial. We have developed and deployed a line of DAA products, but some operators may not be aligned on the specific implementations of DAA and we could lose market share to competitors. Service providers also have the goal of virtualizing CCAP management and control functions as they deploy DAA, and although we are developing a fully virtualized CCAP product, this could potentially enable new competitors to enter the market and reduce operator dependence on our products. In our mobile wireless markets, the shift to 5G includes the deployment of new spectrum in higher frequency bands where larger available bandwidths enable a significant increase in network capacity. In many cases, massive multiple-in-multiple-out (MIMO) technology (active antennas) is the most effective way to deliver coverage in these bands. Consequently, 5G deployments present an inherent headwind to our traditional passive base station antenna business. We are developing technologies and new products to address this shift from passive to active antennas, but we may not be able to completely offset this trend. As there is technology evolution or transformation within the industry, whether it be DOCSIS 4.0, PON, Wi-Fi technology or the shift to 5G, there is a risk that our market position would be weakened. If any of our competitors' products or technologies were to become the industry standard, our business would be negatively affected.

The continued industry shift toward open standards may result in an increase in competition for our products that may adversely impact our future revenues and margins. In addition, many of our customers participate in "technology pools" and increasingly request that we donate a portion of our source code used by customers to these pools, which may impact our ability to recapture the R&D investment made in developing such code. We believe that we will be increasingly required to work with third-party technology providers. As a result, we expect the shift to more open standards may require us to license software and other components indirectly to third parties via various open-source or royalty-free licenses. In some circumstances, our use of such open-source technology may include technology or protocols developed by standard-setting bodies, other industry forums or third-party companies. The terms of the open-source licenses granted by such parties, or the granting of royalty-free licenses, may limit our ability to commercialize products that utilize such technology, which could have a material adverse effect on our results of operations.

In some instances, our customers themselves may also be our competition in other business areas. Some of our customers may develop their own software requiring support within our products and/or may design and develop products of their own that are produced to their own specifications directly by a contract manufacturer. Further, if we are unable to transform our business processes to support changing customer expectations and deliver a superior total customer experience, we may lose sales opportunities in the future. We are also facing significant and increased competition from original design manufacturers (ODMs) and contract manufacturers who are selling and attempting to sell their products directly to service providers.

We cannot assure you that we will continue to compete successfully with our existing competitors or with new competitors. If we are unable to compete in any of our markets at the same level as we have in the past or are forced to reduce the prices of our products in order to continue to be competitive, our business, financial condition, results of operations and cash flows could be materially and adversely affected.

Our ability to sell our products is highly dependent on the quality of our support services after the sale, and our inability to provide adequate support after the sale would have a material adverse effect on our business.

After our products are deployed, our channel partners and end customers depend on our support organization to resolve any issues relating to our products. A high level of support is important for the successful marketing and sale of our products. In many cases, our channel partners provide support directly to our end customers. We do not have complete control over the level or quality of support provided by our channel partners. These channel partners may also provide support for other third-party products, which may potentially distract resources from support for our products. If we and our channel partners do not effectively assist our end customers in deploying our products, quickly resolving post-deployment issues and provide effective ongoing support, it could adversely affect our ability to sell our products to existing end customers and could harm our reputation with potential end customers. In some cases, we guarantee a certain level of performance to our end customers, which could prove to be resource-intensive and expensive for us to fulfill if unforeseen technical problems arise.

Many of our service providers and large enterprise end customers have more complex networks and require higher levels of support than our smaller end customers. In addition, given the extent of our international operations, our support organization faces challenges, including those associated with delivering support, training and documentation in languages other than English. Our failure to maintain high-quality support and services could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Changes to the regulatory environment in which our customers operate and changes in or uncertainty about government funded programs may negatively impact our business.

The telecommunications and cable television industries are subject to significant and changing federal and state regulation, both in the U.S. and other countries. Many of our customers, such as internet service providers, are subject to various rules and regulations, and changes to such rules and regulations could adversely impact our customers' decisions regarding capital spending. We, as well as some of our customers, also participate in and benefit from government funded programs that encourage the development of network infrastructures such as the Infrastructure Investment and Jobs Act (IIJA), Rural Digital Opportunity Fund (RDOF) and American Rescue Plan Act (ARPA). Changes in government programs in our industry or uncertainty regarding future changes could adversely impact our customers' decisions regarding capital spending, which could decrease demand for our products and could materially and adversely affect our business, financial condition, results of operations, cash flows and stock price.

Financial Risks

We may be required to obtain additional financing in the future to address our liquidity needs, and subject to market conditions, we may seek to amend, refinance, restructure or repurchase our outstanding indebtedness and/or raise additional equity financing.

We currently believe that our existing cash and cash equivalents, combined with availability under our asset-based revolving credit facility (Revolving Credit Facility), will be sufficient to meet our presently anticipated future cash needs. However, we may be required to obtain additional financing in the future to address our liquidity needs, and subject to market conditions, we may from time to time seek to amend, refinance, restructure, exchange or repurchase our outstanding indebtedness and/or raise additional equity or other financing. For example, our $1.3 billion of 6.0% senior notes due June 15, 2025 will likely require us to seek refinancing or some other restructuring of our debt in 2024. Any debt we incur in the future may have terms (including cash interest rate, financial covenants and covenants limiting our operating flexibility or ability to obtain additional financings) that are not favorable to us, and any such additional equity financing may dilute the economic and/or voting interests of our existing stockholders, may be preferred in right of payment to our outstanding common stock or confer other privileges to the holders and may contain financial or operational covenants that restrict our operating flexibility or ability to obtain additional financings. Furthermore, our failure to obtain any necessary financing, amendment, refinancing, restructuring, exchange or repurchases could have a material and adverse effect on our results of operations, cash flows, financial condition and liquidity.

We may experience volatility in cash flows between periods due to, among other reasons, variability in the timing of vendor payments and customer receipts. We may, from time to time, seek to obtain alternative sources of financing, by borrowing additional amounts under our Revolving Credit Facility, issuing debt or equity securities or incurring other indebtedness, if market conditions are favorable, utilizing trade credit, selling assets (including businesses or business lines) or securitizing receivables to meet future cash needs or to reduce our borrowing costs. Any issuance of equity or debt may be for cash or in exchange for our outstanding securities or indebtedness, or a combination thereof.

We are aware that our outstanding debt securities and debt under our credit facilities are currently trading at substantial discounts to their respective principal amounts. In order to reduce future cash interest payments, as well as future amounts due at maturity or upon redemption, we may, from time to time, purchase such debt for cash, in exchange for common or preferred stock or debt, or for a combination thereof, in each case in open-market purchases and/or privately negotiated transactions, tender offers or exchange offers and upon such terms and at such prices as we may determine. Any such transactions will be dependent upon several factors, including our liquidity requirements, contractual restrictions, general market conditions and applicable regulatory, legal and accounting factors. Whether or not we engage in any such transactions will be determined at our discretion. The amounts involved in any such transactions, individually or in the aggregate, may be material.

To service our indebtedness and pay dividends on our preferred stock, we will require a significant amount of cash, and our ability to generate sufficient cash depends on many factors beyond our control.

Our operations are conducted through our global subsidiaries, and our ability to make cash payments on our indebtedness and pay cash dividends on our preferred stock will depend on the level of earnings and distributable funds from our subsidiaries. Certain of our subsidiaries may have limitations or restrictions on paying dividends and otherwise transferring funds to us. Our ability to make cash payments on and to refinance our indebtedness will depend upon our financial condition and operating performance, which are subject to prevailing economic and competitive conditions and to financial, business, legislative, regulatory and other factors beyond our control. We might not be able to achieve a level of cash flows from operating activities or transfer sufficient funds from our subsidiaries to permit us to pay the principal, premium, if any, and interest on our indebtedness and dividends on our preferred stock.

If we are unable to generate sufficient cash flow or are otherwise unable to obtain funds necessary to meet required payments of principal, premium, if any, and interest on our indebtedness or if we fail to comply with the various covenants in the instruments governing our indebtedness and we are unable to obtain waivers from the required lenders or noteholders, we could be in default under the terms of the agreements governing such indebtedness. In the event of such default, the holders of our indebtedness could elect to declare all the funds borrowed to be due and payable, together with accrued and unpaid interest. The lenders under our Revolving Credit Facility could elect to terminate their commitments and cease making further loans, and the holders of our secured indebtedness could institute foreclosure proceedings against our assets. As a result, we could be forced into bankruptcy or liquidation.

Our substantial indebtedness could adversely affect our ability to raise additional capital to fund our operations, limit our ability to react to changes in the economy or our industry, expose us to interest rate risk to the extent of our variable rate debt and prevent us from meeting our financial obligations.

As of December 31, 2023, we had approximately $9.3 billion of indebtedness. As of December 31, 2023, we had no outstanding loans under our Revolving Credit Facility and the remaining availability was $688.0 million, reflecting a borrowing base subject to maximum capacity of $1,000.0 million reduced by $97.9 million of outstanding letters of credit. Our ability to borrow under our Revolving Credit Facility depends, in part, on inventory, accounts receivable and other assets that fluctuate from time to time and may further depend on lenders' discretionary ability to impose reserves and availability blocks.

Our interest cost on our senior secured term loan due 2026 (2026 Term Loan) and our Revolving Credit Facility, which make up about $3.1 billion of our indebtedness, is variable and subject to the risk of changes in interest rates. As the Federal Reserve has increased interest rates in 2023, we have seen increased interest cost which has adversely impacted our results of operations and cash flows. This may continue into 2024 if the Federal Reserve continues to maintain higher interest rates or chooses to raise interest rates further. We have entered into certain hedging agreements to reduce our exposure to variable rate debt.

Other consequences our substantial indebtedness has had and could continue to have on our business are as follows:

- limit our ability to obtain additional financing for working capital, capital expenditures, acquisitions, investments and other general corporate purposes;
- require a substantial portion of our cash flows to be dedicated to debt service payments and reduce the amount of cash flows available for working capital, capital expenditures, investments or acquisitions and other general corporate purposes;
- place us at a competitive disadvantage compared to certain of our competitors who have less debt;
- hinder our ability to adjust rapidly to changing market conditions;
- limit our ability to secure adequate bank financing or our ability to refinance existing indebtedness in the future with reasonable terms and conditions, or at all; and
- increase our vulnerability to and limit our flexibility in planning for, or reacting to, a potential downturn in general economic conditions or in one or more of our businesses.

The London Interbank Offered Rate (LIBOR) had historically been the reference interest rate in our variable rate debt agreements, but we transitioned our Revolving Credit Facility to Secured Overnight Financing Rate (SOFR) as the reference rate in 2022 and we amended our 2026 Term Loan to replace LIBOR with SOFR in the first half of 2023. SOFR is calculated differently than LIBOR and they have inherent differences, which could give rise to uncertainties, including limited historical data and volatility. While we do not expect the transition to SOFR to have a material adverse effect on our business, the full effects of the transition to SOFR remain uncertain.

In addition, the indentures and credit agreements governing our indebtedness contain affirmative and negative covenants that limit our ability to engage in activities that may be in our long-term best interests. Our failure to comply with those covenants could result in an event of default which, if not cured or waived, could result in the acceleration of all of our debt and permit our secured creditors to institute foreclosure proceedings against our assets.

Despite current indebtedness levels and restrictive covenants, we may still incur additional indebtedness that could further exacerbate the risks associated with our substantial financial leverage.

We may incur significant additional indebtedness in the future under the agreements governing our indebtedness. Although the indentures and the credit agreements governing our indebtedness contain restrictions on the incurrence of additional indebtedness, these restrictions are subject to a number of thresholds, qualifications and exceptions, and additional indebtedness incurred in compliance with these restrictions could be substantial. Additionally, these restrictions permit us to incur obligations that, although preferential to our common stock in terms of payment, do not constitute indebtedness.

We may need to recognize additional impairment charges related to goodwill, identified intangible assets, fixed assets and right of use assets.

We have substantial balances of goodwill and identified intangible assets. As of December 31, 2023, goodwill and identified intangible assets represented approximately 55% of our total assets. We are required to test goodwill for possible impairment on the same date each year and on an interim basis if there are indicators of a possible impairment. We have recognized substantial impairment charges related to goodwill, including $571.4 million in 2023 and $1,119.6 million in 2022. There were no asset impairment charges related to goodwill in 2021. As of the October 2023 annual impairment test, the fair value of our ANS and Building and Data Center Connectivity (BDCC) reporting units equaled their respective carrying values and slight changes in significant assumptions or business factors could result in material impairment. In the future, if we are unable to improve our results of operations and cash flows, or other indicators of impairment exist, such as a sustained significant decline in our share price and market capitalization, we may incur material charges against earnings relating to our remaining goodwill.

We are also required to evaluate identified intangible assets, fixed assets and right of use assets for impairment if there are indicators of a possible impairment. In the past, due to revisions in financial performance outlooks or deterioration in certain markets, we have recognized significant impairment charges on identified intangible assets and fixed assets. In the future, we may again determine that one or more of our long-lived assets is impaired and additional impairment charges may be recognized that could have a material adverse effect on our financial condition and results of operations.

The IRS may not agree ARRIS was a foreign corporation for U.S. federal income tax purposes.

Following the ARRIS 2016 combination with Pace plc (the “Pace combination”), ARRIS was incorporated under the laws of England and Wales and a tax resident in the United Kingdom (U.K.) for U.K. tax purposes. There is a risk that the IRS does not agree that ARRIS was a foreign corporation for U.S. federal income tax purposes in periods prior to the acquisition of ARRIS by CommScope and we could be subject to substantial additional U.S. taxes. For U.K. tax purposes, ARRIS was expected to be treated as a U.K. tax resident for all periods prior to the acquisition of ARRIS by CommScope and following the Pace combination, regardless of how ARRIS was treated in the U.S. Therefore, if ARRIS was treated as a U.S. corporation for U.S. federal income tax purposes, we could be liable for both U.S. and U.K. taxes in certain periods prior to the acquisition of ARRIS by CommScope, which could have a material adverse effect on our financial condition, results of operations and cash flows.

Supply Chain Risks

We are dependent on certain raw materials and components linked to the commodity markets and utilize a limited number of key suppliers for logistics support of certain of these raw materials and components, subjecting us to cost volatility and supply shortages or delays that could limit our ability to manufacture products.

We are dependent on certain raw materials and components linked to the commodity markets, and our profitability may be materially affected by changes in the market price. The principal raw materials and components we purchase are aluminum, copper, steel, bimetals, optical fiber, plastics and other polymers, capacitors, memory devices and silicon chips. Prices for aluminum, copper, steel, silicon, fluoropolymers and certain other polymers have experienced significant volatility in the past as a result of changes in the levels of global demand, supply disruptions, including port, transportation and distribution delays or interruptions, and other factors. As a result, in the past we saw significant increases in costs that negatively impacted our results of operations. We adjusted our prices for most of our products, but if we see significant increases in costs again, we may have to adjust prices in the future. Delays in implementing price increases or a failure to achieve market acceptance of price increases has in the past, and could in the future, have a material adverse impact on our results of operations. Conversely, in an environment of falling commodities prices, we may be unable to sell higher-cost inventory before implementing price decreases, which could have a material adverse impact on our business, financial condition and results of operations.

We also utilize a limited number of key suppliers for logistics support of certain of our raw material and component purchases, including certain semiconductors, memory and chip capacitors, polymers, copper rod, copper and aluminum tapes, fine aluminum wire, steel wire, optical fiber, circuit boards and other electronic components, subassemblies and modules. Certain of our suppliers are sole source suppliers, and a number of our agreements with suppliers are short-term in nature. Our reliance on sole or limited suppliers and our reliance on subcontractors involves several risks, including a potential inability to obtain an adequate supply of required materials, components and other products, and reduced control over pricing, quality, terms and conditions of purchase and timely delivery. Coming out of the COVID-19 pandemic, we saw shortages in supply of memory devices, capacitors and silicon chips that negatively impacted our ability to deliver on a timely basis and increased our product costs, which unfavorably impacted our results of operations, financial condition and cash flows and increased our risk of excess and obsolescence component inventory.

Our key suppliers have experienced in the past, and could experience in the future, production, operational or financial difficulties, or there may be global shortages and pricing inflation of certain raw materials or components we use. Our inability to find sufficient sources of supply on reasonable terms could impact our ability to manufacture products in a cost-effective manner. We have adjusted our market prices for certain of our products as component prices have changed, but we may not be able to pass along all further cost increases to our customers, which could have a material adverse effect on our gross margin and results of operations, especially in a highly inflationary environment. Our ability to ship products on a timely basis has been and may continue to be unfavorably impacted, which could damage relationships with current and prospective customers and potentially have a material adverse effect on our business.

We also source many of our components from international markets. Any change in the laws and policies of the U.S. or other countries affecting trade is a risk to us. To the extent there are unfavorable changes imposed by the U.S. or other countries and/or retaliatory actions taken by trading partners, such as the addition of new tariffs or trade restrictions, we may experience material adverse impacts on earnings. For a more complete discussion of our risks related to tariffs and trade restrictions, see the risk factor, “Additional tariffs or a global trade war could increase the cost of our products, which could adversely impact the competitiveness of our products” under our “International Risk Factors” in this Item 1A. Risk Factors section.

If our integrated global manufacturing operations, including our contract manufacturers, suffer capacity constraints or production or shipping delays, we may have difficulty meeting customer demands.

Disruption of our ability to produce at or distribute from our manufacturing or contract manufacturing facilities could adversely affect our ability to manufacture products in a cost-effective and timely manner. We experienced lost sales opportunities in the past due to lack of capacity to meet the demand for certain of our products. If we cannot ramp up capacity fast enough to meet customer demand in the future, we may experience lost sales opportunities, lose market share and experience customer relations problems, which could have a material adverse effect on our business, financial condition, results of operations and cash flows.

We rely on unaffiliated contract manufacturers, both domestically and internationally, to produce certain products or key components of products. Our reliance on these contract manufacturers reduces our control over the manufacturing process and exposes us to risks, including reduced control over quality assurance, product supply and costs and timing. Any manufacturing disruption by our contract manufacturers could severely impair our ability to fulfill orders. Our reliance on outsourced manufacturers also increases the potential for infringement or misappropriation of our intellectual property.

If our internal manufacturing operations or contract manufacturers suffer delays or disruptions in production or other operations for any reason, including financial instability of the contract manufacturer, labor disturbances or shortages, fires, electrical outages, cybersecurity incidents, pandemics/epidemics, severe weather events, natural disasters, geopolitical instability, acts of violence or terrorism, shipping interruptions including port distribution delays or interruptions, increased manufacturing lead times, capacity constraints or quality control problems in their manufacturing operations, failure to meet our future requirements for timely delivery or some other catastrophic event, our ability to manufacture products at our manufacturing or contract manufacturer facilities and ship products to our customers in a cost-effective and timely manner could be impaired, which could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Our contract manufacturers typically fulfill our supply requirements on the basis of individual orders. In most cases, we do not have long-term contracts with our contract manufacturers that guarantee capacity, the continuation of particular pricing terms or the extension of credit limits. Accordingly, our contract manufacturers are not always obligated to continue to fulfill our supply requirements, which could result in supply shortages, and the prices we are charged for manufacturing services could be increased on short notice. If our manufacturers are unable or unwilling to continue manufacturing our products in required volumes, we will be required to identify one or more acceptable alternative manufacturers to satisfy our demand. There is no assurance that we would be able to identify suitable alternative manufacturing partners on a timely basis, on terms that are acceptable to us, or at all.

Some of our manufacturing and contract manufacturing facilities rely on aging production equipment and information technology infrastructure, and if we fail or our contract manufacturers fail to properly maintain or update this equipment, it could affect our ability to manufacture or ship products.

Strategic Risks

The successful execution of our CommScope NEXT transformation plan is key to the long-term success of our business.

Over the last several years, we have been executing under a business transformation initiative called CommScope NEXT, designed to drive stakeholder value. CommScope NEXT could result in changes to our business that may result in a number of risks and uncertainties, including the following: lost customers or reduced sales volumes if customers do not accept higher pricing, our new product offerings or if we discontinue or divest of product lines; higher one-time costs such as restructuring costs and transaction, transformation and integration costs; the loss of key management and other employees if we are not successful in getting employee buy-in for CommScope NEXT; and additional supply chain disruptions or higher costs of supplies if we do not successfully execute our projects related to direct and indirect procurement. The implementation of CommScope NEXT may take longer than anticipated, and once implemented, we may not realize, in full or in part, the anticipated benefits or such benefits may be realized more slowly than anticipated. Any failure to realize benefits could have a material adverse effect on our business, financial condition, results of operations, cash flows and stock price.

Difficulties may be encountered in the realignment of manufacturing capacity and capabilities among our global manufacturing facilities and our contract manufacturers that could adversely affect our ability to meet customer demand for our products.

We periodically realign manufacturing capacity among our global facilities and contract manufacturers in order to reduce costs by improving manufacturing efficiency and to strengthen our long-term competitive position. The implementation of these strategic initiatives may include significant shifts of production capacity among facilities and contract manufacturers. We have done this in the past related to the integration of certain acquisitions, including the integration of the ARRIS business. Also, in prior years, with some of the uncertainties in the U.S. trade tariff environment, we transitioned manufacturing for certain impacted products to non-tariff countries. In addition, in response to intermittent shutdowns of our facilities during the COVID-19 pandemic, we transitioned certain manufacturing to less impacted facilities. These changes are time-consuming and costly, and changes in our contract manufacturers or manufacturing locations may cause significant interruptions in supply if the manufacturers have difficulty manufacturing products to our specifications. There are significant risks inherent in the implementation of these initiatives, including our failure to ensure the following: adequate inventory on hand or production capacity to meet customer demand while capacity is being shifted among facilities; maintaining product quality as a result of shifting capacity; adequate raw material and other service providers to meet the needs at the new production locations; ability to successfully remove, transport and re-install equipment; and availability of adequate supervisory, production and support personnel to accommodate the shifted production. In the event manufacturing realignment initiatives are not successfully implemented, we could experience lost future sales and increased operating costs, as well as customer relations problems, any of which could have a material adverse effect on our business, financial condition, results of operations and cash flows.

The separation, discontinuance or divestiture of a business or product line is subject to various risks and uncertainties that could disrupt or adversely affect our business.

To better optimize our portfolio of products, we have divested of the Home Networks business and we may in the future decide to separate, discontinue or divest of other businesses or product lines that we believe are not core to CommScope's business, or where we believe the separation, discontinuation or divestiture will be accretive to stakeholders. A plan to separate, discontinue or divest a business or product line is complex in nature and can be affected by unanticipated developments or changes, including changes in the macroeconomic, regulatory or political environment, changes in credit or equity markets or changes in other market conditions.

If we do choose to separate, discontinue or divest of a business or product line and successfully complete the separation plan, we cannot assure you or any of our stakeholders that we will achieve the expected benefits. Upon completion, we would also be a smaller, less diversified company and may be more vulnerable to changing market conditions. In addition, we will continue to incur ongoing costs some of which may exceed our estimates and may be stranded.

Whether or not a separation plan is completed, our businesses may face risks and uncertainties, including, but not limited to: the diversion of senior management's attention from ongoing business concerns; maintaining employee morale and retaining key management and other employees; retaining existing business and operational relationships, including with customers, suppliers and employees, and attracting new business and operational relationships; foreseen and unforeseen costs and expenses; and potential negative reactions from the financial markets if we fail to complete a separation plan as expected, within the anticipated time frame, or at all. Any of these factors could have a material adverse effect on our business, financial condition, results of operations, cash flows and stock price.

Our business strategy has historically relied, in part, on acquisitions to create growth. We may not fully realize anticipated benefits from past or future acquisitions or investments in other companies.

Our business strategy has historically relied, in part, on acquisitions to create growth, such as our acquisition of ARRIS in 2019 and our acquisition of TE Connectivity's Broadband Network Solutions business (the BNS business) in 2015. We anticipate that a portion of our future growth may be accomplished by acquiring existing businesses, products or technologies. We cannot guarantee that we will be able to identify suitable acquisition opportunities or obtain the necessary financing on acceptable terms to provide these future growth opportunities. We may spend time and money investigating and negotiating with potential acquisition or investment targets without completing the transaction, which may divert or waste resources.

All acquisitions involve risks, such as the assumption of additional liabilities and expenses, issuance of debt, incurrence of transaction and integration costs, diversion of management's attention from other business concerns, assumption of unknown contingent liabilities, unanticipated litigation costs and falling short of growth expectations. There are also significant challenges to integrating an acquired operation into our business, including, but not limited to, successfully managing the operations, manufacturing facilities and technology of the combined business; integrating the sales organizations; maintaining and increasing the customer base; retaining key employees, suppliers and distributors; integrating management information systems, including enterprise resource planning (ERP) systems; integrating inventory management and accounting activities; integrating R&D activities; navigating markets in which we potentially have limited or no prior experience; integrating and implementing effective disclosure controls and procedures and internal controls over financial reporting; and the impact of goodwill or other impairment charges, amortization costs for acquired intangible assets and acquisition accounting treatment, including the loss of deferred revenue and increases in the fair values of inventory and other acquired assets, on our financial condition and results of operations. Furthermore, such acquisitions may be dilutive to our financial results. Although we typically expect to realize strategic, operational and financial benefits as a result of our past and future acquisitions and investments, we cannot predict or guarantee whether and to what extent anticipated cost savings, synergies and growth prospects will be achieved. For example, we have not fully achieved the expected growth prospects associated with the ARRIS acquisition and that has had adverse effects on our financial condition, results of operations, cash flows and stock price.

We may need to undertake additional restructuring actions in the future.

We have previously recognized restructuring charges in conjunction with the implementation of initiatives to reduce costs and improve the efficiency of our operations and to integrate acquisitions. For example, the CommScope NEXT actions to date have included the closure of a manufacturing facility, reduction in our real estate footprint, including the consolidation of distribution facilities, as well as workforce reductions. In prior years, we have also undertaken a number of initiatives to support the integration of acquisitions, such as the 2019 acquisition of the ARRIS business and the 2015 acquisition of the BNS business. These initiatives also included the closure of manufacturing facilities, consolidation of distribution centers and other real estate and various other workforce reductions. As a result of the continued efforts related to CommScope NEXT, changes in business conditions and other developments, we may need to initiate additional restructuring actions that could result in workforce reductions and restructuring charges, which could adversely and materially affect our cash flows.

Carlyle owns a substantial portion of our equity and its interests may not be aligned with yours.

Funding for the acquisition of ARRIS included an investment by Carlyle in our Series A Convertible Preferred Stock. As a result, Carlyle owns approximately 17% of our common stock on an if-converted basis and has the right to designate up to two directors on our Board of Directors (Board). In addition, certain of our existing directors are senior advisors to Carlyle. As a result, Carlyle has significant influence on our business. Circumstances may occur in which the interests of Carlyle could conflict with the interests of our other stockholders.

Business and Operational Risks

Our future success depends on our ability to anticipate and adapt to changes in technology and customer preferences and develop, implement and market innovative solutions.

Many of our markets are characterized by rapid advances in information processing and communications capabilities that require increased transmission speeds and density and greater bandwidth. These advances require significant investments in R&D in order to improve the capabilities of our products and services and develop new offerings or solutions that will meet the needs and preferences of our customers. There can be no assurance that our investments in R&D will yield marketable product or service innovations.

We may not be successful in our ongoing innovation efforts if, among other things, our products and services are not cost effective, brought to market in a timely manner, compliant with evolving industry standards, accepted in the market or recognized as meeting customer requirements. We could experience a material adverse effect on our business, financial condition, results of operations and cash flows if we are not successful in our ongoing innovation efforts.

As our products become more complex and customer preferences continue to change, we may encounter difficulties in meeting such preferences, including performance, service and delivery expectations. Developing our products is expensive, complex and involves uncertainties. Each phase in the development of our products presents serious risks of failure, rework or delay, any one of which could impact the timing and cost-effective development of such product and could jeopardize end customer acceptance of the product. We have experienced in the past, and may in the future experience, design, manufacturing, marketing and other difficulties that could delay or prevent the development, introduction or marketing of new products and enhancements. Any such difficulties or delays could have a material adverse effect on our results of operations, financial condition and cash flows.

If we do not stay current with product life cycle developments, our business may suffer.

To compete successfully, we must continue to innovate in anticipation of both our customers' needs and developing industry trends, which require us to quickly design, develop, manufacture and sell new or enhanced products that provide increasingly higher levels of performance and reliability. If we do not have competitively priced, market-accepted products available to meet our customers' planned roll-out of new technologies, we may miss a significant opportunity and our business, financial condition, results of operations and cash flows could be materially and adversely affected.

The introduction of new or enhanced products requires that we carefully manage the transition from older products to minimize disruption in customer ordering practices and ensure that new products can be timely delivered to meet our customers' demand. If we are not able to support our customers in an effective and cost-efficient manner as they advance from older generation networks or as they expand the capacity of their networks, our business will suffer.

Furthermore, there are several major trends that we expect to continue to impact the enterprise market and product life cycles, including the shift to 5G, enterprise shifts toward mobility indoors and adjustments of in-building cabling designs to support Wi-Fi, more access points and in-building cellular applications. Due to significant increases in data traffic and migrations of applications to the cloud, enterprises are also shifting spending toward multi-tenant data centers and hyperscale cloud service providers, which offer cloud data centers services as a replacement to in-house corporate data centers. As a result, there is growing demand for fiber solutions and decelerating demand for copper solutions. If we are unable to continue to support customers in these transitions, or if sales of copper products decline faster than expected, we could experience a material adverse effect on our business, financial condition, results of operations and cash flows.

In order to stay current with product life cycle developments, we have formed strategic relationships with leading technology companies to provide us with early access to technology that we believe will help keep us at the forefront of our industry. Our strategic alliances are generally based on business relationships that have not been the subject of written agreements expressly providing for the alliance to continue for a significant period of time, and the loss of any such strategic relationship could have a material adverse effect on our business and results of operations.

If our products do not effectively interoperate with cellular networks and mobile devices, future sales of our products could be negatively affected.

Many of our products are designed to interoperate with cellular networks and mobile devices using Wi-Fi technology. These networks and devices have varied and complex specifications. As a result, we must ensure that our products interoperate effectively with these existing and planned networks and devices. To meet these requirements, we must continue development and testing efforts that require significant capital and employee resources. We may not accomplish these development efforts quickly or cost-effectively, or at all. If our products do not interoperate effectively, orders for our products could be delayed or cancelled, which would harm our revenue, operating results and reputation, potentially resulting in the loss of existing and potential end customers. The failure of our products to interoperate effectively with cellular networks or mobile devices may result in significant warranty, support and repair costs, may divert the attention of our engineering personnel from our product development efforts and may cause significant customer relations problems. In addition, our end customers may require our products to comply with new and rapidly evolving security or other certifications and standards. If our products are late in achieving or fail to achieve compliance with these certifications and standards, or our competitors first achieve compliance with these certifications and standards, such end customers may not purchase our products, which would harm our business, operating results, financial condition and cash flows.

If our product or service offerings, including material purchased from our suppliers, have quality or performance issues, our business may suffer.

Our business depends on delivering products and services of consistently high quality. Many of our solutions are highly complex, and testing procedures used by us and our customers are limited to evaluating them under likely and foreseeable failure scenarios. Many of our products include both hardware and software components. It is not unusual for software, especially in earlier versions, to contain bugs that can unexpectedly interfere with expected operations. For various reasons, once deployed, our products may fail to perform as expected. Performance issues could result from faulty design, defective raw materials or components purchased from suppliers, problems in manufacturing or installation errors. We have experienced such performance issues in the past and remain exposed to such performance issues in the future. In some cases, recall of some or all affected products, product redesigns or additional capital expenditures may be required to correct a defect; and depending on the number of products affected, the cost of fixing or replacing such products could have a material impact on our results of operations and cash flows. Our agreements with our contract manufacturers and component suppliers may not cover all costs related to defects.

In some cases, we are dependent on a sole supplier for components used in our products. Defects in sole-sourced components subject us to additional risk of being unable to quickly address any product issues or failures experienced by our customers as a result of the component defect and could delay our ability to deliver new products until the defective components are corrected or a new supplier is identified and qualified. This could increase our costs in resolving the product issue, result in decreased sales of the impacted product or damage our reputation with customers, any of which could negatively impact our results of operations.

Hardware or software defects could also permit unauthorized users to gain access to our customers' networks and/or a consumer's home network. In addition to potentially damaging our reputation with customers, such defects may also subject us to claims for damages under agreements with our customers and fines by regulatory authorities.

We offer warranties on most products, the terms and conditions of which depend upon the product subject to the warranty. In many cases, we also indemnify our customers against damages or losses that might arise from certain claims relating to our products and services. Future claims may have a material adverse effect on our business, financial condition, results of operations and cash flows. Any significant or systemic product or service failure could also result in lost future sales as a result of reputational damage.

Our products have been deployed in many different locations and user environments and are capable of providing services and connectivity to many different types of devices operating a variety of applications. The ability of our products to operate effectively can be negatively impacted by many different elements unrelated to our products. For example, a user's experience may suffer from an incorrect setting in a Wi-Fi device. Although certain technical problems experienced by users may not be caused by our products, users often may perceive them to be the underlying cause of poor performance of the wireless network. This perception, even if incorrect, could harm our business and reputation. Similarly, a high-profile network failure may be caused by improper operation of the network or failure of a network component that we did not supply, but service providers may perceive that our products were implicated, which, even if incorrect, could harm our business, financial condition, results of operations and cash flows.

We depend on cloud computing infrastructure operated by third parties and any disruption in these operations could adversely affect our business.

For certain of our service offerings, in particular our Wi-Fi-related cloud services, we rely on third parties to provide cloud computing infrastructure that offers storage capabilities, data processing and other services. We currently operate our cloud-dependent services using Amazon Web Service (AWS), Google Compute Engine (GCE) or Microsoft Azure (Azure). We cannot easily switch our AWS, GCE or Azure operations to another cloud provider. Any disruption of or interference with our use of these cloud services would impact our operations and our business could be adversely impacted.

Problems faced by our third-party cloud service providers with the telecommunications network providers with whom we or they contract or with the systems by which our telecommunications providers allocate capacity among their customers, including us, could adversely affect the experience of our end customers. If AWS, GCE or Azure are unable to keep up with our needs for capacity, this could have an adverse effect on our business. Any changes in third-party cloud services or any errors, defects, disruptions or other performance problems with our cloud-based applications, could adversely affect our reputation and may damage our end customers' stored files or result in lengthy interruptions in our services. Interruptions in our services might adversely affect our reputation and operating results, cause us to issue refunds or service credits, subject us to potential liabilities or result in contract terminations.

Our business depends on effective management information systems.

We rely on effective management information systems for critical business operations, to support strategic business decisions and to maintain a competitive edge in the marketplace. We rely on our ERP systems to support critical business operations such as processing sales orders and invoicing, manufacturing, shipping, inventory control, purchasing and supply chain management, human resources and financial reporting. In 2020, we began the upgrade and integration of our ERP software to a newer, cloud-based version. The first phase was completed in early 2021 and we completed the second phase in 2023, with limited disruption. These upgrades and integrations do have risks and any future upgrades or integrations could disrupt our operations, divert management's attention and have an adverse effect on our capital resources, financial condition, results of operations or cash flows.

We also rely on management information systems to produce information for business decision-making and planning and to support digital platforms. Failure to maintain an adequate digital platform or to make additional investment in our digital platform to support e-commerce activities and improve our customer experience could have a material adverse impact on our business through lost sales opportunities.

If we are unable to maintain our management information systems, including our IT infrastructure, to support critical business operations, produce information for business decision-making activities and support digital customer experience activities, we could experience a material adverse impact on our business or an inability to timely and accurately report our financial results.

Cybersecurity incidents, including data security breaches, ransomware or computer viruses, could harm our business by exposing us to various liabilities, disrupting our delivery of products and services and damaging our reputation.

We rely extensively on our management information technology systems and those of third parties to operate our business and store proprietary information about our products and intellectual property. Additionally, we and others acting on our behalf receive, process, store and transmit confidential data, including "personally identifiable information," with respect to employees, vendors, customers and others. As the continued rise in cybersecurity incidents around the world indicates, all management information technology systems are vulnerable. We experienced a cybersecurity incident in the first quarter of 2023, but it had limited impact on our business operations. Despite the security controls we have put in place since that incident, our facilities, systems and procedures, and those of our third-party service providers, are still at risk of security breaches, acts of vandalism, ransomware, software viruses, misplaced or lost data, programming and/or human errors or other similar events. In particular, unauthorized access to our computer systems or stored data could result in the theft or improper disclosure of proprietary, confidential, sensitive or personal information, the deletion or modification of records or interruptions in our operations. These cybersecurity risks increase when we transmit information from one location to another, including transmissions over the Internet or other electronic networks. Any future significant compromise or breach of our data security, whether external or internal, or misuse of employee, vendor, customer, or Company data, could result in significant costs, lost sales, fines, lawsuits, lost customers and damage to our reputation. We employ a variety of security breach countermeasures and security controls designed to mitigate these risks, but we cannot guarantee that all breach attempts can be successfully thwarted by these measures as the sophistication of attacks increases. As cyber threats continue to evolve, we may be required to expend additional resources to mitigate new and emerging threats while continuing to enhance our information security capabilities or to investigate and remediate security vulnerabilities.

In addition, defects in some of the hardware or software we develop and sell, including in our engineering or in their implementation by our customers, could result in unauthorized access to our customers' and/or consumers' networks. Such unauthorized access could result in third parties gaining access to the private and personal information and technology of our customers, such as home health information, home cameras or other personal information or technology. Any such events could result in theft of personal information, trade secrets and intellectual property; give rise to legal proceedings; cause us to incur increased costs for insurance premiums, security, remediation and regulatory compliance; subject us to civil and criminal penalties; expose us to liabilities to our customers, employees, vendors, governmental authorities or other third parties; allow others to unfairly compete with us; disrupt our delivery of products and services; expose the confidential information of our clients and others; and have a negative impact on our reputation, all of which could have a material adverse effect on our business, financial condition, results of operations, cash flows and stock price.

Climate change may have a long-term impact on our business.

There are inherent climate change risks wherever business is conducted. The potential physical impacts of climate change on our operations are highly uncertain and would be particular to the geographic areas in which we operate. These may include changes in rainfall and storm patterns and intensities, water shortages, changing sea levels and changing temperatures. These impacts may adversely impact the cost, production levels and financial performance of our operations. Climate-related events, including the increasing frequency and intensity of extreme weather events and their impact on critical infrastructure in the regions in which we operate, have the potential to disrupt our business, our third-party suppliers, and/or the business of our customers and may cause us to experience higher attrition, losses and additional costs to maintain or resume operations. CommScope aligns with the Sustainability Accounting Standards Board (SASB) standards, Global Reporting Initiative (GRI) standards and makes use of the Carbon Disclosure Project (CDP) platform, which is committed to aligning with the Task Force on Climate Related Financial Disclosures (TCFD) recommendations to accurately assess, take potential proactive action and report as appropriate. For additional information, which is not incorporated by reference in this Annual Report on Form 10-K, see our Sustainability report on the CommScope website: https://www.commscope.com/corporate-responsibility-and-sustainability/.

Labor-Related Risks

Failure to attract, develop and maintain a highly skilled and diverse workforce or effectively manage changes in our workforce can have an adverse effect on our business.

Our business requires that we attract, develop and maintain a highly skilled and diverse workforce. Our employees are highly sought after by our competitors and other companies and our continued ability to compete effectively depends on our ability to attract, retain, develop and motivate highly skilled personnel for all areas of our organization. Our ability to do so has been and may continue to be impacted by challenges in the labor market, which has experienced and may continue to experience wage inflation, labor shortages, increased employee turnover, changes in availability of our workforce and a shift toward remote work. Any unplanned turnover, sustained labor shortage or unsuccessful implementation of our succession plans to backfill current leadership positions, including the Chief Executive Officer, or failure to attract, develop and maintain a highly skilled and diverse workforce can deplete our institutional knowledge base, erode our competitive advantage or result in increased costs due to increased competition for employees, higher employee turnover or increased employee benefit costs, all of which could adversely affect our business, financial condition and results of operations.

Disruptions in labor, including strikes or work stoppages, could have a material adverse effect on our business, results of operations and financial condition.

We have experienced and could continue to experience disruption in our manufacturing operations and supply chain, including labor shortages or changes in the availability of our or our business partners' workforce, strikes or work stoppages (including by third parties involved in the manufacture, production and distribution of our products). Although none of our U.S. employees are represented by unions, a significant portion of our international employees are members of unions or subject to works councils or similar statutory arrangements. Strikes or work stoppages or other business interruptions have occurred and may occur in the future if we or the third parties that are involved in the manufacturing, production and distribution of our products are unable to renew or enter into new agreements on satisfactory terms. This can impair the manufacturing and distribution of our products, interrupt product supply, lead to a loss of sales, increase our costs or otherwise affect our ability to fully implement future operational changes to enhance our efficiency or to adapt to changing business needs or strategy, all of which can adversely affect our business.

International Risks

Our significant international operations expose us to economic, political, foreign exchange rate and other risks.

We have significant international sales, manufacturing, distribution and R&D operations. Our major international manufacturing, distribution and R&D facilities are located in China, the Czech Republic, Germany, India, Ireland, Mexico and the U.K. For the year ended December 31, 2023, international sales represented 35% of our consolidated net sales. In general, our international sales have lower gross profit percentages than our domestic sales. To the extent international sales increase as a percentage of our net sales, our overall gross profit percentages may decline.

Our international sales, manufacturing, distribution and R&D operations are subject to the risks inherent in operating abroad, including, but not limited to, coordinating communications among and managing international operations; currency exchange rate fluctuations; economic and political destabilization, including the current risk with China-Taiwan relations, China-U.S. relations and Russia-U.S. relations; restrictive actions by foreign governments; price inflation; volatile interest rates; wage inflation; nationalization of businesses and expropriation of assets; the laws and policies of the U.S. and other countries affecting trade and tariffs, anti-bribery, foreign investment and loans; foreign tax laws, including the ability to recover amounts paid as value-added and similar taxes; potential restrictions on the repatriation of cash; reduced protection of intellectual property; longer customer payment cycles; compliance with local laws and regulations, including the imposition of new data privacy and climate change regulations; volatile geopolitical turmoil, including popular uprisings, regional conflicts, terrorism, and war; shipping interruptions, including shortages of containers or port congestion; major public health or safety concerns, such as pandemics and infectious diseases; natural or man-made disasters; inflexible labor contracts or labor laws in the event of business downturns; and economic boycott for doing business in certain countries. Although the Company maintains insurance coverage for certain types of losses, such insurance coverage may be insufficient to cover all losses that may arise.

A significant portion of our products sold in the U.S. are manufactured outside the U.S. To the extent there are changes in U.S. trade policies, such as significant increases in tariffs or duties for goods brought into the U.S., our competitive position may be adversely impacted and the resulting effect on our earnings could be material. For a more complete discussion of our risks related to trade policies, see the risk factor, "Additional tariffs or a global trade war could increase the cost of our products, which could adversely impact the competitiveness of our products" under "International Risks" in this Item 1A, Risk Factors section.

Risks related to fluctuations in foreign currency rates has impacted in the past and could continue to impact our sales, financial condition, results of operations and cash flows. Our foreign currency risk exposure is mainly concentrated in Chinese yuan, European Union (E.U.) euro, British pound sterling, Mexican peso, Japanese yen, Canadian dollar, Australian dollar, Brazilian real, South African rand, Indian rupee and Czech koruna. We manage our foreign currency rate risks through regular operating and financing activities and use derivative financial instruments such as foreign exchange forward contracts. There can be no assurance that our risk management strategies will be effective or that the counterparties to our derivative contracts will be able to perform. In addition, foreign currency rates in many of the countries in which we operate have at times been extremely volatile and unpredictable. We may choose not to hedge or determine we are unable to effectively hedge the risks associated with this volatility. In such cases, we may experience declines in sales and adverse impacts on earnings and such changes could be material.

Additional tariffs or a global trade war could increase the cost of our products, which could adversely impact the competitiveness of our products.

There is uncertainty about the future relationship between the U.S. and various other countries, most significantly China, with respect to trade policies and tariffs. Past U.S. administrations have called for substantial changes to U.S. foreign trade policy with respect to China and other countries, including the possibility of imposing greater restrictions on international trade and significant increases in tariffs on goods imported into the U.S. The current administration could have a different approach to U.S. foreign trade policy with China as well as other countries, but there remains uncertainty.

This uncertainty about the future relationship between the U.S. and certain of its trading partners may reduce trade between the U.S. and other nations, including countries in which we currently operate, or result in a global economic slowdown with long-term changes to global trade. Changes in policy or continued uncertainty could depress economic activity and restrict our access to suppliers or customers. The tariffs implemented on our products (or on materials, parts or components we use to manufacture our products) by past U.S. administrations increased the cost of our products manufactured in the U.S. and imported into the U.S. If additional tariffs or trade restrictions are implemented on our products (or on materials, parts or components we use to manufacture our products) by the U.S. or other countries, the cost of our products manufactured in China, Mexico or other countries and imported into the U.S. or other countries could increase further. We expect to continue to pass along some of these costs to our customers, but the increased cost could adversely affect the demand for products. We have been successful in the past in shifting the manufacturing locations for the impacted products, but this takes time and results in additional one-time costs and these alternative locations may have higher ongoing manufacturing costs. These cost increases could adversely affect the demand for our products and/or reduce margins, which could have a material adverse effect on our business and our earnings. In addition, a significant percentage of our component parts are manufactured in China and other Southeast Asian countries. The impact of tariffs or other geopolitical instability may limit our access and our manufacturing partners' access to those components which would impact production and could lead to further increases to product costs. Additionally, further escalation of trade tensions could lead to the possible decoupling of the U.S. and China economies. Any or all of these factors could negatively affect demand for our products and our business, financial condition, results of operations and cash flows, and such effects could be material.

Our significant international operations expose us to increased challenges in complying with anti-corruption laws and regulations of the U.S. government and various other international jurisdictions.

We are required to comply with the anti-corruption laws and regulations of the U.S. government and various other international jurisdictions, such as the U.S. Foreign Corrupt Practices Act and the U.K. Bribery Act, and our failure to comply with these laws and regulations may expose us to significant liabilities. These laws and regulations may apply to companies, individual directors, officers, employees and agents, and may restrict our operations, trade practices, investment decisions and partnering activities. Violations of these legal requirements are punishable by significant criminal fines and imprisonment, civil penalties, disgorgement of profits, injunctions, debarment from government contracts and other remedial measures. We have established policies, procedures and internal controls designed to assist us and our personnel in complying with applicable U.S. and international anti-corruption laws and regulations. However, our employees, subcontractors or channel partners could take actions that violate these requirements. In addition, some of the international jurisdictions in which we operate have elevated levels of corruption. As a result, we are exposed to an increased risk of violating anti-corruption laws. Violation of anti-corruption laws could adversely affect our reputation, business, financial condition, results of operations and cash flows, and such effects could be material.

We are subject to governmental export and import controls and sanctions programs that could subject us to liability or impair our ability to compete in international markets.

Certain of our products, including purchased components of such products, are subject to export controls and may be exported only with the required export license or through an export license exemption. In addition, we are required to comply with certain U.S. and foreign import and customs rules, sanctions and embargos such as the U.S. enacted Uyghur Forced Labor Prevention Act (UFLPA) that became effective in 2022. Although we believe the risk of a UFLPA enforcement action against the Company is low at this time, we will continue to monitor the ongoing potential impact as the Customs and Border Protection guidance will continue to evolve. If we were to fail to comply with applicable export licensing, customs regulations, economic sanctions and other laws, we could be subject to substantial civil and criminal penalties, including fines, the incarceration of responsible employees and managers and the possible loss of export or import privileges. In addition, if our distributors fail to obtain appropriate import, export or re-export licenses or permits, we may also be adversely affected through reputational harm and penalties. Obtaining the necessary export license for a particular sale may be time-consuming and may result in a delay or loss of sales opportunities.

Furthermore, export control laws and economic sanctions prohibit the shipment of certain products to embargoed or sanctioned countries, governments and persons. While we train our employees to comply with these regulations and have systems in place designed to prevent compliance failures, we cannot assure you that a violation will not occur, whether knowingly or inadvertently. Any such shipment could have negative consequences, including government investigations, penalties, fines, civil and criminal sanctions and reputational harm.

Any change in export or import regulations, economic sanctions or related legislation, shift in the enforcement or scope of existing regulations or change in the countries, governments, persons or technologies targeted by such regulations could result in our decreased ability to export, import or sell our products to existing or potential customers, particularly those with international operations. Any limitation on our ability to export, import or sell our products could adversely affect our business, financial condition, results of operations and cash flows, and such effects could be material.

Litigation and Regulatory Risks

We may not be successful in protecting our intellectual property and in defending against claims that we are infringing on the intellectual property of others, and any such actions may be costly.

We may encounter difficulties and significant costs in protecting our intellectual property rights or obtaining rights to additional intellectual property to permit us to continue or expand our business. Other companies, including some of our largest competitors, hold intellectual property rights in our industry, and the intellectual property rights of others could inhibit our ability to introduce new products unless we secure necessary licenses on commercially reasonable terms.

In the past, we have initiated litigation in order to enforce patents issued or licensed to us or to determine the scope and/or validity of a third-party's patent or other proprietary rights, and we may initiate similar litigation in the future. We also have been and may in the future be subject to lawsuits by third parties seeking to enforce their own intellectual property rights, including against certain of the products or intellectual property that we have acquired through acquisitions. Any such litigation, regardless of outcome, could be costly and could subject us to significant liabilities or require us to cease using proprietary third-party technology. In addition, the payment of any damages or any necessary licensing fees or indemnification costs associated with a patent infringement claim could be material and could also materially adversely affect our cash flows and operating results. Such litigation can also be a significant distraction to management.

In certain markets, we may be required to address counterfeit versions of our products. We may incur significant costs in pursuing the originators of such counterfeit products and, if we are unsuccessful in eliminating them from the market, we may experience a reduction in the value of our products, harm to our reputation and/or a reduction in our net sales.

Because of the nature of information that may pass through or be stored on certain of our solutions or networks, we, our vendors and our end customers are subject to complex and evolving U.S. and foreign laws and regulations regarding information privacy, data protection, cybersecurity and other related matters.

Globally, there has been an increase in laws and regulatory action concerning privacy-related matters. Generally, these laws create rights for individuals in their personal data as well as impose obligations on businesses regarding the handling of such personal data, including data of employees, consumers and business contacts. Several U.S. states are considering or have adopted legislation requiring companies to disclose the collection of personal data, protect the security of personal information that they hold or respond to individuals' rights regarding their personal data. For example, the California Consumer Privacy Act (CCPA), which went into effect on January 1, 2020, subjects us to stricter obligations, greater fines and more private causes of action related to data security. The California Privacy Rights Act (CPRA), which is effective in 2023, amends and further expands the CCPA. Virginia, Connecticut, Utah and Colorado enacted similar laws in 2023. Many jurisdictions have also enacted or are enacting laws requiring companies to notify regulators or individuals of data security incidents involving certain types of personal data, including the rule issued by the Securities and Exchange Commission in the U.S. in 2023 that requires public disclosure of material security incidents. These mandatory disclosures regarding security incidents often lead to widespread negative publicity. Any security incident, whether actual or perceived, could harm our reputation, erode customer confidence in the effectiveness of our data security measures, negatively impact our ability to attract or retain customers, or subject us to third-party lawsuits, regulatory fines or other action or liability, which could materially and adversely affect our business and operating results.

Foreign data protection, privacy and other laws and regulations can be more restrictive than those in the U.S. For example, the E.U.'s General Data Protection Regulation (GDPR), which became effective in May 2018, was designed to harmonize data privacy laws across Europe, to protect all E.U. citizens' data privacy, empower E.U. citizens with respect to their personal data and to reshape the way organizations across the region approach data privacy. Compliance with GDPR has required changes to products and service offerings, internal and external software systems, including our websites, and changes to many company processes and policies. Failure to comply with GDPR could cause significant penalties and loss of business. Subsequent judicial rulings in Europe about GDPR have invalidated the E.U.-U.S. privacy shield framework, which was the mechanism relied upon by some of our vendors for personal data transfers out of the E.U. Additionally, these rulings require companies like ours to assess their personal data transfers from the E.U. to determine whether the protections in the U.S. or any country without an adequacy determination meet E.U. standards in the context of the specific transfer. A European data protection authority could disagree with our assessment of such transfers, resulting in penalties or required changes in how we transfer data within our company.

In addition, some countries are considering or have passed legislation requiring local storage and processing of data. For example, Brazil and India have each adopted such laws that became effective in January 2020. These new and proposed laws could increase the cost and complexity of offering our solutions or maintaining our business operations in those jurisdictions. The introduction of new solutions or expansion of our activities in certain jurisdictions may subject us to additional laws and regulations. Our channel partners and end customers also may be subject to such laws and regulations in the use of our products and services.

These U.S. federal and state and foreign laws and regulations, which often can be enforced by private parties or government entities, are constantly evolving. In addition, the application and interpretation of these laws and regulations are often uncertain, may be interpreted and applied inconsistently from jurisdiction to jurisdiction and may be contradictory with each other. For example, a government entity in one jurisdiction may demand the transfer of information forbidden from transfer by a government entity in another jurisdiction. If our actions were determined to be in violation of any of these disparate laws and regulations, in addition to the possibility of fines, we could be ordered to change our data practices, which could have an adverse effect on our business and results of operations and financial condition. There is also a risk that we, directly or as the result of a third-party service provider we use, could be found to have failed to comply with the laws or regulations applicable in a jurisdiction regarding the collection, handling, transfer, disposal or consent to the use of personal data, which could subject us to fines or other sanctions, as well as adverse reputational impact.

Some states and countries are considering or have introduced laws and regulations requiring minimum or particular security controls be incorporated into devices that connect to the internet (so called "Internet of Things Security laws"). Where products we manufacture are considered in scope for some of these laws and regulations, compliance obligations or customer contracts may necessitate modification of existing product features and specifications or make inventory obsolete. Inconsistencies in these laws can introduce complexity into our design, manufacturing and inventory management processes.

Compliance with these existing and proposed laws and regulations can be costly and require significant management time and attention, and failure to comply can result in negative publicity and subject us to inquiries or investigations, claims or other remedies, including fines or demands that we modify or cease existing business practices. Customers may demand or request additional functionality in our products or services that they believe are necessary or appropriate to comply with such laws and regulations, which can cause us to incur significant additional costs and can delay or impede the development of new solutions. In addition, there is a risk that failures in systems designed to protect private, personal or proprietary data held by us or our customers using our solutions will allow such data to be disclosed to or seen by others, resulting in application of regulatory penalties, enforcement actions, remediation obligations, private litigation by parties whose data was improperly disclosed or claims from our customers for costs or damages they incur. There can be no assurance that the limitations of liability in our contracts would be enforceable or adequate or would otherwise protect us from any such liabilities or damages with respect to any particular claim. Our existing general liability insurance coverage and coverage for errors and omissions may not continue to be available on acceptable terms or may not be available in sufficient amounts to cover one or more large claims, or our insurers may deny coverage as to any future claim. The successful assertion of one or more large claims against us that exceeds available insurance coverage, or changes in our insurance policies, including premium increases or the imposition of large deductible or co-insurance requirements, could have a material adverse effect on our business, financial condition, results of operations and cash flow.

Compliance with current and future social and environmental laws, regulations, policies and provisions, customer and investor pressures, other efforts to mitigate climate change and potential environmental liabilities may have a material adverse impact on our business, financial condition and results of operations.

We are subject to various federal, state, local and foreign environmental laws and regulations governing, among other things, discharges to air and water, management of regulated materials, energy consumption, handling and disposal of solid and hazardous waste and investigation and remediation of contaminated sites. We are also subject to laws and regulations regarding the types of substances allowable in certain of our products and the handling of our products at the end of their useful life. Because of the nature of our business, we have incurred and will continue to incur costs relating to compliance with or liability under these environmental laws and regulations, and these costs could be material. In addition, there is an increasing focus on corporate social and environmental responsibility in our industry, in which new laws and regulations, new or different interpretations of existing laws and regulations, expansion of existing legal requirements related to our products, the discovery of previously unknown contamination or the imposition of new remediation or discharge requirements could require us to incur costs or could become the basis for new or increased liabilities that could have a material adverse effect on our financial condition.

Certain environmental laws impose strict and, in some circumstances, joint and several liability on current or former owners or operators of a contaminated property, as well as companies that generated, disposed of or arranged for the disposal of hazardous substances at a contaminated property, for the costs of investigation and remediation of such property. Our present and past facilities have been in operation for many years and over that time, in the course of those operations, hazardous substances and wastes have been used, generated and occasionally disposed of at such facilities, and we have disposed of waste products either directly or through third parties at numerous disposal sites. Consequently, it has been necessary to undertake investigation and remediation projects at certain sites and we have been, and may in the future be, held responsible for a portion of the investigation and clean-up costs at these sites and our share of those costs may be material.

Efforts to regulate emissions of greenhouse gases (GHGs), such as carbon dioxide, are continuing to evolve in the U.S. and other countries where we operate, and this could increase the cost of raw materials, production processes and transportation of our products. If we are unable to comply with such regulations or sufficiently increase prices or otherwise reduce costs to offset the increased costs of compliance, GHG regulation could have a material adverse effect on our business, financial condition, results of operations and cash flow.

A number of governments or governmental bodies have also introduced or are contemplating regulatory changes in response to various climate change interest groups and the potential impact of climate change, such as the proposed reporting regulations issued by the Securities and Exchange Commission in the U.S. and final regulations issued in California and the E.U. Increased regulation regarding climate change could impose significant costs on us and our suppliers, including costs related to increased energy requirements, capital equipment, environmental monitoring and reporting, and other costs to comply with such regulations. Any adopted future climate change regulations could also negatively impact our ability to compete with companies situated in areas not subject to such limitations.

Additionally, some of our customers have adopted, or may adopt, procurement policies that include social and environmental responsibility provisions or requirements with which their suppliers should comply. An increasing number of investors are also pushing companies to disclose corporate social and environmental policies, practices and metrics. If we are unable to comply with such policies or meet the requirements of our customers and investors, it may impact the demand for our products, negatively impact our stock price or expose us to potential litigation.

Given the political significance around and uncertainty about how to best mitigate climate change, we cannot predict how legislation, regulation or customer and investor expectations will affect our financial condition, operating performance and ability to compete. Furthermore, even without such regulation, increased awareness and any adverse publicity in the global marketplace about potential impacts on climate change by us or other companies in our industry could harm our reputation.

General Risk Factors

Any future public health crisis could materially adversely affect our business, financial condition, results of operations and cash flows.

Pandemics have had and could have in the future, material and adverse effects on our ability to successfully operate and on our financial condition, results of operations and cash flows due to the following factors, among others:

- health concerns that may lead to a complete or partial closure of, or other operational issues at, our manufacturing facilities or those of our contract manufacturers;
- the reduced economic activity may severely impact our customers' financial condition and liquidity and may lead to decreased demand for our products and services or impact the timing of on-going or planned projects;
- difficulty accessing debt and equity capital on attractive terms, or at all, and a severe disruption and instability in the global financial markets or deteriorations in credit and financing conditions may affect our access to capital necessary to fund business operations or address existing and anticipated liabilities on a timely basis;
- a deterioration in our ability to operate in affected areas or delays in the supply of products or services to us from vendors that are needed for our efficient operations could adversely affect our operations;
- the potential outbreaks among our personnel, particularly if a significant number of them are impacted, could result in a deterioration in our ability to ensure business continuity during a disruption; and
- remote working arrangements may increase our vulnerability to cybersecurity incidents, including breaches of information systems security, which could damage our reputation, disrupt operations and expose us to claims from customers, suppliers, employees and others.

The extent to which any future public health crisis impacts our operations and those of our customers and suppliers will depend on the scope, severity, duration and spread of the health crisis, the actions taken to contain it or mitigate its impact, and the direct and indirect economic effects of the crisis and containment measures, among others, all of which are uncertain and cannot be predicted with confidence.

We do not intend to pay dividends on our common stock and, consequently, the ability of investors to achieve a return on their investment will depend on appreciation in the price of our common stock.

We do not intend to declare and pay dividends on our common stock for the foreseeable future. The payment of future dividends will be at the discretion of our Board; however, the indentures and the credit agreements governing our indebtedness place limitations on our ability to pay dividends. We currently intend to invest our future earnings, if any, to reduce our debt and fund our growth. The success of an investment in our common stock will largely depend upon future appreciation in value, and there can be no guarantee that our common stock will appreciate in value.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 1C. CYBERSECURITY

Like many large, global companies, CommScope relies heavily on digital technology to conduct operations and engage with our customers and business partners. As our engagements become more complex and interdependent, threats from security incidents like ransomware and data breaches increase. To mitigate the threat to our business, we take a comprehensive approach to cybersecurity risk management and make securing our data and the data customers and other stakeholders entrust to us a top priority. Our Board of Directors (Board), through our Audit Committee, and our management team are actively involved in the oversight of our Enterprise Risk Management (ERM) program, of which cybersecurity represents an important component. As described in more detail below, we have established policies, standards, processes and practices for assessing, identifying and managing material risks from cybersecurity threats. We have made investments in resources to implement and maintain security measures to meet regulatory requirements and stakeholder expectations, and we intend to continue to make investments to maintain the security of our data and cybersecurity infrastructure. There can be no guarantee that our policies and procedures will be properly followed in every instance or that those policies and procedures will be effective. We believe that risks from prior cybersecurity threats, including as a result of previous cybersecurity incidents, have not materially affected our business to date. We can provide no assurance that there will not be incidents in the future or that any such incidents will not materially affect us, including our business strategy, results of operations or financial condition. For additional information regarding the risks associated with cybersecurity incidents, see Item 1A. “Risk Factors.”

CommScope’s commitment to cybersecurity begins in the boardroom. The Audit Committee is responsible for oversight of cybersecurity and is actively engaged with our Chief Information Officer (CIO) and Chief Information Security Officer (CISO) at least quarterly, in addition to ad-hoc discussions and our periodic cyber crisis management tabletop exercises. Our CIO and CISO also present on cybersecurity to our full Board at least annually. The commitment extends through our executive leadership team (ELT), who engage continually to review our cybersecurity strategy, planning and execution.

At CommScope, cybersecurity risk is part of our cross-functional Enterprise Risk Management (ERM) program because of the potential for negative impacts of an incident across our business. At least annually we conduct a cybersecurity risk assessment, bringing together threat intelligence, internal assessment of our control posture and third-party opinions. The risk assessment informs our Board and management team and drives the next year’s security strategy and initiatives.

CommScope has implemented a cybersecurity program that is dedicated to protecting our business processes, technology assets and sensitive information entrusted to us by our customers, suppliers, employees and other stakeholders. Drawing on the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF) and Center for Internet Security (CIS) 18 Critical Security Controls, our security program seeks to identify high-value enterprise assets and business processes and manage the cybersecurity threats facing them with layered controls. We practice a “Defense in Depth” methodology, meaning that valuable assets or business processes are generally protected with more than one layered control.

Our cybersecurity program is led by our CIO and our CISO. Our CIO has served in various roles in information technology and information security for over twenty years. Our CISO has over twenty years of technology and security experience and has spent more than fifteen years leading cybersecurity functions. He is also a Certified Information Systems Security Professional (CISSP).

Our CISO leads an in-house information security team responsible for cybersecurity risk and threat evaluation; the writing of relevant policies, control standards, and technical requirements; and the oversight and operation of security controls. The Security Operations team monitors for potential incidents via a global team operating 24 x 7 x 365 in a “follow the sun” model. We also engage outside experts where a third-party opinion or subject matter expertise provides specific value, such as with penetration testing. We use industry-leading security tools, regularly update our technology roadmaps, conduct tabletop exercises and mandate cybersecurity awareness and training for all employees.

We not only focus on cybersecurity threats directly facing CommScope but also those that might affect us through one of the many third parties we do business with, including suppliers and customers. Our procurement and security teams have a shared process to review the cybersecurity risk of our suppliers, performing an assessment during onboarding, requiring them to sign up for contractual security requirements, emplacing security controls and investigating third-party incidents as appropriate.

In the event of a significant cybersecurity incident, we have a detailed Cybersecurity Incident Response Plan (CIRP) in place for informing key stakeholders, ensuring events are properly escalated and for contacting authorities. There are many ways that CommScope might initially learn of a cybersecurity incident, and these potential incidents are escalated, according to decision criteria, to a core team of internal stakeholders comprised of leaders from our information security, legal, business and finance organizations. The core team directs the initial fact-finding and response efforts, and based on their qualitative and quantitative review, may escalate the incident to CommScope’s ELT. The ELT then makes the decision on escalation to the Audit Committee or Board based on the team’s assessment of materiality. Our incident response plan is regularly validated and assessed to consider the types of decisions that would need to be made in the event of a cybersecurity incident.

ITEM 2. PROPERTIES

Our fixed assets include factories and warehouses and a substantial quantity of machinery and equipment. Our factories, warehouses and machinery and equipment are generally in good operating condition, are reasonably maintained and substantially all of our facilities are in regular use. We consider the present level of fixed assets along with planned capital expenditures to be suitable and adequate for operations in the current business environment. At December 31, 2023, we operated approximately 23 manufacturing facilities with approximately 4.8 million square feet, of which approximately 1.3 million square feet were leased. Manufacturing facilities located in the U.S. had approximately 1.7 million square feet, of which approximately 0.1 million square feet were leased. Manufacturing facilities located outside the U.S. had approximately 3.1 million square feet, of which approximately 1.2 million square feet were leased. The square footage by segment related to our manufacturing facilities was approximately 3.7 million square feet, 0.8 million square feet, 0.2 million square feet and 0.1 million square feet for the CCS segment, OWN segment, NICS segment and ANS segment, respectively.

We believe that our facilities are suitable and adequate for our business and are being appropriately utilized for their intended purposes. Utilization of our facilities varies based on demand for the related products. We regularly review our anticipated requirements for facilities and, based on that review, may from time to time acquire or lease additional facilities and/or dispose of existing facilities.

ITEM 3. LEGAL PROCEEDINGS

The Company is party to certain intellectual property claims and also periodically receives notices asserting that its products infringe on another party's patents and other intellectual property rights. These claims and assertions, whether against the Company directly or against its customers, could require the Company to pay damages or royalties, stop offering the relevant products and/or cease other activities. The Company may also be called upon to indemnify certain customers for costs related to products sold to such customers. The outcome of these claims and notices is uncertain and a reasonable estimate of the loss from unfavorable outcomes in certain of these matters either cannot be determined or is estimated at the minimum amount of a range of estimates. The actual loss, through settlement or trial, could be material and may vary significantly from our estimates. From time to time, the Company may also be involved as a plaintiff involving intellectual property claims. Gain contingencies, if any, are recognized when they are realized.

The Company is also either a plaintiff or a defendant in certain other pending legal matters in the normal course of business. Management believes none of these pending legal matters will have a material adverse effect on the Company's business or financial condition upon final disposition.

In addition, the Company is subject to various federal, state, local and foreign laws and regulations governing the use, discharge, disposal and remediation of hazardous materials. Compliance with current laws and regulations has not had, and is not expected to have, a materially adverse effect on the Company's financial condition or results of operations.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Market Information and Holders

Our common stock is traded on the Nasdaq Global Select Market under the symbol "COMM." As of February 14, 2024, all of our outstanding shares of common stock are held by one stockholder of record, Cede & Co., as nominee for the Depository Trust Company. Many brokers, banks and other institutions hold shares of common stock as nominees for beneficial owners that deposit these shares of common stock in participant accounts at the Depository Trust Company.

Issuer Purchases of Equity Securities

The following table summarizes the stock purchase activity for the three months ended December 31, 2023:

Period	Total Number of Shares Purchased [1]	Average Price Paid Per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs	Maximum Value of Shares that May Yet be Purchased Under the Plans or Programs
October 1, 2023 - October 31, 2023	67,935	$ 3.35	—	$ —
November 1, 2023 - November 30, 2023	7,471	$ 1.73	—	$ —
December 1, 2023 - December 31, 2023	2,881	$ 1.89	—	$ —
Total	78,287	$ 3.14	—	

The shares purchased were withheld to satisfy the withholding tax obligations related to restricted stock units and performance share units that vested during the period.

Stock Performance Graph

The following graph compares cumulative total return on $100 invested on December 31, 2018 in each of CommScope's Common Stock, the Standard & Poor's 500 Stock Index (S&P 500 Index) and the Standard & Poor's 1500 Communications Equipment Index (S&P 1500 Communications Equipment). The return of the Standard & Poor's indices is calculated assuming reinvestment of dividends. CommScope has not paid any dividends on its common stock over this period.



	Base Period	INDEXED RETURNS Period Ending				
Company / Index	12/31/2018	12/31/2019	12/31/2020	12/31/2021	12/31/2022	12/31/2023
CommScope Holding Company, Inc.	100	86.58	81.76	67.36	44.84	17.21
S&P 500 Index	100	128.88	149.83	190.13	153.16	190.27
S&P 1500 Communications Equipment Index	100	111.62	109.80	159.85	124.37	142.07

ITEM 6. RESERVED

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

The following discussion and analysis of our financial condition and results of operations is for the year ended December 31, 2023 compared with the year ended December 31, 2022. This comparison should be read in conjunction with our consolidated financial statements and related notes appearing elsewhere in this Annual Report on Form 10-K. This discussion contains forward-looking statements based upon current expectations that involve risks and uncertainties. Our actual results may differ materially from those anticipated in these forward-looking statements as a result of various factors, including those set forth under "Risk Factors" included in Part I, Item 1A or in other parts of this Annual Report on Form 10-K. For a discussion and analysis of our financial condition and results of operations for the year ended December 31, 2022 compared to December 31, 2021, see Part II, Item 7, "Management's Discussion and Analysis of Financial Condition and Results of Operations" included in the 2022 Annual Report on Form 10-K, filed with the Securities and Exchange Commission on February 23, 2023.

OVERVIEW

We are a global provider of infrastructure solutions for communication, data center and entertainment networks. Our solutions for wired and wireless networks enable service providers, including cable, telephone and digital broadcast satellite operators and media programmers, to deliver media, voice, Internet Protocol (IP) data services and Wi-Fi to their subscribers and allow enterprises to experience constant wireless and wired connectivity across complex and varied networking environments. Our solutions are complemented by services including technical support, systems design and integration. We are a leader in digital video and IP television distribution systems, broadband access infrastructure platforms and equipment that delivers data and voice networks to homes. Our global leadership position is built upon innovative technology, broad solution offerings, high-quality and cost-effective customer solutions, and global manufacturing and distribution scale.

CommScope NEXT

Since 2021, we have been engaged in a transformation initiative referred to as CommScope NEXT, which is designed to drive shareholder value through three pillars: profitable growth, operational efficiency and portfolio optimization. We believe these efforts are critical to making us more competitive and allowing us to invest in growth, de-leverage our indebtedness and maximize stockholder and other stakeholder value in the future. In 2022, CommScope NEXT generated positive impacts on net sales, profitability and cash flow from our execution on pricing initiatives, capacity expansion and operational efficiencies. In 2023, we experienced headwinds related to a slow-down in spending by our customers as discussed further below, but we continued to execute under CommScope NEXT to improve our profitability and cash flows by continuing to drive operational efficiencies and focusing on portfolio optimization that should enable us to take advantage of the expected recovery in demand in the second half of 2024. To that end, we incurred $29.7 million, $63.0 million and $85.1 million of net restructuring costs and $27.1 million, $35.1 million and $50.6 million of transaction, transformation and integration costs during the years ended December 31, 2023, 2022 and 2021, respectively, primarily related to CommScope NEXT initiatives. We expect to continue to incur such costs during 2024 as we continue executing on CommScope NEXT initiatives, and the resulting charges and cash requirements could be material.

In 2021, as a step in our CommScope NEXT transformation plan, we announced a plan to separate the Home Networks (Home) segment and began analyzing the financial results of our "Core" business separately from Home. On October 2, 2023, we entered into a Call Option Agreement with Vantiva SA, a société anonyme organized under the Laws of France (Vantiva), pursuant to which we granted Vantiva a binding call option to acquire our Home segment and substantially all of the associated segment assets and liabilities (Home business), which was subsequently exercised and a Purchase Agreement signed on December 7, 2023. The transaction closed on January 9, 2024.

We determined the anticipated sale of our Home business met the "held for sale" criteria and the "discontinued operations" criteria in accordance with Accounting Standards Codification (ASC) No. 360-10, *Impairment and Disposal of Long Lived Assets*, and ASC No. 205-20, *Presentation of Financial Statements: Discontinued Operations* in the fourth quarter of 2023 due to its relative size and strategic rationale. For all periods presented, amounts have been recast to reflect the discontinuation of our Home business in accordance with ASC 205-20. Unless otherwise noted, the following discussions relate solely to our continuing operations. For further discussion of the discontinued operation related to our Home business, see Note 3 in the Notes to Consolidated Financial Statements included elsewhere in the Annual Report on Form 10-K.

The results of our recast continuing operations do not align with our historical "Core" measures, which excluded the Home segment. Our continuing operations results include general corporate costs that were previously allocated to the Home segment. These indirect costs, reflected on the corporate and other line item within our segment information below, are classified as continuing operations, since they were not directly attributable to the discontinued operations of the Home segment. In future years, these costs will be reallocated to our remaining segments and will be at least partially offset by income from our transition services agreement with Vantiva or eliminated with future restructuring actions.

As a result of the divestiture of the Home business, we are now reporting financial performance based on the following remaining four operating segments, which excludes our Home segment: Connectivity and Cable Solutions (CCS), Outdoor Wireless Networks (OWN), Networking, Intelligent Cellular and Security Solutions (NICS) and Access Network Solutions (ANS). For all periods presented, amounts have been recast to reflect these operating segment changes.

Impacts of Current Economic Conditions

In 2023, macroeconomic factors such as higher interest rates and concerns about continued inflation and a global economic slow-down softened demand for our products, with certain customers reducing purchases as they right-sized their inventories and others pausing capital spending. This negatively impacted net sales in our CCS, OWN and ANS segments for the year ended December 31, 2023 and may continue to have a material negative impact on net sales into at least the first half of 2024. Conversely, in our NICS segment, we saw higher demand and favorable pricing impacts that partially offset the impact of lower demand in our other segments for the full year of 2023. However, NICS segment net sales were down in the fourth quarter of 2023 as we saw order rates decline as channel partners paused to digest inventory.

We saw lower input costs across most of our segments as inflation settled during the year. We proactively implemented cost savings initiatives that favorably impacted our profitability for the year ended December 31, 2023 and should enable us to take advantage of the expected recovery in demand in the second half of 2024. If the expected recovery in demand of our products does not occur in 2024, our outlook will be materially impacted.

For more discussion on risks related to our customers, see Part I, Item 1A, "Risk Factors" elsewhere in this Annual Report on Form 10-K.

RESULTS OF OPERATIONS

Comparison of results of operations for the year ended December 31, 2023 with the year ended December 31, 2022

	Year Ended December 31,					
	2023		2022			
	Amount	% of Net Sales	Amount	% of Net Sales	Change	% Change
	(dollars in millions, except per share amounts)					
Net sales	$ 5,789.2	100.0%	$ 7,524.7	100.0%	$ (1,735.5)	(23.1)%
Gross profit	2,148.3	37.1	2,594.0	34.5	(445.7)	(17.2)
Operating loss	(112.9)	(2.0)	(613.1)	(8.1)	500.2	(81.6)
Core segment adjusted EBITDA [1]	1,022.2	17.7	1,250.4	16.6	(228.2)	(18.3)
Non-GAAP adjusted EBITDA [2]	999.0	17.3	1,223.4	16.3	(224.4)	(18.3)
Loss from continuing operations	(851.3)	(14.7)	(1,184.7)	(15.7)	333.4	(28.1)
Diluted loss from continuing operations per share	$ (4.33)		$ (6.00)		$ 1.67	(27.8)

(1)Core segment adjusted EBITDA reflects the results of our CCS, OWN, NICS and ANS segments, in the aggregate, and excludes general corporate costs that were previously allocated to the Home segment and are now classified as continuing operations, since the costs were not directly attributable to the discontinued operations of the Home segment. See the Segment Results section below for illustration of the aggregation of Core segment adjusted EBITDA.

(2)See "Reconciliation of Non-GAAP Measures" in this Management's Discussion and Analysis of Financial Condition and Results of Operations, below.

Net sales

	Year Ended December 31,			
	2023	2022	Change	% Change
	(dollars in millions)			
Net sales	$ 5,789.2	$ 7,524.7	$ (1,735.5)	(23.1)%
Domestic	3,750.0	5,018.1	(1,268.1)	(25.3)
International	2,039.2	2,506.6	(467.4)	(18.6)

Net sales in 2023 decreased $1,735.5 million, or 23.1%, compared to the prior year primarily driven by decreased sales volumes as certain customers reduced purchases as they right-size their inventories and others paused capital spending, partially offset by higher pricing. The decrease was driven by lower net sales in the CCS segment of $1,079.4 million, the OWN segment of $587.9 million and the ANS segment of $246.2 million, partially offset by higher net sales of $178.0 million in the NICS segment. For further details by segment, see the discussion of Segment Results below.

From a regional perspective in 2023, net sales decreased in the U.S. by $1,268.1 million, the Europe, Middle East and Africa (EMEA) region by $184.4 million, the Asia Pacific (APAC) region by $114.0 million, the Caribbean and Latin American (CALA) region by $106.9 million, and Canada by $62.1 million. Net sales to customers located outside of the U.S. comprised 35% of total net sales for 2023 compared to 33% for 2022. Foreign exchange rate changes did not have a material impact on our net sales during 2023. For additional information on regional sales by segment, see discussion of Segment Results below and Note 17 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K.

Gross profit, SG&A expense and R&D expense

	Year Ended December 31,			%
	2023	2022	Change	Change
	(dollars in millions)			
Gross profit	$ 2,148.3	$ 2,594.0	$ (445.7)	(17.2)%
As a percent of sales	37.1 %	34.5 %		
SG&A expense	873.3	1,040.9	(167.6)	(16.1)
As a percent of sales	15.1 %	13.8 %		
R&D expense	459.7	543.6	(83.9)	(15.4)
As a percent of sales	7.9 %	7.2 %		

Gross profit (net sales less cost of sales)

Gross profit decreased in 2023 compared to the prior year primarily due to lower net sales volumes, partially offset by lower freight and material costs and favorable product mix.

Selling, general and administrative expense

For 2023, selling, general and administrative (SG&A) expense decreased by $167.6 million compared to 2022, primarily due to cost saving initiatives, lower variable incentive compensation expense of $59.3 million and lower bad debt expense. Bad debt expense in 2022 was driven by a $20.9 million reserve related to an OWN segment customer. Although our transaction, transformation, and integration costs have decreased year-over-year, we expect to continue to incur these costs in 2024 due to ongoing CommScope NEXT initiatives, and the resulting charges and cash requirements could be material.

We also incurred costs of $5.5 million in 2023 related to the identification, investigation, defense, recovery efforts and litigation claims connected to a cyber incident that occurred in late March of 2023. The incident resulted in minimal impact to our business operations, and we do not expect significant additional costs related to the recovery effort.

Research and development expense

Research and development (R&D) expense for 2023 decreased by $83.9 million primarily due to lower spending across all segments. R&D activities generally involve ensuring that our products are capable of meeting the evolving technological needs of our customers, bringing new products to market and modifying existing products to better serve our customers.

Amortization of purchased intangible assets, Restructuring costs, net and Asset impairments

	Year Ended December 31,			%
	2023	2022	Change	Change
	(dollars in millions)			
Amortization of purchased intangible assets	$ 327.1	$ 440.0	$ (112.9)	(25.7)%
Restructuring costs, net	29.7	63.0	(33.3)	(52.9)
Asset impairments	571.4	1,119.6	(548.2)	(49.0)

Amortization of purchased intangible assets

The amortization of purchased intangible assets was lower in 2023 compared to the prior year because certain of our intangible assets became fully amortized.

Restructuring costs, net

The net restructuring costs recorded in 2023 were primarily related to CommScope NEXT and reflected a gain of $33.9 million, primarily related to selling owned properties, including an international manufacturing facility, along with U.S. office and warehouse facilities. We received proceeds of $67.6 million related to these sales. Excluding the impacts of these facility sales, our restructuring costs were $63.6 million, and we paid $110.8 million to settle restructuring liabilities. We expect to pay an additional $11.7 million in 2024 related to the restructuring actions that have been initiated. Additional restructuring actions related to CommScope NEXT are expected to be identified and the resulting charges and cash requirements could be material.

Asset impairments

We recorded goodwill impairment charges of $472.3 million and $99.1 million in 2023 related to our ANS and Building Data Center Connectivity (BDCC) reporting units, respectively. The ANS reporting unit is the same as our ANS segment and the BDCC reporting unit is in our CCS segment. See the discussion below under "Critical Accounting Policies and Estimates" for more information regarding the goodwill impairment tests performed during 2023. We recorded goodwill impairment charges of $1,119.6 million in 2022 related to our ANS reporting unit.

Other income (expense), net

	Year Ended December 31,			%
	2023	2022	Change	Change
	(dollars in millions)			
Foreign currency loss	$ (13.9)	$ (4.6)	$ (9.3)	202.2%
Other income, net	73.6	4.1	69.5	NM

NM – Not meaningful

Foreign currency loss

Foreign currency loss includes the net foreign currency gains and losses resulting from the settlement of receivables and payables, foreign currency contracts and short-term intercompany advances in a currency other than the subsidiary's functional currency. The change in foreign currency loss in 2023 compared to 2022 was primarily driven by certain unhedged currencies.

Other income, net

The change in other income, net in 2023 compared to 2022 was primarily driven by a gain of $74.3 million on the early extinguishment of debt related to our debt repurchases as further discussed in Note 8 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K.

Interest expense, Interest income and Income taxes

	Year Ended December 31,			%
	2023	2022	Change	Change
	(dollars in millions)			
Interest expense	$ (675.8)	$ (588.9)	$ (86.9)	14.8%
Interest income	11.1	2.8	8.3	296.4
Income tax (expense) benefit	(133.4)	15.0	(148.4)	(989.3)

Interest expense and Interest income

The increase in interest expense in 2023 compared to 2022 was driven by the increased variable interest rate on our senior secured term loan due 2026 (2026 Term Loan) as a result of the Federal Reserve's increases in interest rates. Our weighted average effective interest rate on outstanding borrowings, including the impact of the interest rate swap contracts and the amortization of debt issuance costs and original issue discount, was 7.22% at December 31, 2023 and 6.91% at December 31, 2022. Our interest expense will continue to increase if the Federal Reserve raises interest rates.

Income tax (expense) benefit

For 2023, we recognized income tax expense of $133.4 million on a pretax loss of $717.9 million. Our tax expense on a pretax loss was less than the statutory rate of 21.0% in 2023 primarily due to the unfavorable impact related to a goodwill impairment charge of $571.4 million, for which minimal tax benefits were recorded. Our tax expense was also unfavorably impacted by an additional net $165.4 million of valuation allowance recorded during the year. Our tax expense was also impacted unfavorably by the U.S. anti-deferral provisions and non-creditable withholding taxes, partially offset by tax benefits related to federal tax credits. See Note 13 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K for more discussion of our income tax (expense) benefit.

For 2022, we recognized an income tax benefit of $15.0 million on a pretax loss of $1,199.7 million. Our tax benefit was less than the statutory rate of 21.0% in 2022 primarily due to a goodwill impairment charge of $1,119.6 million, for which minimal tax benefits were recorded. Our tax benefit was also impacted by the unfavorable impacts of U.S. anti-deferral provisions and non-creditable withholding taxes, partially offset by tax benefit related to federal tax credits.

Segment Results

	Year Ended December 31,					
	2023		2022			
	Amount	% of Net Sales	Amount	% of Net Sales	Change	% Change
	(dollars in millions)					
Net sales by segment:						
CCS	$ 2,710.2	46.8 %	$ 3,789.6	50.4 %	$ (1,079.4)	(28.5)%
OWN	880.0	15.2	1,467.9	19.5	(587.9)	(40.1)
NICS	1,117.7	19.3	939.7	12.5	178.0	18.9
ANS	1,081.3	18.7	1,327.5	17.6	(246.2)	(18.5)
Consolidated net sales	$ 5,789.2	100.0 %	$ 7,524.7	100.0 %	$ (1,735.5)	(23.1)%
Operating income (loss) by segment:						
CCS	$ 121.9	4.5 %	$ 438.3	11.6 %	$ (316.4)	(72.2)%
OWN	130.5	14.8	189.0	12.9	(58.5)	(31.0)
NICS	127.0	11.4	(51.2)	(5.4)	178.2	NM
ANS	(462.5)	(42.8)	(1,149.6)	(86.6)	687.1	(59.8)
Core segment operating income (loss) [1]	(83.1)	(1.4)	(573.5)	(7.6)	490.4	(85.5)
Corporate and other [1]	(29.8)	NM	(39.6)	NM	9.8	(24.7)
Consolidated operating loss	$ (112.9)	(2.0)%	$ (613.1)	(8.1)%	$ 500.2	(81.6)%
Adjusted EBITDA by segment:						
CCS	$ 389.6	14.4 %	$ 643.6	17.0 %	$ (254.0)	(39.5)%
OWN	178.1	20.2	269.7	18.4	(91.6)	(34.0)
NICS	225.2	20.1	51.9	5.5	173.3	333.9
ANS	229.3	21.2	285.2	21.5	(55.9)	(19.6)
Core segment adjusted EBITDA [1]	1,022.2	17.7	1,250.4	16.6	(228.2)	(18.3)
Corporate and other [1]	(23.2)	NM	(27.0)	NM	3.8	(14.1)
Non-GAAP consolidated adjusted EBITDA [2]	$ 999.0	17.3 %	$ 1,223.4	16.3 %	$ (224.4)	(18.3)%

NM – Not meaningful

(1)Core financial measures reflect the results of our CCS, OWN, NICS and ANS segments, in the aggregate, and exclude general corporate costs that were previously allocated to the Home segment reflected on the corporate and other line item. These indirect costs are classified as continuing operations since they were not directly attributable to the discontinued operations of the Home segment. In future years, these costs will be reallocated to our remaining segments and will be at least partially offset by income from our transition services agreement with Vantiva or eliminated with future restructuring actions.

(2)See “Reconciliation of Non-GAAP Measures” within this Management’s Discussion and Analysis of Financial Condition and Results of Operations.

Connectivity and Cable Solutions Segment

Net sales for the CCS segment decreased in 2023 compared to the prior year primarily due to lower sales volumes as certain customers paused spending as they right-size their inventory levels. From a regional perspective in 2023, net sales decreased in the U.S. by $783.2 million, the EMEA region by $169.9 million, the APAC region by $50.0 million, Canada by $43.4 million and the CALA region by $32.9 million compared to the prior year. Foreign exchange rate changes did not have a material impact on CCS segment net sales during 2023.

For 2023, CCS segment operating income and adjusted EBITDA were unfavorably impacted by lower sales volumes, partially offset by lower SG&A, material, freight and R&D costs, and favorable product mix. The reductions in SG&A costs impacting both operating income and adjusted EBITDA were primarily due to lower variable incentive compensation expense and cost savings initiatives, partially offset by higher bad debt expense. In 2023, CCS segment operating income was unfavorably impacted by a goodwill impairment charge of $99.1 million, but this was partially offset by a reduction of $24.0 million in amortization expense, a gain on the sale of an international manufacturing facility of $20.2 million that was recorded as a reduction to restructuring costs, a reduction of $8.9 million in transaction, transformation and integration costs and a $2.0 million recovery of accounts receivable that we had previously reserved during the prior year for $2.7 million when we determined they were uncollectible due to the Russia/Ukraine conflict. Goodwill impairment charges, amortization expense, restructuring costs, transaction, transformation and integration costs and the charges and recoveries related to accounts receivable that were determined to be uncollectible as a result of the Russia/Ukraine conflict are not reflected in adjusted EBITDA. See the discussion below under "Critical Accounting Policies and Estimates" for more information regarding the annual goodwill impairment test performed during 2023. Also see "Reconciliation of Segment Adjusted EBITDA" within this Management's Discussion and Analysis of Financial Condition and Results of Operations, below.

Outdoor Wireless Networks Segment

For 2023, OWN segment net sales decreased compared to the prior year primarily due to lower sales volumes as a result of reductions in spending by certain North American operators. From a regional perspective in 2023, OWN segment net sales decreased in the U.S. by $513.0 million, the APAC region by $23.7 million, Canada by $24.5 million, the CALA region by $15.6 million and the EMEA region by $11.1 million compared to the prior year. Foreign exchange rate changes did not have a material impact on OWN segment net sales during 2023.

For 2023, OWN segment operating income and adjusted EBITDA decreased compared to the prior year primarily due to lower sales volumes and unfavorable product mix, partially offset by lower freight, material, SG&A and R&D costs. The reductions in SG&A costs impacting both operating income and adjusted EBITDA were primarily due to lower bad debt expense, lower variable incentive compensation expense and cost savings initiatives. The bad debt expense in 2022 was driven by a $20.9 million reserve related a distributor customer. OWN segment operating income was favorably impacted by a reduction of $15.8 million in restructuring costs, a reduction of $12.0 million in amortization expense and a reduction of $3.9 million in transaction, transformation and integration costs. Amortization expense, restructuring costs and transaction, transformation and integration costs are not reflected in adjusted EBITDA. See "Reconciliation of Segment Adjusted EBITDA" within this Management's Discussion and Analysis of Financial Condition and Results of Operations, below.

Networking, Intelligent Cellular and Security Solutions Segment

Net sales for the NICS segment increased in 2023 compared to the prior year primarily due to higher sales volumes of our Ruckus products as well as our distributed antenna systems products and was favorably impacted by pricing, although to a lesser degree than volume. From a regional perspective in 2023, net sales increased in the U.S. by $124.1 million, the EMEA region by $35.5 million, the APAC region by $13.8 million, Canada by $4.5 million and the CALA region by $0.1 million compared to the prior year. Foreign exchange rate changes did not have a material impact on NICS segment net sales during 2023.

For 2023, NICS segment operating income and adjusted EBITDA increased compared to the prior year primarily due to higher sales volumes, increased pricing and lower freight, R&D, material and SG&A costs. NICS segment operating income was unfavorably impacted by $4.0 million in higher transaction, transformation and integrations cost but was favorably impacted by a gain of $3.5 million related to the settlement of an intellectual property litigation claim. Transaction, transformation and integration costs and intellectual property litigation costs are not reflected in adjusted EBITDA. See "Reconciliation of Segment Adjusted EBITDA" within this Management's Discussion and Analysis of Financial Condition and Results of Operations, below.

Access Network Solutions Segment

Net sales for our ANS segment decreased in 2023 due to lower sales volume as a result of a pause in customer spending. From a regional perspective in 2023, net sales decreased in the U.S. by $96.0 million, the CALA region by $58.5 million, the APAC region by $54.1 million and the EMEA region by $38.9 million but increased in Canada by $1.3 million compared to the prior year. Foreign exchange rate changes did not have a material impact on ANS segment net sales during 2023.

In 2023, ANS segment operating loss and adjusted EBITDA were impacted unfavorably by lower sales volumes but benefited from lower freight, SG&A and R&D costs compared to the prior year. The reductions in SG&A costs impacting both operating income and adjusted EBITDA were primarily due to cost savings initiatives and lower variable incentive compensation expense. For 2023, ANS segment operating loss was negatively impacted by a goodwill impairment charge of $472.3 million, and an increase of $3.3 million in transaction, transformation and integration costs. These negative impacts were partially offset by a decrease of $18.2 million in restructuring costs and reductions of $73.3 million in amortization expense in 2023. The reduction in restructuring costs reflects a gain on the sale of a U.S. warehouse facility of $19.4. Goodwill impairment charges, transaction, transformation and integration costs, amortization expense and restructuring expense are not reflected in adjusted EBITDA. See the discussion below under "Critical Accounting Policies and Estimates" for more information regarding the annual goodwill impairment test performed during 2023. Also see "Reconciliation of Segment Adjusted EBITDA" within this Management's Discussion and Analysis of Financial Condition and Results of Operations, below.

Liquidity and Capital Resources

The following table summarizes certain key measures of our liquidity and capital resources:

	December 31,		$	%
	2023	2022	Change	Change
	(dollars in millions)			
Cash and cash equivalents	$ 543.8	$ 373.0	$ 170.8	45.8 %
Working capital, net of assets and liabilities held for sale [(1)] and excluding cash and cash equivalents and current portion of long-term debt	970.1	1,178.4	(208.3)	(17.7)
Availability under Revolving Credit Facility	688.0	908.8	(220.8)	(24.3)
Long-term debt, including current portion	9,278.6	9,501.6	(223.0)	(2.3)
Total capitalization [(2)]	7,471.9	9,055.9	(1,584.0)	(17.5)
Long-term debt as a percentage of total capitalization	124.2%	104.9%		

(1)Working capital is net of assets and liabilities held for sale and consists of current assets of $2,584.1 million less current liabilities of $1,102.2 million as of December 31, 2023 and current assets of $3,104.3 million less current liabilities of $1,584.9 million as of December 31, 2022.

(2)Total capitalization includes long-term debt, including the current portion, Series A convertible preferred stock (Convertible Preferred Stock) and stockholders' deficit.

Our principal sources of liquidity on a short-term basis are cash and cash equivalents, cash flows provided by operations and availability under our credit facilities. On a long-term basis, our potential sources of liquidity also include raising capital through the issuance of additional equity and/or debt.

In the second quarter of 2023, we amended our 2026 Term Loan to replace LIBOR with an adjusted Secured Overnight Financing Rate as administered by the Federal Reserve Bank of New York (SOFR) as the reference interest rate in anticipation of the cessation of LIBOR. The changes were effective July 1, 2023 and we do not anticipate a material impact on our results of operations or cash flows with the transition to SOFR in our variable rate debt.

The primary uses of liquidity include debt service requirements, voluntary debt repayments, redemptions or purchases on the open market, working capital requirements, capital expenditures, business separation transaction costs, transformation costs, restructuring costs, dividends related to the Convertible Preferred Stock if we elect to pay such dividends in cash, litigation settlements, income tax payments and other contractual obligations.

We currently believe that our existing cash, cash equivalents and cash flows from operations, combined with availability under our Revolving Credit Facility, will be sufficient to meet our presently anticipated future cash needs. However, we may be required to obtain additional financing in the future to address our liquidity needs, and, subject to market conditions, we may from time to time seek to amend, refinance, restructure, exchange or repurchase our outstanding indebtedness and/or raise additional equity or other financing. Any debt we incur in the future may have terms (including cash interest rate, financial covenants and covenants limiting our operating flexibility or ability to obtain additional financings) that are not favorable to us, and any such additional equity financing may dilute the economic and/or voting interests of our existing stockholders, may be preferred in right of payment to our outstanding common stock or confer other privileges to the holders and may contain financial or operational covenants that restrict our operating flexibility or ability to obtain additional financings. Furthermore, our failure to obtain any necessary financing, amendment, refinancing, restructuring, exchange or repurchases could have a material and adverse effect on our results of operations, cash flows, financial condition and liquidity.

We may experience volatility in cash flows between periods due to, among other reasons, variability in the timing of vendor payments and customer receipts. We may, from time to time, seek to obtain alternative sources of financing, by borrowing additional amounts under our senior secured asset-based revolving credit facility (Revolving Credit Facility), issuing debt or equity securities or incurring other indebtedness, if market conditions are favorable, utilizing trade credit, selling assets (including businesses or business lines) or securitizing receivables to meet future cash needs or to reduce our borrowing costs. Any issuance of equity or debt may be for cash or in exchange for our outstanding securities or indebtedness, or a combination thereof.

Our outstanding debt securities and debt under our credit facilities are currently trading at substantial discounts to their respective principal amounts. In order to reduce future cash interest payments, as well as future amounts due at maturity or upon redemption, we may, from time to time, purchase such debt for cash, in exchange for common or preferred stock or debt, or for a combination thereof, in each case in open-market purchases and/or privately negotiated transactions, tender offers or exchange offers and upon such terms and at such prices as we may determine. Any such transactions will be dependent upon several factors, including our liquidity requirements, contractual restrictions, general market conditions and applicable regulatory, legal and accounting factors. Whether or not we engage in any such transactions will be determined at our discretion. The amounts involved in any such transactions, individually or in the aggregate, may be material.

Our interest payments on long-term debt are expected to total $1,831.1 million over the duration of the debt, with $643.0 million due in 2024 (assuming interest rates in effect as of December 31, 2023 on our variable rate debt). In 2023, the interest payments on our variable rate debt increased as a result of the Federal Reserve's increase in interest rates. Our interest payments on our variable rate debt will continue to increase if the Federal Reserve continues to increase rates. For additional information regarding our long-term debt obligations, see Note 8 in the Notes to Consolidated Financial Statements and our discussion of our interest rate risk in Item 7A. Quantitative and Qualitative Disclosures About Market Risk included elsewhere in this Annual Report on Form 10-K. For information on our obligations related to our Convertible Preferred Stock, see Note 14 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K.

In July 2023, we entered into a long-term supply contract with a third-party to secure the supply of certain raw materials. Under the terms of the contract, we will make advance payments through 2026 totaling $120.0 million (undiscounted) and based on meeting certain minimum purchase requirements through 2031, such advance payments will be credited and applied to future orders on a quarterly basis beginning in 2027 through 2031. We have committed to purchases of raw material under this agreement beginning in 2023 and growing to a level of approximately $137 million per year by 2026 and continuing through 2032.

We have $115.7 million in unrecognized tax benefits; however, the timing of the related tax payments is highly uncertain. We anticipate a reduction of up to $8.0 million of unrecognized tax benefits during the next twelve months. See Note 13 in the Notes to Consolidated Financial Statements included elsewhere in the Annual Report on Form 10-K for further discussion.

We are contingently liable under open standby letters of credit issued by our banks to support performance obligations of a third-party contractor that totaled $35.9 million as of December 31, 2023. These amounts represent our estimate of the maximum amounts we would expect to incur upon the contractual non-performance of the contractor, but we also have cross-indemnities in place that may enable us to recover some or all of our losses in the event of the contractor's non-performance. We believe the likelihood of having to perform under these guarantees is remote. There were no material amounts recorded in our consolidated financial statements related to third-party guarantee agreements as of December 31, 2023 or 2022.

Although there are no financial maintenance covenants under the terms of our senior notes, there is a limitation, among other limitations, on certain future borrowings based on an adjusted leverage ratio or a fixed charge coverage ratio. These ratios are based on financial measures similar to non-GAAP adjusted EBITDA as presented in the "Reconciliation of Non-GAAP Measures" section below, but also give pro forma effect to certain events, including acquisitions, synergies and savings from cost reduction initiatives such as facility closures and headcount reductions. For the year ended December 31, 2023, our non-GAAP pro forma adjusted EBITDA, as measured pursuant to the indentures governing our notes, was $1,101.2 million, which included annualized savings expected from cost reduction initiatives of $102.2 million so that the impact of cost reduction initiatives is fully reflected in the twelve-month period used in the calculation of the ratios. In addition to limitations under these indentures, our senior secured credit facilities contain customary negative covenants based on similar financial measures. We believe we are in compliance with the covenants under our indentures and senior secured credit facilities at December 31, 2023.

Cash and cash equivalents increased by $170.8 million during 2023 as described under the Cash Flow Overview section below. As of December 31, 2023, approximately 44% of our cash and cash equivalents were held outside the U.S.

Working capital, net of assets and liabilities held for sale and excluding cash and cash equivalents and the current portion of long-term debt, decreased during 2023 primarily due to lower accounts receivable with unfavorable operating performance driving lower net sales and lower inventory due to inventory reduction initiatives. These were partially offset by reductions in current liabilities with unfavorable operating performance driving lower accounts payable and lower accrued liabilities due to the pay out of our 2022 variable incentive compensation and a reduced 2023 incentive. During 2023, we sold accounts receivable under customer-sponsored supplier financing agreements. This had an impact of approximately $44 million on working capital, excluding cash and cash equivalents and the current portion of long-term debt, as of December 31, 2023. Under these agreements, we are able to sell accounts receivable to a bank, and we retain no interest in and have no servicing responsibilities for the accounts receivable sold. The net reduction in total capitalization during 2023 reflected the net loss for the year.

Cash Flow Overview

The cash flows related to discontinued operations have not been segregated. Accordingly, the following cash flow overview includes the results of continuing and discontinued operations. Cash and cash equivalents increased during 2023 primarily driven by cash generated by operating activities of $289.9 million and proceeds from the sale of property, plant and equipment of $71.2 million, partially offset by cash paid for debt repurchases of $142.6 million and capital expenditures of $53.3 million. The proceeds on the sale of property, plant and equipment primarily relate to selling an international manufacturing facility, along with U.S. office and warehouse facilities that were closed and consolidated into other locations as part of CommScope NEXT.

	Year Ended December 31,		$	%
	2023	2022	Change	Change
	(dollars in millions)			
Net cash generated by operating activities	$ 289.9	$ 190.0	$ 99.9	52.6%
Net cash generated by (used in) investing activities	38.3	(82.1)	120.4	NM
Net cash used in financing activities	(181.7)	(65.0)	(116.7)	179.5

NM – Not meaningful

Operating Activities

	Year Ended December 31,	
	2023	2022
	(in millions)	
Net loss	$ (1,450.9)	$ (1,286.9)
Adjustments to reconcile net loss to net cash generated by operating activities:		
Depreciation and amortization	561.2	696.1
Equity-based compensation	47.3	61.1
Deferred income taxes	(183.3)	(118.4)
Asset impairments	1,217.6	1,119.6
Changes in assets and liabilities:		
Accounts receivable	461.7	(16.0)
Inventories	391.3	(178.8)
Prepaid expenses and other current assets	45.1	30.9
Accounts payable and other accrued liabilities	(723.6)	(43.2)
Other noncurrent assets	(27.4)	8.2
Other noncurrent liabilities	55.0	(88.8)
Other	(104.1)	6.2
Net cash generated by operating activities	$ 289.9	$ 190.0

During 2023, the increase in cash generated by operating activities compared to the prior year was primarily driven by reduced inventory purchases, the impacts of cost saving initiatives and lower cash paid for taxes, partially offset by higher interest payments. For information on significant non-cash operating activities related to our discontinued operations, see Note 3 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K.

Investing Activities

	Year Ended December 31,	
	2023	2022
	(in millions)	
Additions to property, plant and equipment	$ (53.3)	$ (101.3)
Proceeds from sale of property, plant and equipment	71.2	0.1
Other	20.4	19.1
Net cash generated by (used in) investing activities	$ 38.3	$ (82.1)

During 2023, the increase in cash used generated by (used in) investing activities compared to the prior year was primarily driven by $71.2 million of proceeds from the sale of property, plant and equipment and a $48.0 decrease in capital expenditures. Capital expenditures related to our discontinued operations were $2.4 million in 2023 and $4.0 million in 2022. The current period proceeds on the sale of property, plant and equipment primarily relate to the sales of an international manufacturing facility as well as U.S. office and warehouse facilities that were closed as part of CommScope NEXT. Cash generated by other investing activities was favorably impacted in the current year by proceeds of $11.1 million related to the sale of an equity investment and proceeds of $9.3 million on the sale of certain nonfinancial assets. Cash used in other investing activities in the prior year period was favorably impacted by proceeds of $6.9 million related to the sale of an equity method investment, a return of $4.5 million on equity method investments and proceeds of $5.0 million on the sale of certain nonfinancial assets.

Financing Activities

	Year Ended December 31,	
	2023	2022
	(in millions)	
Long-term debt repaid	$ (32.0)	$ (365.0)
Long-term debt repurchases	(142.6)	—
Long-term debt proceeds	—	333.0
Debt issuance costs	—	(7.2)
Dividends paid on Series A convertible preferred stock	—	(14.9)
Tax withholding payments for vested equity-based compensation awards	(9.1)	(14.8)
Other	2.0	3.9
Net cash used in financing activities	$ (181.7)	$ (65.0)

In 2023, we repurchased $133.1 million aggregate principal amount of our 8.25% senior notes due 2027, $58.4 million aggregate principal amount of our 7.125% senior notes due 2028 and $25.4 million aggregate principal amount of our 6.00% senior notes due 2025, for total cash consideration paid of $142.6 million. We also paid quarterly scheduled amortization payments totaling $32.0 million on the 2026 Term Loan. We did not borrow under our Revolving Credit Facility during 2023.

As of December 31, 2023, we had no outstanding borrowings under the Revolving Credit Facility and the remaining availability was $688.0 million, reflecting a borrowing base subject to maximum capacity of $785.9 million reduced by $97.9 million of letters of credit issued under the Revolving Credit Facility. In the future, the availability under our Revolving Credit Facility will likely be lower with the closing of the divestiture transaction for the Home business and the sale of the related assets.

In 2022, we borrowed $333.0 million and repaid $333.0 million under the Revolving Credit Facility. We also paid four quarterly scheduled amortization payments totaling $32.0 million on our 2026 Term Loan during 2022. In connection with the refinancing of our Revolving Credit Facility in October 2022, we paid $7.2 million of debt issuance costs.

In 2023, we paid dividends of $61.8 million in additional shares due under the Convertible Preferred Stock. In 2022, we paid cash dividends of $14.9 million and paid $44.1 million of dividends in additional shares of the Convertible Preferred Stock. During 2023, employees surrendered shares of our common stock to satisfy their tax withholding requirements on vested restricted stock units and performance share units which reduced cash flows by $9.1 million compared to $14.8 million in the prior year.

Reconciliation of Non-GAAP Measures

We believe that presenting certain non-GAAP financial measures enhances an investor's understanding of our financial performance. We further believe that these financial measures are useful in assessing our operating performance from period to period by excluding certain items that we believe are not representative of our core business. We also use certain of these financial measures for business planning purposes and in measuring our performance relative to that of our competitors.

We believe these financial measures are commonly used by investors to evaluate our performance and that of our competitors. However, our use of the term "non-GAAP adjusted EBITDA" may vary from that of others in our industry. This financial measure should not be considered as an alternative to operating income (loss), net income (loss) or any other performance measures derived in accordance with U.S. GAAP as measures of operating performance, operating cash flows or liquidity.

Although there are no financial maintenance covenants under the terms of our senior notes, there is a limitation, among other limitations, on certain future borrowings based on an adjusted leverage ratio or a fixed charge coverage ratio. These ratios are based on financial measures similar to non-GAAP adjusted EBITDA as presented in this section, but also give pro forma effect to certain events, including acquisitions and savings from cost reduction initiatives such as facility closures and headcount reductions.

Consolidated

	Year Ended December 31,		
	2023	2022	2021
	(in millions)		
Loss from continuing operations	$ (851.3)	$ (1,184.7)	$ (350.0)
Income tax expense (benefit)	133.4	(15.0)	(39.2)
Interest income	(11.1)	(2.8)	(1.9)
Interest expense	675.8	588.9	561.2
Other (income) expense, net	(59.7)	0.5	26.5
Operating income (loss)	(112.9)	(613.1)	$ 196.6
Adjustments:			
Amortization of purchased intangible assets	327.1	440.0	510.0
Restructuring costs, net	29.7	63.0	85.1
Equity-based compensation	43.6	55.3	70.8
Asset impairments	571.4	1,119.6	—
Transaction, transformation and integration costs [(1)]	27.1	35.1	50.6
Acquisition accounting adjustments [(2)]	1.2	5.3	9.2
Patent claims and litigation settlements	(3.5)	1.7	3.2
Reserve (recovery) of Russian accounts receivable	(2.0)	2.7	—
Cyber incident costs [(3)]	5.5	—	—
Depreciation	111.8	113.8	117.8
Non-GAAP adjusted EBITDA	$ 999.0	$ 1,223.4	$ 1,043.3

(1)In 2023, primarily reflects transaction costs related to certain CommScope NEXT initiatives. In 2022, primarily reflects transformation costs related to CommScope NEXT and integration costs related to the ARRIS International plc (ARRIS) acquisition. In 2021, primarily reflects transaction separation costs related to the planned separation of the Home segment from CommScope, transformation costs related to CommScope NEXT and integration costs related to the ARRIS acquisition.

(2)In 2023, 2022 and 2021, reflects ARRIS acquisition accounting adjustments related to reducing deferred revenue to its fair value.

(3)In 2023, primarily reflects costs of the identification, investigation, defense, recovery and litigation efforts related to a cyber incident that occurred in late March of 2023.

Reconciliation of Segment Adjusted EBITDA

Segment adjusted EBITDA is provided as a performance measure in Note 17 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K. Below we reconcile segment adjusted EBITDA for each segment individually to operating income (loss) for that segment to supplement the reconciliation of the total segment adjusted EBITDA to consolidated operating income (loss) in that footnote.

The corporate and other line item as presented in Note 17 in the Notes to Consolidated Financial Statements represents general corporate costs that were previously allocated to the Home segment. These indirect costs are classified as continuing operations since they were not directly attributable to the discontinued operations of the Home business. In future years, these costs will be reallocated to our remaining segments and will be at least partially offset by income from our transition services agreement with Vantiva or eliminated with future restructuring actions.

Connectivity and Cable Solutions Segment

	Year Ended December 31,		
	2023	2022	2021
	(in millions)		
Operating income	$ 121.9	$ 438.3	$ 138.4
Adjustments:			
Amortization of purchased intangible assets	75.5	99.5	156.7
Restructuring costs, net	14.0	17.1	62.0
Equity-based compensation	15.6	14.9	19.5
Asset impairments	99.1	—	—
Transaction, transformation and integration costs	1.7	10.6	18.5
Patent claims and litigation settlements	—	1.7	—
Reserve (recovery) of Russian accounts receivable	(2.0)	2.7	—
Cyber incident costs	2.6	—	—
Depreciation	61.3	58.8	53.6
Adjusted EBITDA	$ 389.6	$ 643.6	$ 448.8

Outdoor Wireless Networks Segment

	Year Ended December 31,		
	2023	2022	2021
	(in millions)		
Operating income	$ 130.5	$ 189.0	$ 197.3
Adjustments:			
Amortization of purchased intangible assets	20.4	32.4	33.5
Restructuring costs, net	6.6	22.4	3.6
Equity-based compensation	6.3	7.1	8.4
Transaction, transformation and integration costs	0.6	4.5	8.5
Cyber incident costs	1.1	—	—
Depreciation	12.6	14.3	15.4
Adjusted EBITDA	$ 178.1	$ 269.7	$ 266.8

Networking, Intelligent Cellular and Security Solutions Segment

	Year Ended December 31,		
	2023	2022	2021
	(in millions)		
Operating income (loss)	$ 127.0	$ (51.2)	$ (143.5)
Adjustments:			
Amortization of purchased intangible assets	56.8	59.7	72.0
Restructuring costs, net	12.4	9.9	8.5
Equity-based compensation	10.6	13.5	17.4
Transaction, transformation and integration costs	7.0	3.0	6.2
Acquisition accounting adjustments	1.2	2.0	4.6
Patent claims and litigation settlements	(3.5)	—	0.3
Cyber incident costs	0.7	—	—
Depreciation	13.0	15.0	19.2
Adjusted EBITDA	$ 225.2	$ 51.9	$ (15.3)

Access Network Solutions Segment

	Year Ended December 31,		
	2023	2022	2021
	(in millions)		
Operating income (loss)	$ (462.5)	$ (1,149.6)	$ 71.2
Adjustments:			
Amortization of purchased intangible assets	173.9	247.2	247.0
Restructuring costs (credits), net	(6.0)	12.2	9.2
Equity-based compensation	11.0	15.8	20.9
Asset impairments	472.3	1,119.6	—
Transaction, transformation and integration costs	17.3	14.0	9.4
Acquisition accounting adjustments	0.2	3.3	4.8
Patent claims and litigation settlements	—	—	2.9
Cyber incident costs	1.0	—	—
Depreciation	22.1	22.5	25.8
Adjusted EBITDA	$ 229.3	$ 285.2	$ 391.1

Note: Components may not sum to total due to rounding.

CRITICAL ACCOUNTING POLICIES AND ESTIMATES

Our consolidated financial statements have been prepared in conformity with generally accepted accounting principles (GAAP) in the United States (U.S.). The preparation of these financial statements requires management to make estimates and assumptions that affect the amounts reported in the financial statements and accompanying notes. These estimates and their underlying assumptions form the basis for making judgments about the carrying values of assets and liabilities that are not readily apparent from other objective sources. Management bases its estimates on historical experience and on assumptions that are believed to be reasonable under the circumstances and revises its estimates, as appropriate, when changes in events or circumstances indicate that revisions may be necessary.

The following critical accounting policies and estimates reflected in our financial statements are based on management's knowledge of and experience with past and current events and on management's assumptions about future events. While we have generally not experienced significant deviations from our critical estimates in the past, it is reasonably possible that these estimates may ultimately differ materially from actual results. See Note 2 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K for a description of all our significant accounting policies.

Asset Impairment Reviews

Impairment Reviews of Goodwill

We test goodwill at the reporting unit level for impairment annually as of October 1 and on an interim basis when events occur or circumstances exist that indicate the carrying value may no longer be recoverable. We compare the fair value of our reporting units with the carrying amount, including goodwill. We recognize an impairment charge for the amount by which the reporting unit's carrying amount exceeds its fair value.

We estimate the fair value of a reporting unit using a discounted cash flow (DCF) method or, as appropriate, a combination of the DCF method and a market approach known as the guideline public company method. Under the DCF method, we calculate the fair value of a reporting unit based on the present value of estimated future cash flows. The significant assumptions in the DCF model primarily include, but are not limited to, forecasts of annual revenue growth rates, annual EBITDA margin and the discount rate used to determine the present value of the cash flow projections. When determining these assumptions and preparing these estimates, we consider historical performance trends, terminal growth rates, industry data, insight derived from customers, relevant changes in the reporting unit's underlying business and other market trends that may affect the reporting unit. The discount rate is based on the estimated weighted average cost of capital as of the test date of market participants in the industry in which the reporting unit operates and is commensurate with the risk and uncertainty inherent in each reporting unit and in internally developed forecasts. Under the guideline public company method, we estimate the fair value based upon market multiples of revenue and earnings derived from publicly-traded companies with similar operating and investment characteristics as the reporting unit. The weighting of the fair value derived from the market approach may vary depending on the level of comparability of these publicly-traded companies to the reporting unit. When comparable public companies are not meaningful or not available, we may estimate the fair value of a reporting unit using only the DCF method.

Estimating the fair value of a reporting unit involves uncertainties because it requires management to develop numerous assumptions, including assumptions about the future growth and potential volatility in revenues and costs, capital expenditures, industry economic factors and future business strategy. Changes in projected revenue growth rates, projected EBITDA margins or estimated discount rates due to uncertain market conditions, loss of one or more key customers, changes in our strategy, changes in technology or other factors could negatively affect the fair value in one or more of our reporting units and result in a material impairment charge in the future.

To assess the reasonableness of the calculated fair values of our reporting units, we also compare the sum of the reporting units' fair values to our market capitalization and calculate an implied control premium (the excess of the sum of the reporting units' fair values over the market capitalization). If the implied control premium is not reasonable, we will reevaluate the fair value estimates of the reporting units by adjusting the discount rates and/or other assumptions.

2023 Interim and Annual Goodwill Analysis

Interim Test

Goodwill is tested for impairment annually or at other times if events have occurred or circumstances exist that indicate the carrying value of the reporting unit may exceed its fair value. During the third quarter of 2023, we concluded that a triggering event occurred, primarily due to a sustained decrease in the market value of our debt and common stock affecting the overall business and changes in expected future cash flows due to reduced earnings forecasts and current macroeconomic conditions, including a rising interest rate environment. We performed an interim quantitative goodwill impairment test for our ANS and BDCC reporting units, which were most sensitive to negative performance and outlook, to compare the fair value of the reporting unit to their carrying amounts, including the goodwill. The ANS reporting unit is the same as the ANS segment, and the BDCC reporting unit is part of the CCS segment.

The fair value of the reporting unit was determined using a DCF model and a guideline public company approach, with 75% of the value determined using the DCF model and 25% of the value determined using the guideline public company approach. Under the DCF method, the fair value of a reporting unit is based on the present value of estimated future cash flows. To determine the fair value of our ANS and BDCC reporting units, we developed a revised forecast for 2023 and updated the annual forecasts for the years beyond 2023. We used a discount rate of 15% for the ANS reporting unit, which reflects increased risk from our 2022 annual test due to higher market uncertainty, and 12% for the BDCC reporting unit. Under the guideline public company method, the fair value is based upon market multiples of revenue and earnings derived from publicly-traded companies with similar operating and investment characteristics as the reporting unit.

As a result of our interim goodwill impairment test, we recorded a goodwill impairment charge of $425.9 million during the third quarter of 2023 to partially write down the carrying amount of the ANS reporting unit goodwill. There was no impairment identified in our BDCC reporting unit in the Q3 2023 interim goodwill impairment test.

Annual Test

The annual test of goodwill impairment was performed for each of the reporting units with goodwill balances as of October 1, 2023. For the 2023 annual goodwill test, we determined the fair value of each reporting unit using a DCF model and a guideline public company approach, with 75% of the value determined using the DCF model and 25% of the value determined using the market approach. The range of discount rates used in our annual tests was 11.0% to 15.0% for 2023. The Company determined the goodwill balance in the ANS and BDCC reporting units was impaired and recorded partial impairment charges of $46.3 million and $99.1 million, respectively. The impairment charges resulted from the Company's assessment in the fourth quarter of further lower revenue growth and EBITDA margins, due to adverse impacts of market conditions on the current year profitability and estimated future business results and cash flows. We used a discount rate of 15% for the ANS reporting unit and 12% for the BDCC reporting unit, which were consistent with the Q3 2023 interim test.

Considering the low headroom going forward for each of the ANS and BDCC reporting units, there is a risk for future impairment in the event of further declines in general economic, market or business conditions or any significant unfavorable change in the forecasted cash flows, weighted average cost of capital or growth rates. If current and long-term projections for the ANS and BDCC reporting units are not realized or decrease materially, we may be required to recognize additional goodwill impairment charges, and these charges could be material to our results of operations.

The following table provides summary information regarding our reporting units with goodwill balances as of December 31, 2023 that have the lowest level of headroom. The table presents key assumptions used in our annual goodwill analysis, along with sensitivity analysis showing the effect of a change in certain key assumptions, assuming all other assumptions remain constant, to the resulting fair value using an income approach.

	Key Assumptions		Goodwill		Deficit of Fair Value to Carrying Value			
Reporting Unit	Discount Rate	Terminal Growth Rate	Balance at December 31, 2023	% of Total Assets	Result of Annual Goodwill Test as of October 1, 2023	Decrease of 10% in Cash Flows	Decrease of 0.5% in Long-term Growth Rate	Increase of 0.5% in Discount Rate
ANS	15.0%	1.0%	$ 261.4	2.8%	$ (46.3)	$ (154.6)	$ (62.5)	$ (86.7)
BDCC	12.0%	1.5%	881.5	9.4%	(99.1)	(208.9)	(126.0)	(154.6)

Definite-Lived Intangible Assets and Other Long-Lived Assets

Management reviews definite-lived intangible assets and other long-lived assets for impairment when events or changes in circumstances indicate that their carrying values may not be fully recoverable. This analysis differs from our goodwill impairment analysis in that an intangible or other long-lived asset impairment is only deemed to have occurred if the sum of the forecasted undiscounted future net cash flows related to the assets being evaluated is less than the carrying value of the assets. If the forecasted net cash flows are less than the carrying value, then the asset is written down to its estimated fair value. Other than certain assets impaired as a result of restructuring actions, we did not identify any impairments of definite-lived intangible assets or other long-lived assets in 2023. Changes in the estimates of forecasted net cash flows or changes in classification from held for use may result in future asset impairments that could be material to our results of operations.

Revenue Recognition

We recognize revenue based on the satisfaction of distinct obligations to transfer goods and services to customers. Our revenue is generated primarily from product or equipment sales. We apply a five-step approach as defined in ASC 606, *Revenue from Contracts with Customers*, in determining the amount and timing of revenue to be recognized: (1) identify the contract with a customer; (2) identify the performance obligations in the contract; (3) determine the transaction price; (4) allocate the transaction price to the performance obligations in the contract; and (5) recognize revenue when a corresponding performance obligation is satisfied. Most contracts with customers are to provide distinct products or services within a single contract. However, if a contract is separated into more than one performance obligation, the total transaction price is allocated to each performance obligation in an amount based on the estimated relative standalone selling price.

Product sales, to end-customers or distributors, represent over 90% of our revenue and are generally recognized at the point in time when products have been shipped, right to payment has been obtained and risk of loss has been transferred. Certain of our product performance obligations include proprietary operating system software, which typically is not considered separately identifiable. Therefore, sales of these products and the related software are considered one performance obligation.

License contracts include revenue recognized for the licensing of intellectual property, including software, sold separately without products. Functional intellectual property licenses do not meet the criteria for revenue to be recognized over time, and revenue is most commonly recognized upon delivery of the license/software to the customer.

Revenue is measured based on the consideration to which we expect to be entitled based on customer contracts. Sales are adjusted for variable consideration amounts, including, but not limited to, estimated discounts, rebates, distributor price protection programs and returns. These estimates are determined based upon historical experience, contract terms, inventory levels in the distributor channel and other related factors. Adjustments to variable consideration estimates are recorded when circumstances indicate revisions may be necessary. Variable consideration is primarily related to sales to our distributors, system integrators and value-added resellers.

Contingencies and Litigation

We are a party to lawsuits, claims and proceedings incident to the operation of our business, including intellectual property infringement matters, those pertaining to labor and employment contracts and other matters, some of which allege substantial monetary damages. We assess these matters in order to determine if a contingent liability should be recorded. In making this determination, management may, depending on the nature of the matter, consult with internal and external legal counsel and technical experts. We expense legal fees associated with consultations and defense of lawsuits as incurred. We accrue for loss contingencies when losses become probable and are reasonably estimable. If the reasonable estimate of the loss is a range and no amount within the range is a better estimate, the minimum amount of the range is recorded as a liability. Gain contingencies are recognized when they are realized.

Litigation outcomes are difficult to predict and are often resolved over long periods of time, making our estimates highly judgmental. Estimating probable losses requires the analysis of multiple possible outcomes that often depend on judgments about potential actions by third parties, such as future changes in facts and circumstances, differing interpretations of the law, assessments of the amount of damages and other factors beyond our control. There is the potential for a material adverse effect on our results of operations and cash flows if one or more matters are resolved in a particular period in an amount materially in excess of what we anticipated. Alternatively, if the judgments and estimates made by management are incorrect and a particular contingent loss does not occur, the contingent loss recorded would be reversed, thereby favorably impacting our results of operations.

Inventory Reserves

We maintain reserves to reduce the value of inventory based on the lower of cost or net realizable value, including allowances for excess and obsolete inventory. These reserves are based on management’s assumptions about and analysis of relevant factors including current levels of orders and backlog, forecasted demand, market conditions and new products or innovations that diminish the value of existing inventories. If actual market conditions deteriorate from those anticipated by management, additional allowances for excess and obsolete inventory could be required and may be material to our results of operations.

Tax Valuation Allowances and Liabilities for Unrecognized Tax Benefits

We establish an income tax valuation allowance when available evidence indicates that it is more likely than not that all or a portion of a deferred tax asset will not be realized. In assessing the need for a valuation allowance, we consider the amounts, character, source and timing of expected future deductions or carryforwards as well as sources of taxable income and tax planning strategies that may enable utilization. We maintain an existing valuation allowance until sufficient positive evidence exists to support its reversal. Changes in the amount or timing of expected future deductions or taxable income may have a material impact on the level of income tax valuation allowances. If we determine that we will not be able to realize all or part of a deferred tax asset in the future, an increase to an income tax valuation allowance would be charged to earnings in the period such determination was made.

We also establish allowances related to value-added and similar recoverable taxes when it is considered probable that those assets are not recoverable. Changes in the probability of recovery or in the estimates of the amount recoverable are recognized in the period such determination is made and may be material to our loss from continuing operations.

We recognize income tax benefits related to particular tax positions only when it is considered more likely than not that the tax position will be sustained if examined on its technical merits by tax authorities. The amount of benefit recognized is the largest amount of tax benefit that is evaluated to be greater than 50% likely to be realized. Considerable judgment is required to evaluate the technical merits of various positions and to evaluate the likely amount of benefit to be realized. Lapses in statutes of limitations, developments in tax laws, regulations and interpretations, and changes in assessments of the likely outcome of uncertain tax positions could have a material impact on the overall tax provision.

RECENT ACCOUNTING PRONOUNCEMENTS

See Note 2 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K for a discussion of recent accounting pronouncements.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

We are exposed to market risks related to changes in interest rates, foreign currency exchange rates and commodity prices. We may utilize derivative financial instruments, among other methods, to hedge some of these exposures. We do not use derivative financial instruments for speculative or trading purposes.

Interest Rate Risk

The table below summarizes the expected interest and principal payments associated with our variable rate debt outstanding at December 31, 2023, primarily the 2026 Term Loan and the Revolving Credit Facility. The principal payments presented below are based on scheduled maturities and assume no borrowings under our Revolving Credit Facility. The interest payments presented below assume the interest rates in effect as of December 31, 2023 (see Note 8 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K). The impact of a 1% increase in the interest rate index on projected future interest payments on the variable rate debt is also included in the table below.

	2024	2025	2026	2027	2028	Thereafter
Principal and interest payments on variable rate debt	$ 298.4	$ 292.2	$ 3,032.0	$ —	$ —	$ —
Average cash interest rate	8.83 %	8.72 %	8.72 %	—	—	—
Impact of 1% increase in interest rate index	$ 30.2	$ 29.8	$ 3.7	$ —	$ —	$ —

We also have $6.3 billion aggregate principal amount of fixed rate senior notes. The table below summarizes our expected interest and principal payments related to our fixed rate debt at December 31, 2023.

	2024	2025	2026	2027	2028	Thereafter
Principal and interest payments on fixed rate debt	$ 376.6	$ 1,616.9	$ 1,759.1	$ 1,776.5	$ 746.7	$ 1,279.7
Average cash interest rate	6.06 %	6.06 %	6.08 %	5.91 %	5.24 %	4.75 %

We utilize a hedging strategy to mitigate a portion of the exposure to changes in cash flows resulting from variable interest rates on the 2026 Term Loan. In conjunction with the amendment to our 2026 Term Loan due to reference rate reform, on June 28, 2023, the Company settled its cash flow hedges with a notional value of $300.0 million that were outstanding. During the third quarter of 2023, and following the amendment to the 2026 Term Loan and settlement of the cash flow hedges, we again reenacted our hedging strategy to mitigate the interest rate risk from our variable rate debt, initially associated with our amended 2026 Term Loan and extending to future borrowings or debt issued, to fix a portion of the future interest cash flows by designating qualifying receive-variable and pay-fixed interest rate swaps as a cash flow hedge for accounting and financial reporting purposes. The total notional amount of the interest rate swap derivatives as of December 31, 2023 was $700.0 million with outstanding maturities up to thirty-one months. As of December 31, 2023, the combined fair value of the interest rate swaps was an $8.0 million loss. The table above excludes the impact of these interest rate swap derivatives. See Note 9 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K for further discussion of these contracts.

Foreign Currency Risk

Approximately 35% and 33% of net sales for 2023 and 2022, respectively, were to customers located outside the U.S. Significant changes in foreign currency exchange rates could adversely affect our international sales levels and the related collection of amounts due. In addition, a significant decline in the value of currencies used in certain regions of the world as compared to the U.S. dollar could adversely affect product sales in those regions because our products may become more expensive for those customers to pay for in their local currency. Conversely, significant increases in the value of foreign currencies as compared to the U.S. dollar could adversely affect profitability as certain product costs increase relative to a U.S. dollar-denominated sales price. The foreign currencies to which we have the greatest exposure include the Chinese yuan, euro, British pound sterling, Mexican peso, Japanese yen, Canadian dollar, Australian dollar, Brazilian real, South African rand, Indian rupee and Czech koruna. Local manufacturing provides a partial natural hedge and we continue to evaluate additional alternatives to help us reasonably manage the market risk related to foreign currency exposures.

We use derivative instruments such as forward exchange contracts to manage the risk of fluctuations in the value of certain foreign currencies. As of December 31, 2023, we had foreign exchange contracts with a net unrealized gain of $5.1 million, with maturities of up to nine months and aggregate notional value of $630.1 million (based on exchange rates as of December 31, 2023). These contracts are not designated as hedges for accounting purposes and are marked to market each period through earnings and, as such, there were no unrecognized gains or losses as of December 31, 2023 or 2022. Our derivative instruments are not leveraged and are not held for trading or speculation. See Note 9 in the Notes to Consolidated Financial Statements included elsewhere in this Annual Report on Form 10-K for further discussion of these contracts. We continuously evaluate the amount and type of derivative instruments utilized to manage the market risk related to foreign currency exposures.

Commodity Price Risk

Materials account for a large portion of our cost of sales. These materials, such as aluminum, copper, steel, bimetals, optical fiber, plastics and other polymers, capacitors, memory devices and silicon chips, are subject to changes in market price as they are influenced by commodity markets and supply and demand levels, among other factors. Management attempts to mitigate these risks through effective requirements planning and by working closely with key suppliers to obtain the best possible pricing and delivery terms. We may also enter into agreements with certain suppliers to guarantee our access to certain key components. As of December 31, 2023, we had no forward purchase commitments outstanding under take-or-pay contracts. We continuously evaluate the amount and type of derivative instruments utilized to manage commodity price risk.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

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Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of CommScope Holding Company, Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of CommScope Holding Company, Inc. (the Company) as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive loss, stockholders' deficit and cash flows for each of the three years in the period ended December 31, 2023, and the related notes (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework), and our report dated February 28, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the financial statements that were communicated or required to be communicated to the audit committee and that: (1) relate to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Valuation of Goodwill

Description of the Matter

As more fully described in Note 4 to the consolidated financial statements, at December 31, 2023, the Company's goodwill was $3,514.4 million, of which $261.4 million and $881.5 million relate to the Access Network Solutions (ANS) and Building and Data Center Connectivity (BDCC) reporting units, respectively. The Company's goodwill is initially assigned to its reporting units as of the acquisition date.

Goodwill is tested for impairment at the reporting unit level annually, or more frequently if indicators of potential goodwill impairment exist. The Company performed its goodwill impairment tests using both a discounted cash flow model and a guideline public company approach. During the third quarter of 2023, the Company determined that indicators of goodwill impairment existed for the ANS and BDCC reporting units and performed an interim goodwill impairment test, which indicated that the carrying value of the ANS reporting unit exceeded its fair value. The Company performed its annual goodwill impairment test in the fourth quarter of 2023. As a result of the impairment tests, the Company recorded goodwill impairment charges of $472.3 million and $99.1 million related to the ANS and BDCC reporting units, respectively, which reflects a partial impairment of the goodwill of these reporting units.

Auditing management's goodwill impairment tests was complex and highly judgmental due to the significant estimation required in determining the fair values of the ANS and BDCC reporting units. In particular, the fair value estimates were sensitive to changes in estimated discount rates, projected revenue growth rates and projected EBITDA margin percentages.

How We Addressed the Matter in Our Audit

We evaluated the Company's assessments of the impairment of goodwill for the ANS and BDCC reporting units. We obtained an understanding, evaluated the design and tested the operating effectiveness of controls that address the risks of material misstatement relating to the interim and annual goodwill impairment tests for these reporting units, including controls over management's development and review of the assumptions discussed above.

To test the estimated fair values of the ANS and BDCC reporting units, we performed audit procedures with the assistance of our valuation specialists that included, among others, assessing methodologies and testing the assumptions discussed above and the underlying data used by the Company in its analyses. We evaluated the Company's estimated discount rate methodology and developed independent ranges of reasonable discount rates. We compared the assumptions of projected revenue growth rates and projected EBITDA margin percentages used by management to current industry and economic trends, changes to the Company's business and other relevant factors. We also evaluated the reasonableness of the guideline public companies used to develop the fair value estimates of the ANS and BDCC reporting units. We assessed the historical accuracy of management's estimates and performed sensitivity analyses to evaluate the changes in the fair value of these reporting units that would result from changes in the assumptions discussed above. We also evaluated the related goodwill disclosures included in Note 4 to the consolidated financial statements.

Incomes Taxes - Valuation Allowance

Description of the Matter

As more fully described in Note 13 to the consolidated financial statements, at December 31, 2023, the Company recognized deferred tax assets related to deductible temporary differences and carryforwards of $913.0 million, net of valuation allowances of $838.5 million. Deferred tax assets are reduced by a valuation allowance if, based on the weight of all available evidence, in management's judgment, it is more likely than not that some portion, or all, of the deferred tax assets will not be realized.

Auditing management's assessment of the realizability of deferred tax assets was complex and highly judgmental due to the significant estimation required in determining whether sufficient projected future taxable income supports the realization of the Company's existing deferred tax assets before expiration.

How We Addressed the Matter in Our Audit

We evaluated the Company's assessment of the realizability of deferred tax assets, with the assistance of our income tax professionals. We obtained an understanding, evaluated the design and tested the operating effectiveness of controls that address the risks of material misstatement relating to the realizability of deferred tax assets, including controls over management's development and review of the projected future taxable income.

To test the realizability of deferred tax assets, we performed audit procedures that included, among others, evaluating whether the sources of management's estimated future taxable income were of the appropriate character and would be sufficient to utilize the deferred tax assets under the relevant tax laws. We also evaluated the significant assumptions used by the Company to develop estimates of future taxable income and tested the completeness and accuracy of the underlying data. We also evaluated the related income tax disclosures included in Note 13 to the consolidated financial statements.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 2008.

Charlotte, North Carolina
February 28, 2024

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of CommScope Holding Company, Inc.

Opinion on Internal Control Over Financial Reporting

We have audited CommScope Holding Company, Inc.'s internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, CommScope Holding Company, Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive loss, stockholders' deficit and cash flows for each of the three years in the period ended December 31, 2023, and the related notes and our report dated February 28, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

Charlotte, North Carolina
February 28, 2024

CommScope Holding Company, Inc.
Consolidated Statements of Operations
(In millions, except per share amounts)

	Year Ended December 31,		
	2023	2022	2021
Net sales	$ 5,789.2	$ 7,524.7	$ 6,737.4
Cost of sales	3,640.9	4,930.7	4,297.8
Gross profit	2,148.3	2,594.0	2,439.6
Operating expenses:			
Selling, general and administrative	873.3	1,040.9	1,082.9
Research and development	459.7	543.6	565.0
Amortization of purchased intangible assets	327.1	440.0	510.0
Restructuring costs, net	29.7	63.0	85.1
Asset impairments	571.4	1,119.6	—
Total operating expenses	2,261.2	3,207.1	2,243.0
Operating income (loss)	(112.9)	(613.1)	196.6
Other income (expense), net	59.7	(0.5)	(26.5)
Interest expense	(675.8)	(588.9)	(561.2)
Interest income	11.1	2.8	1.9
Loss from continuing operations before income taxes	(717.9)	(1,199.7)	(389.2)
Income tax (expense) benefit	(133.4)	15.0	39.2
Loss from continuing operations	(851.3)	(1,184.7)	(350.0)
Loss from discontinued operations, net of income tax (expense) benefit of $184.0, $(1.9) and $32.6, respectively	(599.6)	(102.2)	(112.6)
Net loss	(1,450.9)	(1,286.9)	(462.6)
Series A convertible preferred stock dividends	(61.8)	(59.0)	(57.3)
Net loss attributable to common stockholders	$ (1,512.7)	$ (1,345.9)	$ (519.9)
Basic:			
Loss from continuing operations per share	$ (4.33)	$ (6.00)	$ (2.00)
Loss from discontinued operations per share	(2.84)	(0.49)	(0.55)
Loss per share	$ (7.17)	$ (6.49)	$ (2.55)
Diluted:			
Loss from continuing operations per share	$ (4.33)	$ (6.00)	$ (2.00)
Loss from discontinued operations per share	(2.84)	(0.49)	(0.55)
Loss per share	$ (7.17)	$ (6.49)	$ (2.55)
Weighted average shares outstanding:			
Basic	210.9	207.4	203.6
Diluted	210.9	207.4	203.6

See notes to consolidated financial statements.

CommScope Holding Company, Inc.
Consolidated Statements of Comprehensive Loss
(In millions)

	Year Ended December 31,		
	2023	**2022**	**2021**
Comprehensive loss:			
Net loss	$ (1,450.9)	$ (1,286.9)	$ (462.6)
Other comprehensive income (loss), net of tax:			
Foreign currency translation gain (loss)	37.9	(104.5)	(85.3)
Defined benefit plans:			
Change in unrecognized actuarial gain (loss)	(0.5)	(1.5)	22.8
Change in unrecognized net prior service credit	0.4	0.1	0.2
Gain (loss) on hedging instruments	(8.2)	16.0	11.8
Total other comprehensive income (loss), net of tax	29.6	(89.9)	(50.5)
Total comprehensive loss	$ (1,421.3)	$ (1,376.8)	$ (513.1)

See notes to consolidated financial statements.

CommScope Holding Company, Inc.
Consolidated Balance Sheets
(In millions, except share amounts)

	December 31, 2023	2022
Assets		
Cash and cash equivalents	$ 543.8	$ 373.0
Accounts receivable, net of allowance for doubtful accounts of $32.2 and $50.1, respectively	815.2	1,178.2
Inventories, net	1,079.7	1,376.0
Prepaid expenses and other current assets	145.4	177.1
Current assets held for sale	278.6	621.8
Total current assets	2,862.7	3,726.1
Property, plant and equipment, net of accumulated depreciation of $866.1 and $849.3, respectively	500.6	601.0
Goodwill	3,514.4	4,072.4
Other intangible assets, net	1,582.7	1,915.1
Deferred income taxes	615.6	494.6
Other noncurrent assets	295.9	275.4
Noncurrent assets held for sale	—	600.8
Total assets	$ 9,371.9	$ 11,685.4
Liabilities and Stockholders' Deficit		
Accounts payable	$ 435.9	$ 684.3
Accrued and other liabilities	634.3	868.6
Current portion of long-term debt	32.0	32.0
Current liabilities held for sale	307.2	522.6
Total current liabilities	1,409.4	2,107.5
Long-term debt	9,246.6	9,469.6
Deferred income taxes	110.7	115.5
Other noncurrent liabilities	411.9	355.9
Noncurrent liabilities held for sale	—	82.6
Total liabilities	11,178.6	12,131.1
Commitments and contingencies		
Series A convertible preferred stock, $0.01 par value	1,162.1	1,100.3
Stockholders' deficit:		
Preferred stock, $0.01 par value: Authorized shares: 200,000,000; Issued and outstanding shares: 1,162,085 and 1,100,310, respectively, Series A convertible preferred stock	—	—
Common stock, $0.01 par value: Authorized shares: 1,300,000,000; Issued and outstanding shares: 212,108,634 and 208,371,426, respectively	2.3	2.2
Additional paid-in capital	2,550.4	2,542.9
Accumulated deficit	(4,953.1)	(3,502.2)
Accumulated other comprehensive loss	(266.7)	(296.3)
Treasury stock, at cost: 14,424,126 shares and 12,726,695 shares, respectively	(301.7)	(292.6)
Total stockholders' deficit	(2,968.8)	(1,546.0)
Total liabilities and stockholders' deficit	$ 9,371.9	$ 11,685.4

See notes to consolidated financial statements.

CommScope Holding Company, Inc.
Consolidated Statements of Cash Flows [1]
(In millions)

	Year Ended December 31,		
	2023	2022	2021
Operating Activities:			
Net loss	$ (1,450.9)	$ (1,286.9)	$ (462.6)
Adjustments to reconcile net loss to net cash generated by operating activities:			
Depreciation and amortization	561.2	696.1	786.3
Equity-based compensation	47.3	61.1	79.6
Deferred income taxes	(183.3)	(118.4)	(147.5)
Asset impairments	1,217.6	1,119.6	13.7
Changes in assets and liabilities:			
Accounts receivable	461.7	(16.0)	(59.6)
Inventories	391.3	(178.8)	(359.8)
Prepaid expenses and other current assets	45.1	30.9	3.2
Accounts payable and other accrued liabilities	(723.6)	(43.2)	256.0
Other noncurrent assets	(27.4)	8.2	(45.5)
Other noncurrent liabilities	55.0	(88.8)	8.4
Other	(104.1)	6.2	50.1
Net cash generated by operating activities	289.9	190.0	122.3
Investing Activities:			
Additions to property, plant and equipment	(53.3)	(101.3)	(131.4)
Proceeds from sale of property, plant and equipment	71.2	0.1	13.1
Payments upon settlement of net investment hedge	—	—	(18.0)
Other	20.4	19.1	(0.5)
Net cash generated by (used in) investing activities	38.3	(82.1)	(136.8)
Financing Activities:			
Long-term debt repaid	(32.0)	(365.0)	(1,282.0)
Long-term debt repurchases	(142.6)	—	—
Long-term debt proceeds	—	333.0	1,250.0
Debt issuance costs	—	(7.2)	(12.0)
Debt extinguishment costs	—	—	(34.4)
Dividends paid on Series A convertible preferred stock	—	(14.9)	(43.0)
Proceeds from the issuance of common shares under equity-based compensation plans	—	—	5.6
Tax withholding payments for vested equity-based compensation awards	(9.1)	(14.8)	(26.4)
Other	2.0	3.9	2.7
Net cash used in financing activities	(181.7)	(65.0)	(139.5)
Effect of exchange rate changes on cash and cash equivalents	(0.8)	(5.1)	(7.6)
Change in cash and cash equivalents	145.7	37.8	(161.6)
Cash and cash equivalents at beginning of period	398.1	360.3	521.9
Cash and cash equivalents at end of period	$ 543.8	$ 398.1	$ 360.3

(1) The cash flows related to discontinued operations have not been segregated. Accordingly, the Consolidated Statements of Cash Flows include the results of continuing and discontinued operations.

See notes to consolidated financial statements.

CommScope Holding Company, Inc.
Consolidated Statements of Stockholders' Deficit
(In millions, except share amounts)

	Year Ended December 31,		
	2023	**2022**	**2021**
Number of common shares outstanding:			
Balance at beginning of period	208,371,426	204,567,294	200,095,232
Issuance of shares under equity-based compensation plans	5,434,639	5,560,242	6,219,566
Shares surrendered under equity-based compensation plans	(1,697,431)	(1,756,110)	(1,747,504)
Balance at end of period	212,108,634	208,371,426	204,567,294
Common stock:			
Balance at beginning of period	$ 2.2	$ 2.2	$ 2.1
Issuance of shares under equity-based compensation plans	0.1	—	0.1
Balance at end of period	$ 2.3	$ 2.2	$ 2.2
Additional paid-in capital:			
Balance at beginning of period	$ 2,542.9	$ 2,540.7	$ 2,512.9
Issuance of shares under equity-based compensation plans	—	0.1	5.5
Equity-based compensation	47.3	61.1	79.6
Dividends on Series A convertible preferred stock	(61.8)	(59.0)	(57.3)
Other	22.0	—	—
Balance at end of period	$ 2,550.4	$ 2,542.9	$ 2,540.7
Accumulated deficit:			
Balance at beginning of period	$ (3,502.2)	$ (2,215.3)	$ (1,752.7)
Net loss	(1,450.9)	(1,286.9)	(462.6)
Balance at end of period	$ (4,953.1)	$ (3,502.2)	$ (2,215.3)
Accumulated other comprehensive loss:			
Balance at beginning of period	$ (296.3)	$ (206.4)	$ (155.9)
Other comprehensive income (loss), net of tax	29.6	(89.9)	(50.5)
Balance at end of period	$ (266.7)	$ (296.3)	$ (206.4)
Treasury stock, at cost:			
Balance at beginning of period	$ (292.6)	$ (277.8)	$ (251.4)
Net shares surrendered under equity-based compensation plans	(9.1)	(14.8)	(26.4)
Balance at end of period	$ (301.7)	$ (292.6)	$ (277.8)
Total stockholders' deficit	$ (2,968.8)	$ (1,546.0)	$ (156.6)

See notes to consolidated financial statements.

1. BACKGROUND AND DESCRIPTION OF THE BUSINESS

CommScope Holding Company, Inc., along with its direct and indirect subsidiaries (CommScope or the Company), is a global provider of infrastructure solutions for communication, data center and entertainment networks. The Company's solutions for wired and wireless networks enable service providers including cable, telephone and digital broadcast satellite operators and media programmers, to deliver media, voice, Internet Protocol (IP) data services and Wi-Fi to their subscribers and allow enterprises to experience constant wireless and wired connectivity across complex and varied networking environments. The Company's solutions are complemented by services including technical support, systems design and integration. CommScope is a leader in digital video and IP television (IPTV) distribution systems, broadband access infrastructure platforms and equipment that delivers data and voice networks to homes. CommScope's global leadership position is built upon innovative technology, broad solution offerings, high-quality and cost-effective customer solutions, and global manufacturing and distribution scale.

On October 2, 2023, the Company entered into a Call Option Agreement with Vantiva SA, a société anonyme organized under the Laws of France (Vantiva), pursuant to which the Company granted Vantiva a binding call option to acquire the Home Networks (Home) segment and substantially all of the associated segment assets and liabilities (Home business), which was subsequently exercised and a Purchase Agreement signed on December 7, 2023. The Company determined the anticipated sale of the Home business met the "held for sale" criteria and the "discontinued operations" criteria in accordance with Accounting Standards Codification (ASC) No. 360-10, *Impairment and Disposal of Long Lived Assets*, and ASC No. 205-20, *Presentation of Financial Statements: Discontinued Operations,* in the fourth quarter of 2023 due to its relative size and strategic rationale. All prior period amounts in these consolidated financial statements have been recast to reflect the discontinuation of the Home business in accordance with ASC 205-20. The discussions in these consolidated financial statements relate solely to the Company's continuing operations, unless otherwise noted. As a result, the Company is reporting financial performance based on the following remaining four operating segments, which excludes the Home segment: Connectivity and Cable Solutions (CCS), Outdoor Wireless Networks (OWN), Networking, Intelligent Cellular and Security Solutions (NICS) and Access Network Solutions (ANS). For further discussion of the discontinued operation related to the Home business, see Note 3.

2. SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES

Basis of Consolidation

The accompanying consolidated financial statements include CommScope Holding Company, Inc., along with its direct and indirect subsidiaries. All intercompany accounts and transactions are eliminated in consolidation.

Certain prior year amounts have been reclassified to conform to the current year presentation.

Use of Estimates in the Preparation of the Financial Statements

The preparation of the accompanying consolidated financial statements in conformity with accounting principles generally accepted in the United States (U.S.) requires management to make estimates and assumptions that affect the amounts reported in the financial statements and accompanying notes. These estimates and their underlying assumptions form the basis for making judgments about the carrying values of assets and liabilities that are not readily apparent from other objective sources. The Company bases its estimates on historical experience and on assumptions that are believed to be reasonable under the circumstances and revises its estimates, as appropriate, when events or changes in circumstances indicate that revisions may be necessary. Significant accounting estimates reflected in the Company's financial statements include the allowance for doubtful accounts; reserves for sales returns, discounts, allowances, rebates and distributor price protection programs; inventory excess and obsolescence reserves; contingent liabilities; tax valuation allowances; liabilities for unrecognized tax benefits; impairment reviews for investments, property, plant and equipment, goodwill and other intangible assets; and pension and other postretirement benefit costs and liabilities. Although these estimates are based on management's knowledge of and experience with past and current events and on management's assumptions about future events, it is at least reasonably possible that they may ultimately differ materially from actual results.

Cash and Cash Equivalents

Cash and cash equivalents represent deposits in banks and cash invested temporarily in various instruments that are highly liquid and have a maturity of three months or less at the time of purchase.

Accounts Receivable and Allowance for Doubtful Accounts

Trade accounts receivable and contract assets for unbilled receivables are stated at the amount owed by the customer, net of allowances for estimated doubtful accounts, discounts, returns and rebates. The Company measures the allowance for doubtful accounts using an expected credit loss model, which uses a lifetime expected loss allowance for all trade accounts receivable and contract assets. To measure the expected credit losses, trade accounts receivable and contract assets are grouped based on shared credit risk characteristics and the days past due based on the contractual terms of the receivable. Contract assets relate to unbilled work in progress and have substantially the same risk characteristics as trade accounts receivable for the same types of contracts. Therefore, the Company has concluded that the expected loss rates for trade accounts receivable are a reasonable approximation of the loss rates for the contract assets.

In calculating an allowance for doubtful accounts, the Company uses its historical experience, external indicators and forward-looking information to calculate expected credit losses using an aging method. The Company assesses impairment of trade accounts receivable on a collective basis, as they possess shared credit risk characteristics which have been grouped based on the days past due.

The expected loss rates are based on the payment profiles of sales over the preceding thirty-six months and the corresponding historical credit losses experienced within this period. The historical loss rates are adjusted to reflect current and forward-looking information on macroeconomic factors affecting the ability of the customers to settle their trade accounts receivable. Accounts are written off against the allowance account when they are determined to be no longer collectible.

The Company sells certain of its accounts receivable under a customer-sponsored supplier financing agreement. Under this agreement, the Company is able to sell certain accounts receivable to a bank at a discount. The Company sold approximately $183 million and $339 million of trade accounts receivable under this program during the years ended December 31, 2023 and 2022, respectively, and the cost of factoring such receivables was not material. The Company derecognizes the accounts receivable on the Consolidated Balance Sheets once sold to the bank, as it retains no interest in and has no servicing responsibilities for them once they have been sold. The cash received from the bank is classified within the operating activities section in the Consolidated Statements of Cash Flows.

Inventories

Inventories are stated at the lower of cost or net realizable value. Inventory cost is determined on a first-in, first-out (FIFO) basis. Costs such as idle facility expense, excessive scrap and re-handling costs are expensed as incurred. The Company maintains reserves to reduce the value of inventory to the lower of cost or net realizable value, including reserves for excess and obsolete inventory.

Assets Held for Sale and Discontinued Operations

The Company reports the results of operations of a business as discontinued operations if a disposal represents a strategic shift that has or will have a major effect on the Company's operations and financial results when the business is sold, in accordance with the criteria of ASC Topic 205-20, *Presentation of Financial Statements—Discontinued Operations*, and classified as held for sale, in accordance with ASC Topic 360-10, *Impairment and Disposal of Long Lived Assets*. The results of discontinued operations are reported in loss from discontinued operations, net of income taxes in the accompanying Consolidated Statements of Operations for the current and prior periods and include any gain or loss recognized on classification as held for sale, or adjustment of the carrying amount or estimated fair value less cost to sell. If the carrying amount of the business exceeds its estimated fair value less cost to sell, a loss is recognized. Assets and liabilities related to a business classified as held for sale are segregated in the current and prior periods on the Consolidated Balance Sheets. If a business is classified as held for sale after the balance sheet date but before the financial statements are issued or are available to be issued, the business continues to be classified as held and used in those financial statements when issued or when available to be issued.

Derivative Instruments and Hedging Activities

CommScope is exposed to risks resulting from adverse fluctuations in commodity prices, interest rates and foreign currency exchange rates. CommScope’s risk management strategy includes the use of derivative financial instruments whenever management determines their use to be reasonable and practical. This strategy does not permit the use of derivative financial instruments for trading or speculation.

The Company periodically uses forward contracts to hedge a portion of its balance sheet foreign exchange re-measurement risk and to hedge certain planned foreign currency expenditures. Unrealized gains and losses resulting from these contracts are recognized in other income (expense), net and partially offset corresponding foreign exchange gains and losses on the balances and expenditures being hedged. These instruments are not designated as hedges for hedge accounting purposes and are marked to market each period through earnings.

The Company also utilizes a hedging strategy to mitigate the interest rate risk from its variable rate debt, initially associated with its amended senior secured term loan due 2026 (2026 Term Loan) and extending to future borrowings or debt issued, to fix a portion of the future interest cash flows by designating qualifying receive-variable and pay-fixed interest rate swaps as a cash flow hedge for accounting and financial reporting purposes. Hedge effectiveness is assessed each quarter, and for hedges that meet the effectiveness requirements, changes in fair value are recorded as a component of other comprehensive income (loss), net of tax, and are reclassified to interest expense as interest payments are made on the Company's variable rate debt.

The Company has elected and documented the use of the normal purchases and sales exception for normal purchase and sales contracts that meet the definition of a derivative financial instrument. See Note 9 for further disclosure related to the derivative instruments and hedging activities.

Leases

The Company determines if a contract is a lease or contains a lease at inception. Right of use assets related to operating type leases are reported in other noncurrent assets and the present value of remaining lease obligations is reported in accrued and other liabilities and other noncurrent liabilities on the Consolidated Balance Sheets. For the periods presented, CommScope does not have any financing type leases.

Operating lease liabilities are recognized based on the present value of the future minimum lease payments over the lease term at commencement date. The majority of the Company's leases do not provide an implicit rate; therefore, the Company uses the incremental borrowing rates applicable to the economic environment and the duration of the lease, based on the information available at commencement date, in determining the present value of future payments. The right of use asset for operating leases is measured using the lease liability adjusted for the impact of lease payments made prior to commencement, lease incentives received, initial direct costs incurred and any asset impairments. Lease terms may include options to extend or terminate the lease when it is reasonably certain that the option will be exercised. Lease expense for minimum lease payments is recognized on a straight-line basis over the lease term.

The Company remeasures and reallocates the consideration in a lease when there is a modification of the lease that is not accounted for as a separate contract. The lease liability is remeasured when there is a change in the lease term or a change in the assessment of whether the Company will exercise a lease option. The Company assesses right of use assets for impairment in accordance with its long-lived asset impairment policy.

The Company accounts for lease agreements with contractually required lease and non-lease components on a combined basis. Lease payments made for cancellable leases, variable amounts that are not based on an observable index and lease agreements with an original duration of less than twelve months are recorded directly to lease expense.

Property, Plant and Equipment

Property, plant and equipment are stated at cost. Upon application of acquisition accounting, property, plant and equipment are measured at estimated fair value as of the acquisition date to establish a new historical cost basis. Provisions for depreciation are based on estimated useful lives of the assets using the straight-line method. Useful lives generally range from 10 to 35 years for buildings and improvements and 3 to 10 years for machinery and equipment. Expenditures for repairs and maintenance are expensed as incurred. Assets that management intends to dispose of and that meet "held for sale" criteria are carried at the lower of the carrying value or fair value less costs to sell.

Goodwill and Other Intangible Assets

Goodwill is assigned to reporting units based on the difference between the purchase price as allocated to the reporting units and the estimated fair value of the identified net assets acquired as allocated to the reporting units. Purchased intangible assets with finite lives are carried at their estimated fair values at the time of acquisition less accumulated amortization and any impairment charges. Amortization is recognized on a straight-line basis over the estimated useful lives of the respective assets, which approximates the pattern that the economic benefits are realized by the Company.

Asset Impairments

Goodwill is tested for impairment annually or at other times if events have occurred or circumstances exist that indicate the carrying value of the reporting unit may exceed its fair value. Property, plant and equipment, intangible assets with finite lives and right of use assets are reviewed for impairment whenever events or changes in circumstances indicate that the carrying value of the assets may not be recoverable, based on the undiscounted cash flows expected to be derived from the use and ultimate disposition of the assets. Assets identified as impaired are adjusted to estimated fair value. Equity investments without readily determinable fair values are evaluated each reporting period for impairment based on a qualitative assessment and are then measured at fair value if an impairment is determined to exist. See Notes 4 and 10 for discussion of asset impairment charges.

Revenue Recognition

The Company recognizes revenue based on the satisfaction of distinct obligations to transfer goods and services to customers. The Company's revenue is generated primarily from product or equipment sales. The Company also generates revenue from custom design and installation services as well as bundled sales arrangements that include product, software and services. The Company applies a five-step approach as defined in ASC 606, *Revenue from Contracts with Customers,* in determining the amount and timing of revenue to be recognized: (1) identify the contract with a customer; (2) identify the performance obligations in the contract; (3) determine the transaction price; (4) allocate the transaction price to the performance obligations in the contract; and (5) recognize revenue when a corresponding performance obligation is satisfied. Most contracts with customers are to provide distinct products or services within a single contract. However, if a contract is separated into more than one performance obligation, the total transaction price is allocated to each performance obligation in an amount based on the estimated relative standalone selling price.

Product sales to end-customers or distributors represent over 90% of the Company's revenue and are recognized at a point-in-time, which is generally at the point in time when products have been shipped, right to payment has been obtained and risk of loss has been transferred. Certain of the Company's product performance obligations include proprietary operating system software, which typically is not considered separately identifiable. Therefore, sales of these products and the related software are considered one performance obligation.

License contracts include revenue recognized for the licensing of intellectual property, including software, sold separately without products. Functional intellectual property licenses do not meet the criteria for revenue to be recognized over time, and revenue is most commonly recognized upon delivery of the license/software to the customer.

The Company has service arrangements where net sales are recognized over time. These arrangements include a variety of post-contract support service offerings, which are generally recognized over time as the services are provided, including the following: maintenance and support services provided under annual service-level agreements; "Day 2" professional services to help customers maximize their utilization of deployed systems; and installation services related to the routine installation of equipment ordered by the customer at the customer's site.

Revenue is measured based on the consideration the Company expects to be entitled based on customer contracts. Sales are adjusted for variable consideration amounts, including, but not limited to, estimated discounts, rebates, distributor price protection programs and returns. These estimates are determined based upon historical experience, contract terms, inventory levels in the distributor channel and other related factors. Adjustments to variable consideration estimates are recorded when circumstances indicate revisions may be necessary. Variable consideration is primarily related to the Company's sales to distributors, system integrators and value-added resellers.

A contract liability for deferred revenue is recorded when consideration is received or is unconditionally due from a customer prior to transferring control of goods or services to the customer under the terms of a contract. Deferred revenue balances typically result from advance payments received from customers for product contracts or from billings in excess of revenue recognized on services arrangements.

Unbilled receivables are recorded when revenues are recognized in advance of invoice issuance. These assets are presented on a combined basis with accounts receivable and are converted to accounts receivable once the Company's right to the consideration becomes unconditional, which varies by contract but is generally based on achieving certain acceptance milestones. The Company recognizes the incremental costs of obtaining a contract as an expense when incurred if the amortization period of the asset would be one year or less.

Shipping and Handling Costs

The Company includes shipping and handling costs billed to customers in net sales and includes the costs incurred to transport product to customers as well as certain internal handling costs, which relate to activities to prepare goods for shipment, as cost of sales. Shipping and handling costs incurred after control is transferred to the customer are accounted for as fulfillment costs and are not accounted for as separate revenue obligations.

Tax Collected from Customers

Taxes assessed by a governmental authority that are both imposed on and concurrent with a specific revenue-producing transaction, which are collected by the Company from customers, are excluded from net sales.

Advertising Costs

Advertising costs are expensed in the period in which they are incurred and are reflected in selling, general and administrative expense on the Consolidated Statements of Operations. Advertising expense was $22.9 million, $32.1 million and $28.4 million for the years ended December 31, 2023, 2022 and 2021, respectively.

Research and Development

Research and development (R&D) costs are expensed in the period in which they are incurred. R&D costs include materials and equipment that have no alternative future use, depreciation on equipment and facilities currently used for R&D purposes, personnel costs, contract services and reasonable allocations of indirect costs, if clearly related to an R&D activity. Expenditures in the pre-production phase of an R&D project are recorded as R&D expense. However, costs incurred in the pre-production phase that are associated with output actually used in production are recorded in cost of sales. A project is considered finished with pre-production efforts when management determines that it has achieved acceptable levels of scrap and yield, which vary by project. Expenditures related to ongoing production are recorded in cost of sales.

Restructuring

The Company records restructuring charges associated with management-approved restructuring plans, which could include the elimination of job functions, closure or relocation of facilities, reorganization of operations, changes in management structure, workforce reductions or other actions. Restructuring charges may include ongoing and enhanced termination benefits related to employee separations, contract termination costs, impairment of certain assets and other related costs associated with exit or disposal activities. Severance benefits are provided to employees primarily under the Company's ongoing benefit arrangements. These severance costs are accrued once management commits to a plan of termination and it becomes probable that employees will be separated and entitled to benefits at amounts that can be reasonably estimated. In some instances, the Company enhances its ongoing termination benefits with one-time termination benefits, which are recognized when employees are notified of their enhanced termination benefits.

Foreign Currency Translation

For the years ended December 31, 2023, 2022 and 2021, approximately 35%, 33% and 40%, respectively, of the Company's net sales were to customers located outside the U.S. A portion of these sales was denominated in currencies other than the U.S. dollar, particularly sales from the Company's foreign subsidiaries. The financial position and results of operations of certain of the Company's foreign subsidiaries are measured using the local currency as the functional currency. Revenues and expenses of these subsidiaries have been translated into U.S. dollars at average exchange rates prevailing during the period. Assets and liabilities of these subsidiaries have been translated at the exchange rates as of the balance sheet date. Translation gains and losses are recorded in accumulated other comprehensive loss. Upon sale or liquidation of an investment in a foreign subsidiary, the amount of net translation gains or losses that have been accumulated in other comprehensive loss attributable to that investment are reported as a gain or loss in earnings in the period in which the sale or liquidation occurs.

Aggregate foreign currency gains and losses, such as those resulting from the settlement of receivables or payables, foreign currency contracts and short-term intercompany advances in a currency other than the subsidiary's functional currency, are recorded currently in earnings (included in other income (expense), net) and resulted in losses of $13.9 million, $4.6 million and $4.1 million during the years ended December 31, 2023, 2022 and 2021, respectively. Foreign currency remeasurement gains and losses related to certain long-term intercompany loans that are not expected to be settled in the foreseeable future are recorded in accumulated other comprehensive loss.

Equity-Based Compensation

The estimated fair value of stock awards is recognized as expense over the requisite service periods. Forfeitures of stock awards are recognized as they occur. The Company records deferred tax assets related to compensation expense for awards that are expected to result in future tax deductions for the Company, based on the amount of compensation cost recognized and the Company's statutory tax rate in the jurisdiction in which it expects to receive a deduction. Differences between the deferred tax assets recognized for financial reporting purposes and actual tax deductions reported on the Company's income tax return are recorded in the Consolidated Statements of Operations within income tax expense benefit.

Income Taxes

Deferred income taxes reflect the future tax consequences of differences between the financial reporting and tax basis of assets and liabilities. The Company records a valuation allowance, when appropriate, to reduce deferred tax assets to an amount that is more likely than not to be realized.

Tax benefits that result from uncertain tax positions may be recognized only if they are considered more likely than not to be sustainable, based on their technical merits. The amount of benefit to be recognized is the largest amount of tax benefit that is at least 50% likely to be realized.

In addition, the Company does not provide for U.S. taxes related to the foreign currency remeasurement gains and losses on its long-term intercompany loans with foreign subsidiaries. These loans are not expected to be repaid in the foreseeable future, and the foreign currency gains and losses are therefore recorded to accumulated other comprehensive loss.

The Company records the income tax effects related to the activity of its defined benefit plans and hedging instruments in accumulated other comprehensive loss at the currently enacted tax rate and reclassifies it to net income (loss) in the same period that the related pre-tax accumulated comprehensive income (loss) reclassifications are recognized.

Earnings (Loss) Per Share

Basic earnings (loss) per share (EPS) is computed by dividing net income (loss), less any dividends related to the Series A convertible preferred stock (Convertible Preferred Stock), by the weighted average number of common shares outstanding during the period. The numerator in diluted EPS is based on the basic EPS numerator, adjusted to add back any dividends related to the Convertible Preferred Stock, subject to antidilution requirements. The denominator used in diluted EPS is based on the basic EPS computation plus the effect of potentially dilutive common shares related to the Convertible Preferred Stock and equity-based compensation plans, subject to antidilution requirements.

For the years ended December 31, 2023, 2022 and 2021, 17.9 million, 11.3 million and 12.2 million shares, respectively, of outstanding equity-based compensation awards were not included in the computation of diluted EPS because either the effect was antidilutive or the performance conditions were not met. Of those amounts, for the years ended December 31, 2023, 2022 and 2021, 1.5 million, 2.9 million and 4.9 million shares, respectively, would have been considered dilutive if the Company had not been in a net loss attributable to common stockholders position.

For the years ended December 31, 2023, 2022 and 2021, 40.8 million, 39.1 million and 37.9 million, respectively, of as-if converted shares related to the Convertible Preferred Stock were excluded from the diluted share count because they were anti-dilutive; however, they may have been considered dilutive if the Company had not been in a net loss attributable to common stockholders position.

The following table presents the basis for the EPS computations (in millions, except per share data):

	Year ended December 31,		
	2023	2022	2021
Numerator:			
Loss from continuing operations	$ (851.3)	$ (1,184.7)	$ (350.0)
Loss from discontinued operations, net of tax	(599.6)	(102.2)	(112.6)
Net loss	$ (1,450.9)	$ (1,286.9)	$ (462.6)
Dividends on Series A convertible preferred stock	(61.8)	(59.0)	(57.3)
Net loss attributable to common stockholders	$ (1,512.7)	$ (1,345.9)	$ (519.9)
Denominator:			
Weighted average common shares outstanding – basic	210.9	207.4	203.6
Dilutive effect of as-if converted Series A convertible preferred stock	—	—	—
Dilutive effect of equity-based awards	—	—	—
Weighted average common shares outstanding – diluted	210.9	207.4	203.6
Basic:			
Loss from continuing operations per share	$ (4.33)	$ (6.00)	$ (2.00)
Loss from discontinued operations per share	(2.84)	(0.49)	(0.55)
Loss per share	$ (7.17)	$ (6.49)	$ (2.55)
Diluted			
Loss from continuing operations per share	$ (4.33)	$ (6.00)	$ (2.00)
Loss from discontinued operations per share	(2.84)	(0.49)	(0.55)
Loss per share	$ (7.17)	$ (6.49)	(2.55)

Concentrations of Risk

Non-derivative financial instruments used by the Company in the normal course of business include letters of credit and commitments to extend credit, primarily accounts receivable. The Company generally does not require collateral on its accounts receivable. These financial instruments involve risk, including the credit risk of nonperformance by the counterparties to those instruments, and the actual loss may exceed the reserves provided in the Company's Consolidated Balance Sheets. See Note 17 for further discussion of customer-related concentrations of risk.

The Company manages its exposures to credit risk associated with accounts receivable using tools such as credit approvals, credit limits and monitoring procedures. CommScope estimates the allowance for doubtful accounts based on the actual payment history and individual circumstances of significant customers as well as the age of receivables. In management's opinion, as of December 31, 2023, the Company did not have significant unreserved risk of credit loss due to the non-performance of customers or other counterparties related to amounts receivable. However, an adverse change in financial condition of a significant customer or group of customers or in the telecommunications industry could materially affect the Company's estimates related to doubtful accounts.

The principal raw materials and components purchased by CommScope (aluminum, copper, steel, bimetals, optical fiber, plastics and other polymers, capacitors, memory devices and silicon chips) are subject to changes in market price as these materials are linked to various commodity markets. The Company attempts to mitigate these risks through effective requirements planning and by working closely with its key suppliers to obtain the best possible pricing and delivery terms.

The Company relies on sole suppliers or a limited group of suppliers for certain key components (memory devices, capacitors and silicon chips), subassemblies and modules and a limited group of contract manufacturers to manufacture a significant portion of its products. Any disruption or termination of these arrangements could have a material adverse impact on the Company's results of operations.

Recent Accounting Pronouncements

Adopted in 2023

On January 1, 2023, the Company adopted Accounting Standards Update (ASU) No. 2022-04, *Liabilities—Supplier Finance Programs (Subtopic 405-50): Disclosure of Supplier Finance Program Obligations.* The new guidance improves the transparency of supplier finance programs by requiring that a buyer in a supplier finance program disclose sufficient qualitative and quantitative information about the program to allow a user of its financial statements to understand the program's nature, activity during the period, changes from period to period and potential effect on an entity's financial statements. This guidance has been applied retrospectively to all periods in which a balance sheet is presented, except for the requirement to disclose rollforward information, which is effective prospectively for the Company as of January 1, 2024. The impact of adopting this new guidance was not material to the consolidated financial statements.

In June 2023, the Company adopted ASU No. 2020-04, *Reference Rate Reform (Topic 848): Facilitation of the Effects of Reference Rate Reform on Financial Reporting and the related updates*, ASU No. 2021-01, *Reference Rate Reform (Topic 848): Scope* and ASU No. 2022-06, *Reference Rate Reform (Topic 848): Deferral of the Sunset Date of Topic 848.* Together, the ASUs provide temporary optional guidance to ease the potential burden in accounting for reference rate reform. The guidance provides optional expedients and exceptions for applying generally accepted accounting principles (GAAP) to transactions affected by reference rate reform if certain criteria are met. The Company adopted the reference rate reform guidance in connection with its amendment of the 2026 Term Loan on June 8, 2023 to transition from the Eurodollar Rate based on the London Interbank Offered Rate (LIBOR) to a Secured Overnight Financing Rate (SOFR), as the reference interest rate, effective as of July 1, 2023, in anticipation of the cessation of LIBOR in 2023 (see Note 8 for further discussion). The impact of adopting this guidance was not material to the consolidated financial statements. The Company continues to evaluate and monitor developments and its assessment of this guidance during the LIBOR transition period.

Issued but Not Adopted

In December 2023, the Financial Accounting Standards Board (FASB) issued ASU No. 2023-09, *Income Taxes (Topic 740): Improvements to Income Tax Disclosures*. The new guidance is expected to improve income tax disclosures primarily related to the rate reconciliation and income taxes paid information by requiring 1) consistent categories and greater disaggregation of information in the rate reconciliation and 2) income taxes paid disaggregated by jurisdiction. The guidance is effective for the Company on a prospective basis, although retrospective application is permitted, as of January 1, 2025 for the annual period. Early adoption is permitted. The Company is currently evaluating the impact of the new guidance on the consolidated financial statements.

In November 2023, the FASB issued ASU No. 2023-07, *Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures.* The new guidance is expected to improve reportable segment disclosure requirements, primarily through enhanced disclosures for significant segment expenses. The guidance is effective for the Company on a retrospective basis as of January 1, 2024 for the annual period and January 1, 2025 for the interim periods. Early adoption is permitted. The Company is currently evaluating the impact of the new guidance on the consolidated financial statements.

3. DISCONTINUED OPERATIONS

On October 2, 2023, the Company entered into a Call Option Agreement with Vantiva SA, a société anonyme organized under the Laws of France (Vantiva), pursuant to which the Company granted Vantiva a binding call option to acquire the Home segment and substantially all of the associated segment assets and liabilities, which was subsequently exercised and a Purchase Agreement signed on December 7, 2023, in exchange for (i) 134,704,669 shares of Vantiva common stock representing a 24.73% equity stake in Vantiva (determined on a fully diluted basis), (ii) $250,465 in cash (in addition to cash paid in exchange for the cash on the Home business companies' balance sheets) and (iii) an earn-out of up to $100 million in the aggregate. The $250,465 in cash paid in connection with the closing was expected to be used to acquire additional shares of Vantiva common stock, following which the Company was expected to own a 25% equity stake in Vantiva (on a fully diluted basis). The earn-out payments are contingent on Vantiva achieving adjusted EBITDA equal to or greater than €400 million for one or more of Vantiva's first five fiscal years following the closing of the transaction. The earn-out payment with respect to any fiscal year will be subject to an additional annual cap, the amount of which will depend on certain elections made by the Company following Vantiva reaching the €400 million adjusted EBITDA threshold for the first time, and on Vantiva's maintenance of certain liquidity levels (after giving effect to such payment).

The Home business qualified as "held for sale" per ASC 360-10 in the fourth quarter of 2023 and was classified as a discontinued operation per ASC 205-20, as the Company determined, qualitatively and quantitatively, that this transaction represents a strategic shift that has or will have a major effect on the Company's operations and financial results. As such, activity directly attributable to the Home business has been removed from continuing operations and presented in loss from discontinued operations, net of income taxes on the Consolidated Statements of Operations for all periods presented. In addition, all assets and liabilities of the Home business have been classified as assets and liabilities held for sale on the Consolidated Balance Sheets as of December 31, 2023 and 2022. In connection with the "held for sale" classification, the Company recognized a loss on classification as held for sale of $177.0 million, which has been included in loss from discontinued operations, net of income taxes on the Consolidated Statements of Operations for the year ended December 31, 2023.

Following the closing of this transaction, the Company's ownership interest in Vantiva is expected to be reported as an equity method investment. In addition, the Company has also entered into a Supply Agreement, whereby the Company will sell certain retained inventory to Vantiva for a period of two years following the close of the transaction at market prices.

The following table presents the summarized components of loss from discontinued operations, net of income taxes:

	Year Ended December 31,		
	2023	2022	2021
Net sales	$ 1,210.2	$ 1,703.4	$ 1,849.4
Cost of sales	1,060.2	1,493.3	1,604.5
Gross profit	150.0	210.1	244.9
Operating expenses:			
Selling, general and administrative	100.7	94.2	150.9
Research and development	98.3	113.8	118.2
Amortization of purchased intangible assets	83.4	103.0	103.0
Restructuring costs (credits), net	6.0	(0.1)	6.8
Asset impairments	469.2	—	13.7
Total operating expenses	757.6	310.9	392.6
Operating loss	(607.6)	(100.8)	(147.7)
Other income, net	1.0	0.5	2.5
Loss from discontinued operations before income taxes	(606.6)	(100.3)	(145.2)
Loss on classification as held for sale	(177.0)	—	—
Income tax (expense) benefit	184.0	(1.9)	32.6
Loss from discontinued operations, net of income taxes	$ (599.6)	$ (102.2)	$ (112.6)

The following table presents balance sheet information for assets and liabilities held for sale:

	December 31,	
	2023	2022
Cash and cash equivalents	$ —	$ 25.1
Accounts receivable, net	253.6	345.5
Inventories, net	118.4	212.1
Prepaid expenses and other	23.9	39.1
Other assets	59.7	—
Total current assets	455.6	621.8
Loss on classification of held for sale	(177.0)	—
Total current assets held for sale	278.6	621.8
Property, plant, and equipment, net	—	8.6
Other intangible assets, net	—	558.5
Other noncurrent assets	—	33.7
Total noncurrent assets	—	600.8
Total assets held for sale	$ 278.6	$ 1,222.6
Accounts payable	$ 192.5	$ 341.2
Accrued and other liabilities	101.7	181.4
Other liabilities	13.0	—
Total current liabilities held for sale	307.2	522.6
Deferred income taxes	—	58.0
Other noncurrent liabilities	—	24.6
Total noncurrent liabilities	—	82.6
Total liabilities held for sale	$ 307.2	$ 605.2

The cash flows related to discontinued operations have not been segregated in the Consolidated Statements of Cash Flows, and accordingly, they include the results from continuing and discontinued operations. The following table summarizes significant non-cash operating items and capital expenditures of the discontinued operation included in the Consolidated Statements of Cash Flows:

	Year Ended December 31,		
	2023	2022	2021
Significant non-cash operating items:			
Depreciation and amortization	$ 94.3	$ 116.5	$ 121.9
Equity-based compensation	3.7	5.8	8.8
Asset impairments [(1)]	646.2	—	13.7
Capital expenditures	2.4	4.0	5.6

(1)Includes loss of $177.0 million on classification as held for sale and $469.2 million on impairment of long-lived assets.

The loss on classification as held for sale of $177.0 million recorded in the fourth quarter of 2023 reduced the carrying value of the Home business disposal group, comprised of all the assets and liabilities of the Home business, to its fair value less cost to sell. The Company estimated the fair value of the disposal group using Company-specific inputs, including estimates of consideration to be exchanged upon the close of the divestiture. The determination of the loss on classification as held for sale was based on Level 3 valuation inputs.

The Company also recorded a partial impairment of $469.2 million related to the Home segment's intangible assets during the third quarter of 2023. The Company estimated the fair value of the asset group using Company-specific inputs, including estimates of consideration to be received upon the impending divestiture, as well as other market participant assumptions. The determination of the impairment charge was based on Level 3 valuation inputs.

4. GOODWILL AND OTHER INTANGIBLE ASSETS

The following table presents details of the Company's intangible assets other than goodwill:

	December 31, 2023			December 31, 2022		
	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount
Customer base	$ 2,951.5	$ 1,945.7	$ 1,005.8	$ 2,946.5	$ 1,814.9	$ 1,131.6
Trade names and trademarks	937.5	531.8	405.7	936.6	474.2	462.4
Patents and technologies	1,694.9	1,523.7	171.2	1,693.6	1,372.5	321.1
Other	32.3	32.3	—	32.3	32.3	—
Total intangible assets	$ 5,616.2	$ 4,033.5	$ 1,582.7	$ 5,609.0	$ 3,693.9	$ 1,915.1

There were no impairments of finite-lived intangible assets identified during the years ended December 31, 2023, 2022 or 2021.

Amortization expense for intangible assets was $327.1 million, $440.0 million and $510.0 million for the years ended December 31, 2023, 2022 and 2021, respectively. Future amortization expense as of December 31, 2023 is as follows:

	Estimated Amortization Expense
2024	$ 273.5
2025	230.2
2026	187.9
2027	157.9
2028	119.4
Thereafter	613.8

The table above includes amortization expense of $19.5 million in 2024 and $4.4 million in 2025 related to retained patents in the Home business, which will be classified as discontinued operations.

The following table presents the activity in goodwill by reportable segment.

	December 31, 2022			Activity			December 31, 2023		
	Goodwill	Accumulated Impairment Losses	Total	Additions (Deductions)	Impairment	Foreign Exchange and Other	Goodwill	Accumulated Impairment Losses	Total [1]
CCS	$ 2,280.9	$ (51.5)	$ 2,229.4	$ —	$ (99.1)	$ 10.1	$ 2,291.0	$ (150.6)	$ 2,140.4
OWN	660.3	(159.5)	500.8	—	—	2.4	662.7	(159.5)	503.2
NICS	649.4	(41.2)	608.2	—	—	1.2	650.6	(41.2)	609.4
ANS	1,995.7	(1,261.7)	734.0	—	(472.3)	(0.3)	1,995.4	(1,734.0)	261.4
Total	$ 5,586.3	$ (1,513.9)	$ 4,072.4	$ —	$ (571.4)	$ 13.4	$ 5,599.7	$ (2,085.3)	$ 3,514.4

(1) As of December 31, 2023, the CCS reportable segment goodwill balance of $2,140.4 million includes goodwill balances of $881.5 million and $1,258.9 million for the Building and Data Center Connectivity (BDCC) and Network Cable and Connectivity reporting units, respectively.

	December, 31, 2021			Activity			December 31, 2022		
	Goodwill	Accumulated Impairment Losses	Total	Additions (Deductions)	Impairment	Foreign Exchange and Other	Goodwill	Accumulated Impairment Losses	Total
CCS	$ 2,307.3	$ (51.5)	$ 2,255.8	$ —	$ —	$ (26.4)	$ 2,280.9	$ (51.5)	$ 2,229.4
OWN	666.6	(159.5)	507.1	—	—	(6.3)	660.3	(159.5)	500.8
NICS	653.0	(41.2)	611.8	—	—	(3.6)	649.4	(41.2)	608.2
ANS	1,999.1	(142.1)	1,857.0	—	(1,119.6)	(3.4)	1,995.7	(1,261.7)	734.0
Total	$ 5,626.0	$ (394.3)	$ 5,231.7	$ —	$ (1,119.6)	$ (39.7)	$ 5,586.3	$ (1,513.9)	$ 4,072.4

During the annual impairment test performed in the fourth quarter of 2023 and in conjunction with the development of the Company's 2024 and long-range plans, CommScope continued to evaluate its ANS and BDCC reporting units' expected cash flows and market valuations. The Company determined the goodwill balances in the ANS and BDCC reporting units were impaired and recorded partial impairment charges of $46.4 million and $99.1 million, respectively, in asset impairments in the Consolidated Statements of Operations. The impairment charges resulted from the Company's assessment of further lower revenue growth and EBITDA margins in the fourth quarter of 2023, due to adverse impacts of market conditions on the current year profitability and estimated future business results and cash flows. The ANS reporting unit is the same as the ANS segment, and the BDCC reporting unit is part of the CCS reportable segment. See Note 10 for further discussion of the assumptions used in the valuation.

During the third quarter of 2023, the Company concluded that a triggering event occurred, primarily due to a sustained decrease in the market value of the Company's debt and common stock affecting the overall business and changes in expected future cash flows due to reduced earnings forecasts and current macroeconomic conditions, including a rising interest rate environment. The Company performed an interim quantitative goodwill impairment test for its ANS and BDCC reporting units, which were most sensitive to negative performance and outlook, to compare the fair values of the reporting units to their carrying amounts, including the goodwill. As a result, the Company recorded a goodwill impairment charge of $425.9 million in asset impairments in the Consolidated Statements of Operations to partially write down the carrying amount of the ANS reporting unit goodwill in the ANS segment. There was no impairment identified in the BDCC reporting unit.

During the annual impairment test performed in the fourth quarter of 2022 and in conjunction with the development of its 2023 and long-range plans, the Company identified changes in the ANS reporting unit's expected future cash flows due to various market trends expected to affect the business, including technology shifts affecting hardware sales, trends affecting bandwidth growth and other operational challenges, as well as an increase in the cost of capital. As a result, the Company determined the goodwill balance in the ANS reporting unit was impaired and recorded a $1,119.6 million impairment charge.

Estimating the fair value of a reporting unit involves uncertainties because it requires management to develop numerous assumptions, including assumptions about the future growth and potential volatility in revenues and costs, capital expenditures, industry economic factors and future business strategy. Changes in projected revenue growth rates, projected EBITDA margin percentages or estimated discount rates due to uncertain market conditions, terminal growth rates, lower market multiples, loss of one or more key customers, changes in the Company's strategy, changes in technology or other factors could negatively affect the fair value in one or more of the Company's reporting units and result in a material impairment charge in the future.

5. REVENUE FROM CONTRACTS WITH CUSTOMERS

Disaggregated Net Sales

See Note 17 for the presentation of net sales by segment and geographic region.

Allowance for Doubtful Accounts

	Year Ended December 31,		
	2023	2022	2021
Allowance for doubtful accounts, beginning of period	$ 50.1	$ 29.8	$ 36.9
Provision (benefit)	3.1	22.0	(4.4)
Write-offs	(21.9)	(0.5)	(1.1)
Foreign exchange and other	0.9	(1.2)	(1.6)
Allowance for doubtful accounts, end of period	$ 32.2	$ 50.1	$ 29.8

During the year ended December 31, 2022, the Company recorded an allowance for $20.9 million to reserve the balance due from a distributor in the OWN segment based on deterioration in the customer's risk profile which is included in the provision line in the table above and in selling, general and administrative expense on the Consolidated Statements of Operations. During the year ended December 31, 2023, the Company subsequently wrote off $18.5 million related to the balance.

Customer Contract Balances

The following table provides the balance sheet location and amounts of contract assets, or unbilled accounts receivable, and contract liabilities, or deferred revenue, from contracts with customers as of December 31, 2023 and December 31, 2022.

Contract Balance Type	Balance Sheet Location	December 31, 2023	2022
Unbilled accounts receivable	Accounts receivable, net of allowance for doubtful accounts	$ 16.8	$ 26.8
Deferred revenue - current	Accrued and other liabilities	$ 85.4	$ 88.9
Deferred revenue - noncurrent	Other noncurrent liabilities	70.6	57.1
Total contract liabilities		$ 156.0	$ 146.0

There were no material changes to contract asset balances for the year ended December 31, 2023 as a result of changes in estimates or impairments. The change in the contract liability balance from December 31, 2022 to December 31, 2023 was primarily due to upfront support billings to be recognized over the support term. During the year ended December 31, 2023, the Company recognized $82.8 million of revenue related to contract liabilities recorded as of December 31, 2022.

6. LEASES

The Company has operating type leases for real estate, equipment and vehicles both in the U.S. and internationally. As of December 31, 2023 and 2022, the Company had no finance type leases. Operating lease expense was $87.0 million, $91.5 million and $95.4 million for the years ended December 31, 2023, 2022 and 2021, respectively. Operating lease expense included period cost for short-term, cancellable and variable leases that were not included in lease liabilities of $32.5 million, $36.2 million and $33.2 million for the years ended December 31, 2023, 2022 and 2021, respectively.

The Company occasionally subleases all or a portion of certain unutilized real estate facilities. As of December 31, 2023, the Company's sublease arrangements were classified as operating type leases and the income amounts were not material for the years ended December 31, 2023, 2022 and 2021, respectively.

Supplemental cash flow information related to operating leases, which includes both continuing operations and discontinued operations:

	Year Ended December 31, 2023	2022	2021
Operating cash paid to settle lease liabilities	$ 57.6	$ 59.4	$ 71.5
Right of use asset additions in exchange for lease liabilities	63.2	43.5	25.3

Supplemental balance sheet information related to operating leases:

	Balance Sheet Location	December 31, 2023	2022
Right of use assets	Other noncurrent assets	$ 157.2	$ 141.9
Lease liabilities - current	Accrued and other liabilities	$ 36.5	$ 44.4
Lease liabilities - noncurrent	Other noncurrent liabilities	139.9	119.4
Total lease liabilities		$ 176.4	$ 163.8

Weighted average remaining lease term (in years)	6.1
Weighted average discount rate	8.2%

Future minimum lease payments under non-cancellable leases as of December 31, 2023 are as follows:

	Operating Leases
2024	49.8
2025	$ 44.0
2026	33.4
2027	22.8
2028	17.5
Thereafter	66.5
Total minimum lease payments	234.0
Less: imputed interest	(57.6)
Total	$ 176.4

7. SUPPLEMENTAL FINANCIAL STATEMENT INFORMATION

Accounts Receivable

	December 31, 2023	2022
Accounts receivable - trade	$ 826.9	$ 1,197.0
Accounts receivable - other	20.5	31.3
Allowance for doubtful accounts	(32.2)	(50.1)
Total accounts receivable, net	$ 815.2	$ 1,178.2

Inventories

	December 31, 2023	2022
Raw materials	$ 551.4	$ 530.1
Work in process	165.6	212.7
Finished goods	362.7	633.2
Total inventories, net	$ 1,079.7	$ 1,376.0

Property, Plant and Equipment

	December 31, 2023	2022
Land and land improvements	$ 38.6	$ 52.2
Buildings and improvements	313.6	339.5
Machinery and equipment	987.9	1,006.7
Construction in progress	26.6	51.9
	1,366.7	1,450.3
Accumulated depreciation	(866.1)	(849.3)
Total property, plant and equipment, net	$ 500.6	$ 601.0

Depreciation expense was $111.8 million, $113.8 million and $117.8 million during the years ended December 31, 2023, 2022 and 2021, respectively. No interest was capitalized during the years ended December 31, 2023, 2022 or 2021.

Accrued and Other Liabilities

	December 31, 2023	2022
Compensation and employee benefit liabilities	$ 178.6	$ 285.9
Accrued interest	113.2	118.1
Deferred revenue	85.4	88.9
Operating lease liabilities	36.5	44.4
Product warranty accrual	26.0	31.5
Restructuring liabilities	11.7	58.9
Other	182.9	240.9
Total accrued and other liabilities	$ 634.3	$ 868.6

Accumulated Other Comprehensive Loss

The following table presents changes in accumulated other comprehensive loss (AOCL), net of tax:

	Year Ended December 31, 2023	2022
Foreign currency translation		
Balance at beginning of period	$ (270.3)	$ (165.8)
Other comprehensive income (loss)	37.9	(104.3)
Amounts reclassified from AOCL	—	(0.2)
Balance at end of period	$ (232.4)	$ (270.3)
Defined benefit plan activity		
Balance at beginning of period	$ (14.8)	$ (13.4)
Other comprehensive loss	(0.1)	(1.4)
Amounts reclassified from AOCL	—	—
Balance at end of period	$ (14.9)	$ (14.8)
Hedging instruments		
Balance at beginning of period	$ (11.2)	$ (27.2)
Other comprehensive income (loss)	(8.2)	16.0
Balance at end of period	$ (19.4)	$ (11.2)
Net AOCL at end of period	$ (266.7)	$ (296.3)

Amounts reclassified from net AOCL related to foreign currency translation and defined benefit plans are recorded in other income (expense), net in the Consolidated Statements of Operations.

Cash Flow Information

	Year Ended December 31, 2023	2022	2021
Cash paid during the period for:			
Income taxes, net of refunds	$ 101.1	$ 130.7	$ 79.4
Interest	654.0	563.2	525.9

8. FINANCING

	December 31, 2023	December 31, 2022
7.125% senior notes due July 2028	$ 641.6	$ 700.0
5.00% senior notes due March 2027	750.0	750.0
8.25% senior notes due March 2027	866.9	1,000.0
6.00% senior notes due June 2025	1,274.6	1,300.0
4.75% senior secured notes due September 2029	1,250.0	1,250.0
6.00% senior secured notes due March 2026	1,500.0	1,500.0
Senior secured term loan due April 2026	3,064.0	3,096.0
Senior secured revolving credit facility	—	—
Total principal amount of debt	9,347.1	9,596.0
Less: Original issue discount, net of amortization	(11.5)	(15.9)
Less: Debt issuance costs, net of amortization	(57.0)	(78.5)
Less: Current portion	(32.0)	(32.0)
Total long-term debt	$ 9,246.6	$ 9,469.6

Senior Notes

As of December 31, 2023, the Company had outstanding two series of senior secured notes: (1) $1,250.0 million of 4.75% senior secured notes due September 1, 2029 (the 2029 Secured Notes) issued by CommScope, LLC (f/k/a CommScope, Inc.) in August 2021; and (2) $1.5 billion of 6.00% senior secured notes due March 1, 2026 issued by CommScope, LLC in February 2019 (the 2026 Secured Notes and, together with the 2029 Secured Notes, the Secured Notes). As of December 31, 2023, the Company had outstanding four series of senior unsecured notes: (1) $700.0 million initial aggregate principal amount of 7.125% senior notes due July 1, 2028 (the 2028 Notes) issued by CommScope, LLC in July 2020; (2) $ 750.0 million initial aggregate principal amount of 5.00% senior notes due March 15, 2027 issued by CommScope Technologies LLC (CommScope Technologies), a wholly owned subsidiary of the Company, in March 2017 (the 5.00% 2027 Notes); (3) $1.3 billion initial aggregate principal amount of 6.00% senior notes due June 15, 2025 issued by CommScope Technologies in June 2015 (the 2025 Notes, and together with the 5.00% 2027 Notes, the CommScope Technologies Notes); and (4) $1.0 billion initial aggregate principal amount of 8.25% senior notes due March 1, 2027 issued by CommScope, LLC in February 2019 (the 8.25% 2027 Notes and, together with the 2028 Notes, the CommScope, LLC Notes; the Secured Notes, the CommScope Technologies Notes and the CommScope, LLC Notes, collectively, the Senior Notes).

The indentures governing the Senior Notes contain covenants that restrict the ability of CommScope, LLC and its restricted subsidiaries to, among other things, incur additional debt, make certain payments, including payment of dividends (except, in the case of the CommScope, LLC Notes and the Secured Notes, with respect to the Convertible Preferred Stock) or repurchases of equity interests of CommScope, LLC or the applicable issuer, make loans or acquisitions or capital contributions and certain investments, incur certain liens, sell assets, merge or consolidate or liquidate other entities and enter into certain transactions with affiliates.

There are no financial maintenance covenants in the indentures governing the Senior Notes. Events of default under the indentures governing the Senior Notes include, among others, non-payment of principal or interest when due, covenant defaults, bankruptcy and insolvency events and cross acceleration to material debt.

4.75% Senior Secured Notes due 2029 and 6.00% Senior Secured Notes due 2026 (the Secured Notes)

The 2029 Secured Notes mature on September 1, 2029 and the 2026 Secured Notes mature on March 1, 2026. Interest is payable on the Secured Notes semi-annually in arrears on March 1 and September 1 of each year. The Secured Notes are guaranteed on a senior secured basis by the Company and each of CommScope, LLC's existing and future wholly owned domestic restricted subsidiaries that is an obligor under the senior secured credit facilities or certain other debt, subject to certain exceptions. The Secured Notes and the related guarantees are secured on a first-priority basis by security interests in all of the assets that secure indebtedness under the 2026 Term Loan on a first-priority basis, and on a second-priority basis in all assets that secure the Revolving Credit Facility (as defined below) on a first-priority basis and the 2026 Term Loan on a second-priority basis. The Secured Notes and the related guarantees rank senior in right of payment to all of CommScope, LLC's and the guarantors' subordinated indebtedness and equally in right of payment with all of CommScope, LLC's and the guarantors' senior indebtedness (without giving effect to collateral arrangements), including the senior secured credit facilities and the other Senior Notes. The Secured Notes and the related guarantees are effectively senior to all of CommScope, LLC's and the guarantors' unsecured indebtedness and debt secured by a lien junior to the liens securing the Secured Notes, in each case to the extent of the value of the collateral, and effectively equal to all of CommScope, LLC's and the guarantors' senior indebtedness secured on the same priority basis as the Secured Notes, including the 2026 Term Loan. The Secured Notes and the related guarantees are effectively subordinated to any of CommScope, LLC's or the guarantors' indebtedness that is secured by assets that do not constitute collateral for the Secured Notes and effectively subordinated to any of CommScope, LLC's or the guarantors' indebtedness that is secured by a senior-priority lien, including under the Revolving Credit Facility, in each case to the extent of the value of the assets securing such indebtedness. In addition, the Secured Notes and related guarantees are structurally subordinated to all existing and future liabilities (including trade payables) of CommScope, LLC's subsidiaries that do not guarantee the Secured Notes.

The Secured Notes may be redeemed prior to maturity under certain circumstances. Upon certain change of control events, the Secured Notes may be redeemed at the option of the holders at 101% of their face amount, plus accrued and unpaid interest. The 2029 Secured Notes may be redeemed on or after September 1, 2024 by CommScope, LLC at the redemption prices specified in the indenture governing the 2029 Secured Notes. Prior to September 1, 2024, the 2029 Secured Notes may be redeemed by CommScope, LLC at a redemption price equal to 100% of their principal amount, plus a make-whole premium (as specified in the indenture governing the 2029 Secured Notes), plus accrued and unpaid interest. Prior to September 1, 2024, under certain circumstances, CommScope, LLC may also redeem up to 40% of the aggregate principal amount of the 2029 Secured Notes at a redemption price of 104.750%, plus accrued and unpaid interest, using the proceeds of certain equity offerings. At any time prior to September 1, 2024, CommScope, LLC may redeem during each calendar year up to 10.0% of the aggregate principal amount of the 2029 Secured Notes at a redemption price equal to 103.0% of the aggregate principal amount of the 2029 Secured Notes to be redeemed, plus accrued and unpaid interest, if any, to, but not including, the date of redemption. The 2026 Secured Notes may be redeemed by CommScope, LLC at the redemption prices specified in the indenture governing the 2026 Secured Notes.

7.125% Senior Notes due 2028 and 8.25% Senior Notes due 2027 (the CommScope, LLC Notes)

The 2028 Notes mature on July 1, 2028 and the 8.25% 2027 Notes mature on March 1, 2027. Interest is payable semi-annually in arrears on the 2028 Notes on July 1 and January 1 of each year and on the 8.25% 2027 Notes on March 1 and September 1 of each year. The CommScope, LLC Notes are guaranteed on a senior unsecured basis by each of CommScope, LLC's existing and future wholly owned domestic restricted subsidiaries that is an obligor under the senior secured credit facilities or certain other capital markets debt, subject to certain exceptions. The CommScope, LLC Notes and the related guarantees rank senior in right of payment to all of CommScope, LLC's and the guarantors' subordinated indebtedness and equally in right of payment with all of CommScope, LLC's and the guarantors' senior indebtedness (without giving effect to collateral arrangements), including the senior secured credit facilities and the other Senior Notes. The CommScope, LLC Notes and the related guarantees are effectively junior to all of CommScope, LLC's and the guarantors' existing and future secured indebtedness, including the Secured Notes and the senior secured credit facilities, to the extent of the value of the assets securing such secured indebtedness. In addition, the CommScope, LLC Notes and related guarantees are structurally subordinated to all existing and future liabilities (including trade payables) of CommScope, LLC's subsidiaries that do not guarantee the CommScope, LLC Notes.

The CommScope, LLC Notes may be redeemed prior to maturity under certain circumstances. Upon certain change of control events, the CommScope, LLC Notes may be redeemed at the option of the holders at 101% of their principal amount, plus accrued and unpaid interest. The 2028 Notes may be redeemed by CommScope, LLC at the redemption prices specified in the indenture governing the 2028 Notes. The 8.25% 2027 Notes may be redeemed by CommScope, LLC at the redemption prices specified in the indenture governing the 8.25% 2027 Notes.

During the year ended December 31, 2023, the Company repurchased $58.4 million and $133.1 million aggregate principal amounts of the 2028 Notes and 2027 Notes, respectively, which resulted in gains on the early extinguishment of debt of $73.1 million reflected in other income (expense), net, and the write-off of debt issuance costs of $1.8 million reflected in interest expense, in the Consolidated Statements of Operations. See the "Other Matters" section below for the summarized debt repurchase activity for the year ended December 31, 2023.

5.00% Senior Notes due 2027 and 6.00% Senior Notes due 2025 (the CommScope Technologies Notes)

The 5.00% 2027 Notes mature on March 15, 2027 and the 2025 Notes mature on June 15, 2025. Interest is payable on the 5.00% 2027 Notes semi-annually in arrears on March 15 and September 15 of each year and on the 2025 Notes on June 15 and December 15 of each year.

The CommScope Technologies Notes are guaranteed on a senior unsecured basis by CommScope, LLC and each of CommScope, LLC's existing and future wholly owned domestic restricted subsidiaries (other than CommScope Technologies) that is an obligor under the senior secured credit facilities or certain other capital markets debt, subject to certain exceptions. The CommScope Technologies Notes and the related guarantees rank senior in right of payment to all of CommScope Technologies' and the guarantors' subordinated indebtedness and equally in right of payment with all of CommScope Technologies' and the guarantors' senior indebtedness (without giving effect to collateral arrangements), including the senior secured credit facilities and the other Senior Notes. The CommScope Technologies Notes and the related guarantees are effectively junior to all of CommScope Technologies' and the guarantors' existing and future secured indebtedness, including the Secured Notes and the senior secured credit facilities, to the extent of the value of the assets securing such secured indebtedness. In addition, the CommScope Technologies Notes and related guarantees are structurally subordinated to all existing and future liabilities (including trade payables) of CommScope, LLC's subsidiaries that do not guarantee the CommScope Technologies Notes.

The CommScope Technologies Notes may be redeemed prior to maturity under certain circumstances. Upon certain change of control events, the CommScope Technologies Notes may be redeemed at the option of the holders at 101% of their principal amount, plus accrued and unpaid interest. The 5.00% 2027 Notes may be redeemed by CommScope Technologies at the redemption prices specified in the indenture governing the 5.00% 2027 Notes. The 2025 Notes may be redeemed by CommScope Technologies at the redemption prices specified in the indenture governing the 2025 Notes.

During the year ended December 31, 2023, the Company repurchased $25.4 million aggregate principal amount of the 2025 Notes, which resulted in gains on the early extinguishment of debt of $1.2 million reflected in other income (expense), net, and the write-off of debt issuance costs of $0.2 million reflected in interest expense, in the Consolidated Statements of Operations. See the "Other Matters" section below for the summarized debt repurchase activity for the year ended December 31, 2023.

Senior Secured Credit Facilities

Senior Secured Term Loan due 2026

The 2026 Term Loan has scheduled amortization payments of $32.0 million per year due in equal quarterly installments, which began with the quarter ended December 31, 2019, with the balance due at maturity (April 2026). On June 8, 2023, the Company amended its 2026 Term Loan to transition from the Eurodollar Rate based on LIBOR to SOFR, effective as of July 1, 2023, as the reference interest rate in anticipation of the cessation of LIBOR in 2023. As a result, the 2026 Term Loan bears interest at (1) an adjusted Term SOFR rate, subject to certain adjustments, plus an applicable margin of 3.25% or (2) at the option of the Company, a base rate plus an applicable margin of 2.25%.

Subject to certain conditions, the 2026 Term Loan may be increased or a new incremental term loan facility may be added to increase the capacity by up to the sum of the greater of $950.0 million and 50% of Consolidated EBITDA, as defined in the credit agreement governing the 2026 Term Loan (the Credit Agreement), plus an unlimited amount as long as on a pro forma basis the Company meets certain net leverage ratios or fixed charge ratios as defined in the Credit Agreement.

CommScope, LLC may voluntarily prepay loans under the 2026 Term Loan, subject to minimum amounts, with prior notice but without premium or penalty. CommScope, LLC must prepay the 2026 Term Loan with the net cash proceeds of certain asset sales, the incurrence or issuance of specified refinancing indebtedness and 50% of excess cash flow (such percentage subject to reduction based on the achievement of specified Consolidated First Lien Net Leverage Ratios), in each case, subject to certain reinvestment rights and other exceptions.

CommScope, LLC's obligations under the 2026 Term Loan are guaranteed by the Company and each of CommScope, LLC's direct and indirect wholly owned U.S. subsidiaries (subject to certain permitted exceptions based on immateriality thresholds of aggregate assets and revenues of excluded U.S. subsidiaries). The 2026 Term Loan is secured by a lien on substantially all of CommScope, LLC's and the guarantors' current and fixed assets (subject to certain exceptions), and the 2026 Term Loan has a first-priority lien on all fixed assets and a second-priority lien on all current assets (second in priority to the liens securing the Revolving Credit Facility), in each case, subject to other permitted liens.

The 2026 Term Loan contains customary negative covenants consistent with those applicable to the 2026 Secured Notes, including, but not limited to, restrictions on the ability of CommScope, LLC and its subsidiaries to merge and consolidate with other companies, incur indebtedness, grant liens or security interests on assets, pay dividends (except with respect to the Convertible Preferred Stock) or make other restricted payments, sell or otherwise transfer assets or enter into certain transactions with affiliates.

The 2026 Term Loan provides that, upon the occurrence of certain events of default, the obligations thereunder may be accelerated. Such events of default will include payment defaults, material inaccuracies of representations and warranties, covenant defaults, cross-defaults to other material indebtedness, voluntary and involuntary bankruptcy proceedings, material money judgments, material pension-plan events, change of control and other customary events of default.

During the year ended December 31, 2023, the Company made scheduled amortization payments totaling $32.0 million due in equal quarterly installments on the 2026 Term Loan. The current portion of long-term debt reflects $32.0 million of repayments due under the 2026 Term Loan.

No portion of the 2026 Term Loan was reflected as a current portion of long-term debt as of December 31, 2023 related to the potentially required excess cash flow payment because no such payment is expected to be required. There was no excess cash flow payment required in 2023 related to 2022.

Senior Secured Revolving Credit Facility

On October 19, 2022, the Company completed the refinancing (the Refinancing) of the Company's asset-based revolving credit facility (Revolving Credit Facility) which continues to provide borrowing capacity of up to $1.0 billion, subject to certain limitations, available to CommScope, LLC and its U.S. subsidiaries designated as co-borrowers (the U.S. Revolving Borrowers). The Refinancing, among other things, (i) refinanced in full all existing loans outstanding under the Revolving Credit Facility immediately prior to the Refinancing, (ii) extended the maturity of the Revolving Credit Facility from April 2024 to the earliest of (x) September 30, 2027, (y) the date the commitments of the lenders under the agreement governing the Revolving Credit Facility are reduced to zero and (z) 91 days prior to the maturity date of any other indebtedness of a "Credit Party" (as defined in the credit agreement governing the Revolving Credit Facility) that has a scheduled maturity or weighted average life to maturity that is prior to September 30, 2027 (subject to certain exceptions) and (iii) replaced the existing revolving loan commitments outstanding under the Revolving Credit Facility immediately prior to the Refinancing with a new tranche of commitments (the Tranche A Revolving Commitments) for borrowings denominated in U.S. dollars, euros and pounds sterling made to the U.S. Revolving Borrowers and a separate tranche of commitments (the Tranche B Revolving Commitments) for borrowings denominated in euros, pounds sterling and Swiss francs made to the U.S. Revolving Borrowers and certain of the Company's wholly owned Irish, English and Swiss subsidiaries that are joined as borrowers under the Revolving Credit Facility (such subsidiaries, the European Revolving Borrowers and, together with the U.S. Revolving Borrowers, the Revolving Borrowers). Prior to the joinder of any European Revolving Borrower, Tranche A Loans are available to the U.S. Revolving Borrowers in an aggregate amount equal to (i) the lesser of (x) $1.0 billion and (y) the borrowing base of the U.S. Revolving Borrowers minus (ii) the aggregate amount of all "Tranche A Revolving Credit Outstandings" (as defined in the credit agreement governing the Revolving Credit Facility). From and after the joinder of any European Revolving Borrower, the Revolving Borrowers may reallocate an amount of the Tranche A Revolving Commitments to Tranche B Revolving Commitments, and Tranche B Loans will then be available to the Revolving Borrowers in an amount equal to (i) the lesser of (x) the Tranche B Revolving Commitments and (y) the sum of the borrowing base of the European Revolving Borrowers minus (ii) the aggregate amount of all "Tranche B Revolving Credit Outstandings" (as defined in the credit agreement governing the Revolving Credit Facility). At no time will the aggregate commitments of the lenders under the Revolving Credit Facility exceed $1.0 billion. Borrowing base calculations are based on the sum of specific percentages of eligible accounts receivable and eligible inventory, minus the amount of any applicable reserves. The ability to draw under the Revolving Credit Facility or issue letters of credit is conditioned upon, among other things, delivery of prior written notice of a borrowing or issuance, as applicable, the ability of the Revolving Borrowers to reaffirm the representations and warranties contained in the credit agreement governing the Revolving Credit Facility and the absence of any default or event of default. As of December 31, 2023, the Company had no outstanding borrowings under the Revolving Credit Facility and had availability of $688.0 million, after giving effect to borrowing base limitations and outstanding letters of credit.

Letters of credit under the Revolving Credit Facility are limited to the lesser of (x) $250.0 million and (y) the aggregate unused amount of commitments under the Revolving Credit Facility then in effect. Subject to certain conditions, the Revolving Credit Facility may be expanded by up to $400.0 million in additional commitments. Loans under the Revolving Credit Facility may be denominated, at the option of the Revolving Borrowers, (i) with respect to Tranche A Loans, in U.S. dollars, euros or pounds sterling, and (ii) with respect to Tranche B Loans, U.S. dollars, euros, pounds sterling or Swiss francs.

Borrowings under the Revolving Credit Facility will bear interest at a floating rate, which can be either (1) an adjusted Term SOFR rate (for borrowings denominated in U.S. dollars), (2) the Euro Interbank Offered Rate (EURIBOR) (for borrowings denominated in euros), (3) the Sterling Overnight Index Average (SONIA) (for borrowings denominated in pounds sterling) or (4) the Swiss Average Rate Overnight (SARON) (for borrowings denominated in Swiss francs), in each case, subject to certain adjustments plus an applicable margin of 1.25% to 1.50% or, at the option of the Revolving Borrowers, a base rate plus an applicable margin of 0.25% to 0.50%.

The obligations of the U.S. Revolving Borrowers under the Revolving Credit Facility are guaranteed by the Company, CommScope, LLC and each of CommScope, LLC's direct and indirect wholly owned U.S. subsidiaries (subject to certain permitted exceptions based on immateriality thresholds of aggregate assets and revenues of excluded U.S. subsidiaries). The Revolving Credit Facility is secured by a lien on substantially all of the U.S. Revolving Borrowers' and the guarantors' current and fixed assets (subject to certain exceptions). The Revolving Credit Facility has a first-priority lien on all current assets and a second-priority lien on all fixed assets (second in priority to the liens securing the 2029 Secured Notes, the 2026 Secured Notes and the 2026 Term Loan), in each case, subject to other permitted liens.

The following fees are applicable under the Revolving Credit Facility: (i) an unused line fee of (x) 0.250% per annum of the unused portion of the Revolving Credit Facility when the average unused portion of the facility is less than 50% of the aggregate commitments under the Revolving Credit Facility or (y) 0.375% per annum of the unused portion of the Revolving Credit Facility when the average unused portion of the facility is equal to or greater than 50% of the aggregate commitments under the Revolving Credit Facility; (ii) a letter of credit participation fee on the aggregate stated amount of each letter of credit equal to the applicable margin for adjusted Term SOFR, EURIBOR, SONIA and SARON loans, as applicable; (iii) a letter of credit fronting fee of 0.125% per annum, multiplied by the average aggregate daily maximum amount available to be drawn under all applicable letters of credit issued by such letter of credit issuer; and (iv) certain other customary fees and expenses of the lenders and agents thereunder.

The Revolving Borrowers will be required to make prepayments under the Revolving Credit Facility at any time when, and to the extent that, the aggregate amount of the outstanding loans and letters of credit under the Revolving Credit Facility exceeds the lesser of the aggregate amount of commitments in respect of the Revolving Credit Facility and the borrowing base.

The Revolving Credit Facility contains customary covenants, including, but not limited to, restrictions on the ability of CommScope, LLC and its subsidiaries to merge and consolidate with other companies, incur indebtedness, grant liens or security interests on assets, make acquisitions, loans, advances or investments, pay dividends (except with respect to the Convertible Preferred Stock), sell or otherwise transfer assets, optionally prepay or modify terms of any junior indebtedness, enter into certain transactions with affiliates or change lines of business. The Revolving Credit Facility contains a Covenant Fixed Charge Coverage Ratio (as defined in the credit agreement governing the Revolving Credit Facility) of 1.00 to 1.00. The credit agreement governing the Revolving Credit Facility provides that the Covenant Fixed Charge Coverage Ratio must be tested and must exceed the level set forth above only in the event that excess availability under the Revolving Credit Facility is less than the greater of $80 million and 10% of the maximum credit as of the end of the most recent fiscal quarter. As of December 31, 2023, the Company's excess availability and Covenant Fixed Charge Coverage Ratio were in excess of the Revolving Credit Facility's requirements.

The Revolving Credit Facility provides that, upon the occurrence of certain events of default, the obligations thereunder may be accelerated and the lending commitments terminated. Such events of default include payment defaults, material inaccuracies of representations and warranties, covenant defaults, cross-defaults to other material indebtedness, voluntary and involuntary bankruptcy proceedings, material money judgments, material pension plan events, certain change of control events and other customary events of default.

Other Matters

The table below summarizes the debt repurchase activity during the year ended December 31, 2023:

	Aggregate Principal Repurchased	Reacquisition Cost
7.125% senior notes due July 2028	$ 58.4	$ 28.7
8.25% senior notes due March 2027	133.1	89.7
6.00% senior notes due June 2025	25.4	24.2
Total	$ 216.9	$ 142.6

For the year ended December 31, 2023, the repurchase of debt resulted in gains on the early extinguishment of debt of $74.3 million reflected in other income (expense), net, and the write-off of debt issuance costs of $2.0 million reflected in interest expense, in the Consolidated Statements of Operations.

The following table summarizes scheduled maturities of long-term debt as of December 31, 2023:

	2024	2025	2026	2027	2028	Thereafter
Scheduled maturities of long-term debt	$ 32.0	$ 1,306.6	$ 4,500.0	$ 1,616.9	$ 641.6	$ 1,250.0

The Company's non-guarantor subsidiaries held $2,828 million, or 30%, of total assets and $623 million, or 6%, of total liabilities as of December 31, 2023 and accounted for $2,144 million, or 31%, of net sales for the year ended December 31, 2023. All amounts presented exclude intercompany balances.

The Company is dependent upon the earnings and cash flow of its subsidiaries to make certain payments, including debt and interest payments. Certain subsidiaries may have limitations or restrictions on transferring funds to other subsidiaries that may be necessary to meet those requirements.

The weighted average effective interest rate on outstanding borrowings, including the impact of the interest rate swap contracts and the amortization of debt issuance costs and original issue discount, was 7.22% at December 31, 2023 and 6.91% at December 31, 2022.

9. DERIVATIVES AND HEDGING ACTIVITIES

Derivatives Not Designated as Hedging Instruments

The Company uses forward contracts to hedge a portion of its balance sheet foreign exchange re-measurement risk and to hedge certain planned foreign currency expenditures. As of December 31, 2023, the Company had foreign exchange contracts outstanding with maturities of up to nine months and aggregate notional values of $630.1 million (based on exchange rates as of December 31, 2023). Unrealized gains and losses resulting from these contracts are recognized in other income (expense), net and partially offset corresponding foreign exchange gains and losses on the balances and expenditures being hedged.

The following table presents the balance sheet location and fair value of the Company's derivatives not designated as hedging instruments:

		December 31,	
Contract Type	**Location of Asset (Liability)**	**2023**	**2022**
Foreign currency contracts	Prepaid expenses and other current assets	$ 8.7	$ 9.9
Foreign currency contracts	Accrued and other liabilities	(3.6)	(6.5)
Total derivatives not designated as hedging instruments		$ 5.1	$ 3.4

The pretax impact of the foreign currency forward contracts, both matured and outstanding, on the Consolidated Statements of Operations is as follows:

	Year Ended December 31,		
Location of Gain (Loss)	**2023**	**2022**	**2021**
Other income (expense), net	$ 2.8	$ (19.0)	$ (2.6)

Derivative Instruments Designated as Cash Flow Hedges of Interest Rate Risk

The Company has utilized a hedging strategy to mitigate a portion of the exposure to changes in cash flows resulting from variable interest rates on the 2026 Term Loan. In conjunction with the amendment to its 2026 Term Loan due to reference rate reform, on June 28, 2023, the Company settled its cash flow hedges with a notional value of $300.0 million and received cash of $6.8 million which is included within operating cash flow activity in accordance with the Company's accounting policy. The cash flow hedges were derecognized, and the gain of $6.1 million remaining as a component of accumulated comprehensive loss in the Consolidated Balance Sheets continues to be reclassified to earnings through interest expense as the interest payments are made on the 2026 Term Loan unless the forecasted hedged transaction becomes probable of not occurring. See Note 8 for further discussion of the amendment to the 2026 Term Loan. Following the amendment to the 2026 Term Loan and settlement of the cash flow hedges noted above, during the third quarter of 2023, the Company reenacted its hedging strategy to mitigate the interest rate risk from its variable rate debt, initially associated with its amended 2026 Term Loan and extending to future borrowings or debt issued, to fix a portion of the future interest cash flows by designating qualifying receive-variable and pay-fixed interest rate swaps as a cash flow hedge for accounting and financial reporting purposes. The total notional amount of the interest rate swap derivatives as of December 31, 2023 was $700.0 million with outstanding maturities up to thirty-one months. There was no ineffectiveness on the instruments designated as cash flow hedges for the years ended December 31, 2023, 2022 or 2021.

The following table presents the balance sheet location and fair value of the derivative instruments designated as cash flow hedges of interest rate risk:

Contract Type	Location of Asset (Liability)	December 31, 2023	December 31, 2022
Interest rate swap contracts	Other noncurrent assets	$ —	$ 8.6
Interest rate swap contracts	Other noncurrent liabilities	(8.0)	—
Total derivatives designated as cash flow hedging instruments		$ (8.0)	$ 8.6

The impact of the effective portion of the interest rate swap contracts designated as cash flow hedging instruments on the Consolidated Statements of Comprehensive Loss is as follows:

Location of Gain (Loss)	Year Ended December 31, 2023	2022	2021
Other comprehensive income (loss), net of tax	$ (8.2)	$ 16.0	$ 14.4

10. FAIR VALUE MEASUREMENTS

The Company's financial instruments consist primarily of cash and cash equivalents, trade receivables, trade payables, debt instruments, interest rate swap contracts and foreign currency contracts. For cash and cash equivalents, trade receivables and trade payables, the carrying amounts of these financial instruments as of December 31, 2023 and 2022 were considered representative of their fair values due to their short terms to maturity. The fair values of the Company's debt instruments, interest rate swap contracts and foreign currency contracts were based on indicative quotes.

Fair value measurements using quoted prices in active markets for identical assets and liabilities fall within Level 1 of the fair value hierarchy, measurements using significant other observable inputs fall within Level 2, and measurements using significant unobservable inputs fall within Level 3.

The carrying amounts, estimated fair values and valuation input levels of the Company's debt instruments, interest rate derivatives and foreign currency contracts as of December 31, 2023 and 2022, are as follows:

	December 31, 2023		December 31, 2022		
	Carrying Amount	Fair Value	Carrying Amount	Fair Value	Valuation Inputs
Assets:					
Foreign currency contracts	$ 8.7	$ 8.7	$ 9.9	$ 9.9	Level 2
Interest rate swap contracts	—	—	8.6	8.6	Level 2
Liabilities:					
7.125% senior notes due 2028	$ 641.6	$ 301.6	$ 700.0	$ 502.6	Level 2
5.00% senior notes due 2027	750.0	312.2	750.0	513.4	Level 2
8.25% senior notes due 2027	866.9	454.9	1,000.0	780.8	Level 2
6.00% senior notes due 2025	1,274.6	1,038.8	1,300.0	1,183.4	Level 2
4.75% senior secured notes due 2029	1,250.0	840.6	1,250.0	1,000.0	Level 2
6.00% senior secured notes due 2026	1,500.0	1,327.5	1,500.0	1,383.3	Level 2
Senior secured term loan due 2026	3,064.0	2,742.3	3,096.0	2,925.7	Level 2
Foreign currency contracts	3.6	3.6	6.5	6.5	Level 2
Interest rate swap contracts	8.0	8.0	—	—	Level 2

Non-Recurring Fair Value Measurements

During the annual impairment test in the fourth quarter of 2023, a goodwill impairment charge of $46.4 million was recorded related to the ANS reporting unit which is the same as the ANS segment, and $99.1 million was recorded related to the BDCC reporting unit which is part of the CCS segment. The fair value of the reporting unit was determined using a discounted cash flow (DCF) model and a guideline public company approach, with 75% of the value determined using the DCF model and 25% of the value determined using the guideline public company approach. Under the DCF method, the fair value of a reporting unit is based on the present value of estimated future cash flows. Under the guideline public company method, the fair value is based upon market multiples of revenue and earnings derived from publicly-traded companies with similar operating and investment characteristics as the reporting unit. The inputs to both the DCF model and the guideline public company analysis are Level 3 valuation inputs. Changes in any of these inputs, among other factors, could negatively affect the fair value of the ANS and BDCC reporting units and result in a material impairment charge in the future.

In the third quarter of 2023, a goodwill impairment charge of $425.9 million was recorded related to the ANS reporting unit. The fair value of each reporting unit was determined using Level 3 valuation inputs with an approach consistent with the 2023 annual impairment test.

During the annual impairment test in the fourth quarter of 2022, a goodwill impairment charge of $1,119.6 million was recorded related to the ANS reporting unit. The fair value of each reporting unit was determined using Level 3 valuation inputs with an approach consistent with the 2023 annual impairment test.

These fair value estimates are based on pertinent information available to management as of the valuation date. Although management is not aware of any factors that would significantly affect these fair value estimates, such amounts have not been comprehensively revalued for purposes of these financial statements since those dates, and current estimates of fair value may differ significantly from the amounts presented.

11. RESTRUCTURING COSTS

The Company incurs costs associated with restructuring initiatives intended to improve overall operating performance and profitability. The costs related to restructuring actions are generally cash-based and primarily consist of employee-related costs, which include severance and other one-time termination benefits.

In addition to the employee-related costs, the Company records other costs associated with restructuring actions such as the gain or loss on the sale of facilities and impairment costs arising from unutilized real estate or equipment. The Company attempts to sell or lease this unutilized space but additional impairment charges may be incurred related to these or other excess assets.

The Company's net pretax restructuring activity included in restructuring costs, net on the Consolidated Statements of Operations, by segment, was as follows:

	Year Ended December 31,		
	2023	2022	2021
CCS	$ 14.0	$ 17.1	$ 62.0
OWN	6.6	22.4	3.6
NICS	12.4	9.9	8.5
ANS	(6.0)	12.2	9.2
Corporate and other	2.7	1.4	1.8
Total	$ 29.7	$ 63.0	$ 85.1

The corporate and other line item above reflects general corporate restructuring costs that were previously allocated to the Home segment. These indirect expenses have been classified as continuing operations for all periods presented since the costs were not directly attributable to the discontinued operations of the Home business. These costs will be reallocated to the Company's remaining segments in future years.

Restructuring liabilities were included in the Company's Consolidated Balance Sheets as follows:

	December 31,	
	2023	2022
Accrued and other liabilities	$ 11.7	$ 58.9
Other noncurrent liabilities	0.1	0.5
Total restructuring liabilities	$ 11.8	$ 59.4

CommScope NEXT Restructuring Actions

In the first quarter of 2021, the Company announced and began implementing a business transformation initiative called CommScope NEXT. This initiative is designed to drive shareholder value through three pillars: profitable growth, operational efficiency and portfolio optimization. The activity within the liability established for CommScope NEXT restructuring actions was as follows:

	Employee-Related Costs	Other	Total
Balance at December 31, 2021	$ 60.3	$ —	$ 60.3
Additional expense, net	50.8	8.6	59.4
Cash paid	(48.7)	(0.5)	(49.2)
Foreign exchange and other non-cash items	(3.7)	(8.1)	(11.8)
Balance at December 31, 2022	58.7	—	58.7
Additional expense (income), net	63.6	(33.9)	29.7
Cash received (paid)	(110.8)	67.6	(43.2)
Foreign exchange and other non-cash items	0.2	(33.7)	(33.5)
Balance at December 31, 2023	$ 11.7	$ —	$ 11.7

CommScope NEXT actions to date have included the closure of manufacturing, administration and warehouse facilities as well as headcount reductions in other manufacturing locations and engineering, marketing, sales and administrative functions and asset impairments associated with restructuring related actions. During the year ended December 31, 2023, additional expenses were recorded for employee related costs for severance. Other costs included asset impairments for certain leased real estate and equipment, and sales of several owned properties, including an international manufacturing facility, along with U.S. office and warehouse facilities. The Company recorded net proceeds of $67.6 million related primarily to facility sales, resulting in a gain of $33.9 million. Employee related and other costs are recorded as restructuring costs, net on the Consolidated Statements of Operations.

Concurrent with the international manufacturing facility sale, the Company entered into a sale leaseback transaction whereby a minor portion of the building used to support research and development operations was leased back for a term of nine years with annual payments that range from €800 thousand to €1.5 million. The Company determined the lease to be an operating lease, and upon entering into the lease, recognized a right of use asset and operating lease liability of €7.5 million based on the present value of the minimum lease payments discounted using an incremental borrowing rate of 7.15%.

The Company has recognized restructuring charges of $174.5 million to date related to CommScope NEXT actions. The Company expects to make cash payments of $11.7 million in 2024 to settle CommScope NEXT restructuring actions. Additional restructuring actions related to CommScope NEXT are expected to be identified, and the resulting charges and cash requirements could be material.

12. EMPLOYEE BENEFIT PLANS

Defined Contribution Plans

The Company and certain of its subsidiaries have defined contribution retirement savings plans, the most significant of which is a 401(k) plan in the U.S. These plans allow employees meeting certain requirements to contribute a portion of their compensation on a pretax and/or after-tax basis in accordance with guidelines established by the plans and the Internal Revenue Service or other tax authorities. The Company matches a percentage of the employee contributions up to certain limits. During the years ended December 31, 2023, 2022 and 2021, the Company made contributions to defined contribution retirement savings plans of $27.9 million, $48.1 million and $49.0 million, respectively. The Company temporarily paused its employer contribution as of July 1, 2023 and incurred no expenses for contributions through the end of 2023.

The Company also maintains noncontributory and contributory deferred compensation plans. During the years ended December 31, 2023, 2022 and 2021, the Company recognized pretax costs (benefits) of $2.1 million, $(2.7) million and $1.3 million, respectively, related to these plans. The liability related to these plans was $22.7 million and $22.5 million as of December 31, 2023 and 2022, respectively.

Pension Plans

The Company sponsors defined benefit pension plans covering certain active and former domestic and foreign employees. Both funded and unfunded plans are included in the defined benefit pension plans. The following table summarizes information for the defined benefit pension plans:

	December 31,			
	U.S. Plans		Non-U.S. Plans	
	2023	2022	2023	2022
Change in benefit obligation:				
Benefit obligation, beginning	$ 9.4	$ 12.5	$ 183.3	$ 275.5
Service cost	—	—	5.6	5.7
Interest cost	0.4	0.3	7.9	3.7
Actuarial loss (gain)	0.4	(2.6)	2.2	(68.7)
Benefits paid	(0.8)	(0.8)	(4.1)	(4.5)
Settlements	—	—	(9.7)	(6.2)
Curtailment	—	—	(0.1)	—
Foreign exchange and other	—	—	7.1	(22.2)
Benefit obligation, ending	$ 9.4	$ 9.4	$ 192.2	$ 183.3
Change in plan assets:				
Fair value of plan assets, beginning	$ —	$ —	$ 181.2	$ 279.4
Employer and plan participant contributions	0.8	0.8	7.3	6.4
Return on plan assets	—	—	7.6	(69.9)
Benefits paid	(0.8)	(0.8)	(4.1)	(4.5)
Settlements	—	—	(9.7)	(6.2)
Foreign exchange and other	—	—	7.1	(24.0)
Fair value of plan assets, ending	$ —	$ —	$ 189.4	$ 181.2
Funded status, net liability	$ 9.4	$ 9.4	$ 2.8	$ 2.1

The following table presents the balance sheet location of the Company's pension liabilities and assets:

	December 31,			
	U.S. Plans		Non-U.S. Plans	
	2023	2022	2023	2022
Accrued and other liabilities	$ (0.9)	$ (0.9)	$ (0.9)	$ (0.7)
Other noncurrent liabilities	(8.5)	(8.5)	(16.5)	(15.3)
Other noncurrent assets		—	14.6	13.9

The accumulated benefit obligation for the Company’s U.S. defined benefit pension plans was $9.4 million for both periods as of December 31, 2023 and 2022, and the accumulated benefit obligation for the Company’s non-U.S. defined benefit pension plans was $181.5 million and $159.9 million as of December 31, 2023 and 2022, respectively.

The following table summarizes information for the Company's pension plans with an accumulated benefit obligation in excess of plan assets:

	December 31,			
	U.S. Plans		Non-U.S. Plans	
	2023	2022	2023	2022
Projected benefit obligation	$ 9.4	$ 9.4	$ 46.7	$ 44.1
Accumulated benefit obligation	9.4	9.4	44.2	41.3
Fair value of plan assets	—	—	29.4	28.4

The following table summarizes pretax amounts included in accumulated other comprehensive loss:

	December 31,			
	U.S. Plans		Non-U.S. Plans	
	2023	2022	2023	2022
Unrecognized net actuarial gain (loss)	$ 0.6	$ 1.0	$ (17.8)	$ (16.5)
Unrecognized prior service cost	—	—	0.5	—
Total	$ 0.6	$ 1.0	$ (17.3)	$ (16.5)

Actuarial gains and losses are amortized using a corridor approach. The corridor is equal to 10% of the greater of the benefit obligation and the fair value of the assets. Gains and losses in excess of the corridor are generally amortized over the average remaining life of the plan participants. Pretax amounts for net periodic benefit cost and other amounts included in other comprehensive income (loss) for the defined benefit pension plans consisted of the following components:

	Year Ended December 31,					
	U.S. Plans			Non-U.S. Plans		
	2023	2022	2021	2023	2022	2021
Service cost	$ —	$ —	$ —	$ 5.6	$ 5.7	$ 6.8
Interest cost	0.4	0.3	0.3	7.9	3.7	3.1
Recognized actuarial loss	—	0.1	0.1	0.6	0.1	1.6
Expected return on plan assets	—	—	—	(7.2)	(4.9)	(6.4)
Settlement loss	—	—	—	0.9	1.6	0.3
Curtailment gain	—	—	—	(0.1)	—	(2.5)
Net periodic benefit cost	$ 0.4	$ 0.4	$ 0.4	$ 7.7	$ 6.2	$ 2.9
Changes in plan assets and benefit obligations included in other comprehensive income (loss):						
Change in unrecognized net actuarial loss (gain)	$ 0.4	$ (2.7)	$ (0.6)	$ 1.3	$ 2.9	$ (28.0)
Change in unrecognized prior service cost	—	—	—	(0.5)	(0.1)	(0.4)
Curtailment and settlements	—	—	—	—	—	(1.8)
Total included in other comprehensive income (loss)	$ 0.4	$ (2.7)	$ (0.6)	$ 0.8	$ 2.8	$ (30.2)
Total recognized in net periodic benefit cost and included in other comprehensive income (loss)	$ 0.8	$ (2.3)	$ (0.2)	$ 8.5	$ 9.0	$ (27.3)

The Company reports the service cost component of net periodic benefit cost in the same line item as other compensation costs arising from the services rendered by the employee and records the other components of net periodic benefit cost in other income (expense), net.

Assumptions

Significant weighted average assumptions used in determining benefit obligations and net periodic benefit cost are as follows:

	U.S. Plans			Non-U.S. Plans		
	2023	2022	2021	2023	2022	2021
Benefit obligations:						
Discount rate	4.77 %	4.99 %	2.55 %	4.01 %	4.37 %	1.47 %
Rate of compensation increase	— %	— %	— %	3.31 %	3.36 %	3.79 %
Net periodic benefit cost:						
Discount rate	4.99 %	2.55 %	2.07 %	4.37 %	1.47 %	1.02 %
Rate of return on plan assets	— %	— %	— %	4.57 %	4.03 %	1.96 %
Rate of compensation increase	— %	— %	— %	3.36 %	3.79 %	3.59 %

The Company considered the available yields on high-quality fixed-income investments with maturities corresponding to the Company's expected benefit obligations to determine the discount rates at each measurement date.

Plan Assets

In developing the expected rate of return on plan assets, the Company considered the expected long-term rate of return on individual asset classes. Expected return on plan assets is based on the market value of the assets. The majority of the non-U.S. pension assets are managed by independent investment advisors. In general, the investment strategy is designed to accumulate a diversified portfolio among markets, asset classes or individual securities in order to reduce market risk and assure that the pension assets are available to pay benefits as they come due.

Mutual funds classified as Level 1 are valued at net asset value, which is based on the fair value of the funds' underlying securities. Certain mutual funds are classified as Level 2 because a portion of the funds' underlying assets are valued using significant other observable inputs. Other assets are primarily composed of fixed income investments (including insurance and real estate products) and are valued based on the investment's stated rate of return, which approximates market interest rates.

The Company had no U.S. defined benefit pension plan assets as of December 31, 2023 or 2022. The estimated fair values and the valuation input levels of the Company's non-U.S. defined benefit pension plan assets are as follows:

	December 31, 2023 Non-U.S. Plans	
	Level 1 Fair Value	Level 2 Fair Value
Mutual funds:		
International equity	$ 30.6	$ —
International debt	30.6	95.1
Other	3.0	30.1
Total	$ 64.2	$ 125.2

	December 31, 2022 Non-U.S. Plans	
	Level 1 Fair Value	Level 2 Fair Value
Mutual funds:		
International equity	$ 19.7	$ 7.7
International debt	36.8	76.0
Absolute return	—	3.8
Other	7.7	29.5
Total	$ 64.2	$ 117.0

Expected Cash Flows

The Company expects to contribute $0.9 million to U.S. defined benefit pension plans and $4.9 million to non-U.S. defined benefit pension plans during 2024.

The following table summarizes projected benefit payments from pension plans through 2033, including benefits attributable to estimated future service (in millions):

	U.S. Plans	Non-U.S. Plans
2024	$ 0.9	$ 17.1
2025	0.9	10.1
2026	0.9	14.2
2027	0.8	11.4
2028	0.8	11.0
2029-2033	3.7	61.4

13. INCOME TAXES

Loss from continuing operations before income taxes includes the results from domestic and international operations as follows:

	Year Ended December 31,		
	2023	2022	2021
U.S. companies	$ (811.6)	$ (1,253.5)	$ (409.0)
Non-U.S. companies	93.7	53.8	19.8
Loss from continuing operations before income taxes	$ (717.9)	$ (1,199.7)	$ (389.2)

The components of income tax expense (benefit) were as follows:

	Year Ended December 31,		
	2023	2022	2021
Current:			
Federal	$ 80.7	$ 38.3	$ (16.4)
Foreign	68.0	32.4	74.8
State	0.4	15.6	7.7
Current income tax expense	$ 149.1	$ 86.3	$ 66.1
Deferred:			
Federal	$ 0.3	$ (83.7)	$ (97.1)
Foreign	(23.3)	(7.7)	(1.8)
State	7.3	(9.9)	(6.4)
Deferred income tax benefit	(15.7)	(101.3)	(105.3)
Total income tax expense (benefit)	$ 133.4	$ (15.0)	$ (39.2)

The reconciliation of income taxes calculated at the statutory U.S. federal income tax rate to the Company's expense (benefit) for income taxes was as follows:

	Year Ended December 31,		
	2023	2022	2021
Benefit for income taxes at federal statutory rate	$ (150.8)	$ (251.9)	$ (81.8)
State income taxes, net of federal tax effect	(1.3)	(9.9)	(18.0)
Other permanent items	4.9	11.6	6.9
Equity-based compensation	8.1	(5.6)	7.0
Other changes in tax laws and tax rulings	(4.1)	4.7	37.4
Goodwill related items	113.4	232.0	—
Base erosion and anti-abuse tax	—	—	10.2
Foreign-derived intangible income deduction	(1.9)	(7.4)	(7.5)
Federal tax credits	(20.0)	(24.3)	(19.5)
Change in unrecognized tax benefits	(4.1)	(7.6)	(13.8)
Withholding taxes and Subpart F income, net of foreign tax credits	15.2	26.7	19.3
Foreign earnings taxed at other than federal rate	5.9	6.5	5.2
Tax provision adjustments and revisions to prior years' returns	2.7	(3.2)	(5.3)
Change in valuation allowances	165.4	13.4	20.7
Total expense (benefit) for income taxes	$ 133.4	$ (15.0)	$ (39.2)

The components of deferred income tax assets and liabilities and the classification of deferred tax balances on the balance sheet were as follows:

	December 31,	
	2023	2022
Deferred tax assets:		
Accounts receivable, inventory and warranty reserves	$ 210.3	$ 136.8
Employee benefits	42.1	59.0
Foreign net operating loss and tax credit carryforwards	593.4	565.0
Federal net operating loss and tax credit carryforwards	24.2	22.0
State net operating loss and tax credit carryforwards	105.7	103.6
Unrecognized tax benefits	27.9	30.8
Interest limitation	158.8	75.4
Capitalized research and development costs	506.2	471.6
Other	82.9	92.1
Total deferred tax assets	1,751.5	1,556.3
Valuation allowance	(838.5)	(642.0)
Total deferred tax assets, net of valuation allowance	$ 913.0	$ 914.3
Deferred tax liabilities:		
Intangible assets	$ (353.9)	$ (485.3)
Property, plant and equipment	(14.3)	(16.4)
Undistributed foreign earnings	(21.4)	(20.6)
Other	(18.5)	(12.9)
Total deferred tax liabilities	(408.1)	(535.2)
Net deferred tax asset	$ 504.9	$ 379.1
Deferred taxes recognized on the balance sheet:		
Noncurrent deferred tax asset	$ 615.6	$ 494.6
Noncurrent deferred tax liability	(110.7)	(115.5)
Net deferred tax asset	$ 504.9	$ 379.1

The deferred tax asset for foreign net operating loss and tax credit carryforwards as of December 31, 2023 includes foreign net operating loss carryforwards (net of federal tax effects) of $582.4 million, which begin to expire in 2024, and foreign tax credit carryforwards (net of federal tax effects) of $11.0 million, which begin to expire in 2024. Certain of these foreign net operating loss carryforwards are subject to local restrictions limiting their utilization. Valuation allowances of $569.3 million have been established related to these foreign deferred tax assets.

The deferred tax asset for federal net operating loss and tax credit carryforwards as of December 31, 2023 relates to $4.2 million of net operating loss carryforwards, which begin to expire in 2031, and $20.0 million of U.S. foreign tax credit carryforwards, which begin to expire in 2029. A valuation allowance of $20.0 million has been established against these deferred tax assets.

The deferred tax asset for state net operating loss and tax credit carryforwards as of December 31, 2023 includes state net operating loss carryforwards (net of federal tax impact) of $38.3 million, which begin to expire in 2024, and state tax credit carryforwards (net of federal tax impact) of $67.4 million, which begin to expire in 2024. A valuation allowance of $87.8 million has been established against these and other state income tax related deferred tax assets.

The deferred tax asset for federal and state interest limitation carryforwards as of December 31, 2023 are $158.8 million which have an indefinite carryforward. During 2023, a valuation allowance of $158.8 million has been established against these federal and state deferred tax assets.

In addition to the valuation allowances detailed above, the Company has also established a valuation allowance of $2.6 million against other deferred tax assets.

Under current U.S. tax regulations, repatriation of foreign earnings to the U.S. can generally be completed with no incremental U.S. tax. However, repatriation of foreign earnings could subject the Company to U.S. state and non-U.S. jurisdictional taxes (including withholding taxes) on distributions. As of December 31, 2023, the Company has a deferred tax liability of $21.4 million for the estimated foreign and state tax costs associated with the expected repatriation of the Company's undistributed foreign earnings. The unrecorded deferred tax liability for foreign and state tax costs associated with earnings considered permanently reinvested is not material as of December 31, 2023.

The following table reflects a reconciliation of the beginning and end of period amounts of gross unrecognized tax benefits, excluding interest and penalties:

	Year Ended December 31,		
	2023	2022	2021
Balance at beginning of period	$ 145.5	$ 176.6	$ 190.5
Increase related to prior periods	0.1	1.1	0.7
Decrease related to prior periods	(7.9)	(23.3)	(0.3)
Increase related to current periods	5.5	5.1	5.9
Decrease related to settlements with taxing authorities	(0.3)	(13.4)	(7.5)
Decrease related to lapse in statutes of limitations	(3.2)	(0.6)	(12.7)
Balance at end of period	$ 139.7	$ 145.5	$ 176.6

The Company's liability for unrecognized tax benefits that, if recognized, would favorably affect the effective tax rate in future periods was $110.7 million as of December 31, 2023. The Company operates in numerous jurisdictions worldwide and is subject to routine tax audits on a regular basis. The determination of the Company's unrecognized tax benefits involves significant management judgment regarding interpretation of relevant facts and tax laws in each of these jurisdictions.

Unrecognized tax benefits are reviewed and evaluated on an ongoing basis and may be adjusted for changing facts and circumstances including the lapse of applicable statutes of limitation and closure of tax examinations. Although the timing and outcome of such events are difficult to predict, the Company estimates that the balance of unrecognized tax benefits, excluding the impact of accrued interest and penalties, may be reduced by up to $8.0 million within the next twelve months.

The Company provides for interest and penalties related to unrecognized tax benefits as income tax expense. The Company accrued $5.4 million and $9.4 million for interest and penalties as of December 31, 2023 and 2022, respectively. During the years ended December 31, 2023, 2022 and 2021, the net expense (benefit) for interest and penalties recognized through income tax expense (benefit) was $(4.0) million, $0.1 million and $(0.1) million, respectively.

The Company files federal, state and local tax returns with statutes of limitation generally ranging from 3 to 4 years. The Company is currently undergoing a U.S. federal income tax audit for the 2019 tax year and is generally no longer subject to state and local tax examinations for years prior to 2020. Tax returns filed by the Company's significant foreign subsidiaries are generally subject to statutes of limitation of 3 to 7 years and are generally no longer subject to examination for years prior to 2018. In many jurisdictions, tax authorities retain the ability to review prior years' tax returns and to adjust any net operating loss or tax credit carryforwards from these years that are available to be utilized in subsequent periods. During 2023, the Company recognized $(2.3) million (net of payments) related to the lapse of applicable statutes of limitations and the conclusion of various domestic and foreign examinations.

The following table presents income tax expense (benefit) related to amounts presented in other comprehensive income (loss):

	Year Ended December 31,		
	2023	2022	2021
Foreign currency translation	$ (2.9)	$ 1.2	$ 1.2
Defined benefit plans	(0.3)	0.8	6.6
Total	$ (3.2)	$ 2.0	$ 7.8

14. SERIES A CONVERTIBLE PREFERRED STOCK

On April 4, 2019, the Company issued and sold 1,000,000 shares of the Convertible Preferred Stock for $1.0 billion, or $1,000 per share, pursuant to an Investment Agreement between the Company and The Carlyle Group (Carlyle), dated November 8, 2018 (the Investment Agreement). As of December 31, 2023, the Company had authorized 1,200,000 shares of Series A Convertible Preferred Stock.

Dividend Rights

The Convertible Preferred Stock ranks senior to the shares of the Company's common stock with respect to dividend rights and rights on the distribution of assets on any voluntary or involuntary liquidation, dissolution or winding up of the affairs of the Company. The Convertible Preferred Stock has a liquidation preference of $1,000 per share. Holders of the Convertible Preferred Stock are entitled to a cumulative dividend at the rate of 5.5% per year, payable quarterly in arrears. If CommScope does not declare and pay a dividend, the dividend rate will increase by 2.5% to 8.0% per year (and that rate will increase by an additional 0.50% every three months until such unpaid dividend is declared and paid, subject to a cap of 11.0% per year) until all accrued but unpaid dividends have been paid in full. Dividends can be paid in cash, in-kind through the issuance of additional shares of Convertible Preferred Stock or any combination of the two, at the Company's option.

During the years ended December 31, 2023, 2022 and 2021, the Company paid dividends in-kind of $61.8 million, $44.2 million and $14.3 million, respectively, which were recorded as additional Convertible Preferred Stock on the Consolidated Balance Sheets. The Company did not pay cash dividends for the year ended December 31, 2023 but did pay cash dividends of $14.9 million and $43.0 million for the years ended December 31, 2022 and 2021, respectively.

Conversion Features

The Convertible Preferred Stock is convertible at the option of the holders at any time into shares of CommScope common stock at an initial conversion rate of 36.3636 shares of common stock per share of the Convertible Preferred Stock (equivalent to $27.50 per share of common stock). The conversion rate is subject to customary anti-dilution and other adjustments. At any time after the third anniversary of the issuance of the Convertible Preferred Stock, if the volume weighted average price of CommScope's common stock exceeds the conversion price of $49.50, as may be adjusted pursuant to the Certificate of Designations, for at least thirty trading days in any period of forty-five consecutive trading days (including the final five trading days of any such forty-five-trading day period) all of the Convertible Preferred Stock may be converted at the election of CommScope into the relevant number of shares of CommScope common stock.

Redemption Rights

On any date during the three months following the eight year and six-month anniversary of the Investment Agreement closing date and the three months following each anniversary thereafter, holders of the Convertible Preferred Stock will have the right to require CommScope to redeem all or any portion of the Convertible Preferred Stock at 100% of the liquidation preference thereof plus all accrued and unpaid dividends. The redemption price is payable, at the Company's option, in cash or a combination of cash and common stock, subject to certain restrictions.

Upon certain change of control events involving CommScope, CommScope has the right, subject to the holder's right to convert prior to such redemption, to redeem all of the Convertible Preferred Stock for the greater of (i) an amount in cash equal to the sum of the liquidation preference of the Convertible Preferred Stock, all accrued but unpaid dividends and, if the applicable redemption date is prior to the fifth anniversary of the first dividend payment date, the present value, discounted at a rate of 10%, of any remaining scheduled dividends through the five year anniversary of the first dividend payment date, assuming CommScope chose to pay such dividends in cash and (ii) the consideration the holders would have received if they had converted their shares of the Convertible Preferred Stock into CommScope common stock immediately prior to the change of control event.

To the extent that CommScope does not exercise the redemption right described above, following the effective date of any such change of control event, the holders of the Convertible Preferred Stock can require CommScope to repurchase the Convertible Preferred Stock at the greater of (i) an amount in cash equal to 100% of the liquidation preference thereof plus all accrued but unpaid dividends and (ii) the consideration the holders would have received if they had converted their shares of the Convertible Preferred Stock into CommScope common stock immediately prior to the change of control event.

Voting Rights

Holders of the Convertible Preferred Stock are entitled to vote with the holders of the Company's common stock on an as-converted basis. Holders of the Convertible Preferred Stock are entitled to a separate class vote with respect to, among other things, amendments to CommScope's organizational documents that have an adverse effect on the Convertible Preferred Stock, issuances by CommScope of securities that are senior to, or equal in priority with, the Convertible Preferred Stock and issuances of shares of the Convertible Preferred Stock after the closing date of the ARRIS International plc (ARRIS) acquisition, other than shares issued as dividends with respect to shares of the Convertible Preferred Stock.

15. STOCKHOLDERS' EQUITY

Equity-Based Compensation Plans

In 2019, the Company's stockholders approved the 2019 Long-Term Incentive Plan authorizing 8.0 million shares for issuance, plus additional shares underlying awards outstanding under the predecessor plans. Subsequently, in each of the years 2020 through 2023, the Company's stockholders approved the Amended and Restated 2019 Long-Term Incentive Plan (the 2019 Plan) and authorized an additional aggregate 24.5 million shares for issuance. All future equity awards will be made from the 2019 Plan. Awards under the 2019 Plan may include stock options, stock appreciation rights, restricted stock, stock units (including restricted stock units (RSUs) and deferred stock units), performance awards (represents any of the awards already listed with a performance-vesting component), other stock-based awards and cash-based awards. As of December 31, 2023, there were 1.7 million shares available for future grants under the 2019 Plan.

As of December 31, 2023, $71.4 million of total unrecognized compensation expense related to unvested stock options, RSUs and performance share units (PSUs) is expected to be recognized over a remaining weighted average period of 1.9 years. There were no significant capitalized equity-based compensation costs at December 31, 2023.

The following table shows a summary of the equity-based compensation expense included in the Consolidated Statements of Operations:

	Year Ended December 31,		
	2023	2022	2021
Selling, general and administrative	$ 28.7	$ 31.8	$ 37.0
Research and development	9.7	16.0	21.8
Cost of sales	5.2	7.5	12.0
Total equity-based compensation expense	$ 43.6	$ 55.3	$ 70.8

The Company believes the valuation techniques and the approaches utilized to develop the underlying assumptions are appropriate in estimating the fair values of its equity-based compensation. Estimates of fair value are not intended to predict actual future events or the value ultimately realized by employees who receive equity awards. Subsequent events are not indicative of the reasonableness of the original estimates of fair value made by the Company.

Stock Options

Stock options are awards that allow the recipient to purchase shares of the Company's common stock at a fixed price. Stock options are granted at an exercise price equal to the Company's stock price at the date of grant. The Company uses the Black-Scholes model to estimate the fair value of stock options at the date of grant. These awards generally vest over five years following the grant date and have a contractual term of ten years. There were 2.0 million options outstanding as of December 31, 2023 with no intrinsic value and the majority were vested. There were no stock options granted during the years ended December 31, 2023, 2022 or 2021. The intrinsic value of options exercised during the years ended December 31, 2023 and 2022 was not material. The intrinsic value of options exercised during the year ended December 31, 2021 was $5.3 million.

Restricted Stock Units

RSUs entitle the holder to shares of common stock after a vesting period of generally three years. The fair value of the awards is determined on the grant date based on the Company's stock price.

The following table summarizes the RSU activity (in millions, except per share data):

	Restricted Stock Units	Weighted Average Grant Date Fair Value Per Share
Non-vested share units at December 31, 2022	10.6	$ 10.47
Granted	7.4	$ 4.57
Vested and shares issued	(5.0)	$ 10.84
Forfeited	(1.5)	$ 9.58
Non-vested share units at December 31, 2023	11.5	$ 6.62

The weighted average grant date fair value per unit of these awards granted during the years ended December 31, 2023, 2022 and 2021 was $4.57, $8.05 and $20.16, respectively. The total fair value of RSUs that vested during the years ended December 31, 2023, 2022 and 2021 was $53.8 million, $67.1 million and $74.1 million, respectively.

Performance Share Units

PSUs are stock awards in which the number of shares ultimately received by the employee depends on achievement towards a performance measure. Certain of CommScope's PSUs have an internal performance measure and the awards vest at the end of three years. The number of shares issued under these awards can vary between 0% and 300% of the number of PSUs granted. The fair value of these awards is determined on the date of grant based on the Company's stock price.

CommScope also has PSUs with a market condition performance measure based on stock price milestones over a three-year period. The number of shares issued under these awards can vary between 0% to 100% of the number of PSUs granted. In addition, the Company has PSUs with a market condition based on the Company's total stockholder return (TSR) ranking relative to the S&P 500 TSR for a three-year period. The number of shares issued under these awards can vary between 0% to 200% of the number of PSUs granted. The Company uses a Monte Carlo simulation model to estimate the fair value of PSUs with a market condition performance measure at the date of grant. Key assumptions used in the model include the risk-free interest rate, which reflects the yield on zero-coupon U.S. treasury securities, and stock price volatility, which is derived based on the historical volatility of the Company's stock.

The following table presents the weighted average assumptions used to estimate the fair value of these awards granted:

	Year Ended December 31,		
	2023	2022	2021
Risk-free interest rate	4.4%	1.6%	0.4%
Expected volatility	67.2%	61.1%	55.9%
Weighted average fair value at grant date	$ 9.16	$ 11.27	$ 12.17

The following table summarizes the PSU activity (in millions, except per share data):

	Performance Share Units	Weighted Average Grant Date Fair Value Per Share
Non-vested share units at December 31, 2022	2.8	$ 8.03
Granted	3.9	$ 5.73
Forfeited	(0.3)	$ 9.05
Non-vested share units at December 31, 2023	6.4	$ 6.59

The weighted average grant date fair value per unit of these awards granted during the years ended December 31, 2023, 2022 and 2021 was $5.73, $9.54 and $14.55, respectively. The total fair value of PSUs that vested during the year ended December 31, 2023, was not material. The total fair value of PSUs that vested during the years ended December 31, 2022 and 2021 was $3.1 million, and $1.0 million, respectively.

16. COMMITMENTS AND CONTINGENCIES

The following table summarizes the activity in the product warranty accrual, included in accrued and other liabilities and other noncurrent liabilities:

	Year Ended December 31,		
	2023	2022	2021
Product warranty accrual, beginning of period	$ 33.7	$ 36.6	$ 28.4
Provision for warranty claims	13.7	17.5	20.2
Warranty claims paid	(20.1)	(19.9)	(11.6)
Foreign exchange	0.1	(0.5)	(0.4)
Product warranty accrual, end of period	$ 27.4	$ 33.7	$ 36.6

Third-Party Guarantees

The Company was contingently liable under open standby letters of credit issued by its banks to support performance obligations of a third-party contractor that totaled $35.9 million as of December 31, 2023. These amounts represent an estimate of the maximum amounts the Company would expect to incur upon the contractual non-performance of the third-party contractor, but the Company also has cross-indemnities in place that may enable it to recover amounts in the event of non-performance by the third-party contractor. The Company believes the likelihood of having to perform under these guarantees is remote. There were no material amounts recorded in the consolidated financial statements related to third-party guarantee agreements as of and for the years ended December 31, 2023 or 2022. As of December 31, 2023, these instruments reduced the available borrowings under the Revolving Credit Facility.

Non-cancellable Purchase Obligations

In July 2023, the Company entered into a long-term supply contract with a third-party to secure the supply of certain raw materials. Under the terms of the contract, the Company will make advance payments through 2026 totaling $120.0 million (undiscounted) and based on meeting certain minimum purchase requirements through 2031, such advance payments will be credited and applied to future orders on a quarterly basis beginning in 2027 through 2031. Advance payments of $30.0 million are recorded as other noncurrent assets in the Consolidated Balance Sheets as of December 31, 2023. The Company has committed to purchases of raw materials under this agreement beginning in 2023, growing to a level of approximately $137 million per year by 2026 and continuing through 2032.

Legal Proceedings

The Company is a party to certain intellectual property claims and also periodically receives notices asserting that its products infringe on another party's patents and other intellectual property rights. These claims and assertions, whether against the Company directly or against its customers, could require the Company to pay damages or royalties, stop offering the relevant products and/or cease other activities. The Company may also be called upon to indemnify certain customers for costs related to products sold to such customers. The outcome of these claims and notices is uncertain, and a reasonable estimate of the loss from unfavorable outcomes in certain of these matters either cannot be determined or is estimated at the minimum amount of a range of estimates. The actual loss, through settlement or trial, could be material and may vary significantly from the Company's estimates. From time to time, the Company may also be involved as a plaintiff in intellectual property claims. Gain contingencies, if any, are recognized when they are realized.

During the year ended December 31, 2021, the Company received $17.1 million in the settlement of a warranty indemnification matter that was assumed in the acquisition of ARRIS in 2019. The recovery was recorded as a reduction of cost of sales in the Consolidated Statements of Operations and is reflected in the results of the ANS segment.

The Company is either a plaintiff or a defendant in certain other pending legal matters in the normal course of business. Management believes none of these other pending legal matters will have a material adverse effect on the Company's business or financial condition upon final disposition.

The Company is subject to various federal, state, local and foreign laws and regulations governing the use, discharge, disposal and remediation of hazardous materials. Compliance with current laws and regulations has not had, and is not expected to have, a materially adverse effect on the Company's financial condition or results of operations.

17. INDUSTRY SEGMENTS, MAJOR CUSTOMERS, RELATED PARTY TRANSACTIONS AND GEOGRAPHIC INFORMATION

Segment Information

As a result of the pending divestiture of the Home business, the Company is now reporting financial performance based on the following remaining four operating segments, which excludes the Home segment: CCS, OWN, NICS and ANS. For all periods presented, amounts have been recast to reflect these operating segment changes.

The Connectivity and Cable Solutions (CCS) segment provides fiber optic and copper connectivity and cable solutions for use in telecommunications, cable television, residential broadband networks, data centers and business enterprises. The CCS portfolio includes network solutions for indoor and outdoor network applications. Indoor network solutions include optical fiber and twisted pair structured cable solutions, intelligent infrastructure management hardware and software and network rack and cabinet enclosures. Outdoor network solutions are used in both local-area and wide-area networks and "last mile" fiber-to-the-home installations, including deployments of fiber-to-the-node, fiber-to-the-premises and fiber-to-the-distribution point to homes, businesses and cell sites.

The Outdoor Wireless Networks (OWN) segment focuses on the macro and metro cell markets. The segment includes base station antennas, radio frequency (RF) filters, tower connectivity, microwave antennas, metro cell products, cabinets, steel, accessories and the wireless spectrum management business, Comsearch.

The Networking, Intelligent Cellular and Security Solutions (NICS) segment provides wireless networks for enterprises and service providers. Product offerings include indoor cellular solutions such as distributed antenna systems, public key infrastructure solutions, indoor and outdoor Wi-Fi and long-term evolution (LTE) access points, access and aggregation switches; an Internet of Things suite, on-premises and cloud-based control and management systems; and software and software-as-a-service applications addressing security, location, reporting and analytics.

The Access Network Solutions (ANS) segment's product solutions include cable modem termination systems, video infrastructure, distribution and transmission equipment and cloud solutions that enable facility-based service providers to construct a state-of-the-art residential and metro distribution network.

The following table provides summary financial information by reportable segment:

	December 31,	
	2023	2022
Identifiable segment-related assets:		
CCS	$ 3,608.2	$ 4,263.8
OWN	993.8	1,166.8
NICS	1,271.5	1,338.1
ANS	1,946.7	2,632.6
Corporate and other	113.7	194.0
Total identifiable segment-related assets	7,933.9	9,595.3
Reconciliation to total assets:		
Cash and cash equivalents	543.8	373.0
Deferred income tax assets	615.6	494.5
Assets held for sale	278.6	1,222.6
Total assets	$ 9,371.9	$ 11,685.4

The corporate and other line item above reflects certain assets retained by the Company related to the Home business, with the primary asset being net inventory in the amount of $88.0 million as of December 31, 2023. The Company has entered into a Supply Agreement, whereby the Company will sell the retained inventory to Vantiva for a period of two years following the close of the transaction at market prices.

The Company's measurement of segment performance is adjusted EBITDA (earnings before interest, income taxes, depreciation and amortization). The Company defines adjusted EBITDA as operating income (loss), adjusted to exclude depreciation, amortization of intangible assets, restructuring costs, asset impairments, equity-based compensation, transaction, transformation and integration costs and other items that the Company believes are useful to exclude in the evaluation of operating performance from period to period because these items are not representative of the Company's core business.

The following table provides net sales, adjusted EBITDA, depreciation expense and additions to property, plant and equipment by reportable segment:

	Year Ended December 31,		
	2023	2022	2021
Net sales:			
CCS	$ 2,710.2	$ 3,789.6	$ 3,053.8
OWN	880.0	1,467.9	1,417.1
NICS	1,117.7	939.7	861.9
ANS	1,081.3	1,327.5	1,404.6
Consolidated net sales	$ 5,789.2	$ 7,524.7	$ 6,737.4
Segment adjusted EBITDA:			
CCS	$ 389.6	$ 643.6	$ 448.8
OWN	178.1	269.7	266.8
NICS	225.2	51.9	(15.3)
ANS	229.3	285.2	391.1
Corporate and other	(23.2)	(27.0)	(48.1)
Total segment adjusted EBITDA	999.0	1,223.4	1,043.3
Amortization of intangible assets	(327.1)	(440.0)	(510.0)
Restructuring costs, net	(29.7)	(63.0)	(85.1)
Equity-based compensation	(43.6)	(55.3)	(70.8)
Asset impairments	(571.4)	(1,119.6)	—
Transaction, transformation and integration costs	(27.1)	(35.1)	(50.6)
Acquisition accounting adjustments	(1.2)	(5.3)	(9.2)
Patent claims and litigation settlements	3.5	(1.7)	(3.2)
Recovery (reserve) of Russian accounts receivable	2.0	(2.7)	—
Cyber incident costs	(5.5)	—	—
Depreciation	(111.8)	(113.8)	(117.8)
Consolidated operating income (loss)	$ (112.9)	$ (613.1)	$ 196.6
Depreciation expense:			
CCS	$ 61.3	$ 58.8	$ 53.6
OWN	12.6	14.3	15.4
NICS	13.0	15.0	19.2
ANS	22.1	22.5	25.8
Corporate and other	2.8	3.2	3.8
Consolidated depreciation expense	$ 111.8	$ 113.8	$ 117.8
Additions to property, plant and equipment:			
CCS	$ 29.1	$ 64.6	$ 81.5
OWN	3.8	9.9	11.0
NICS	4.3	7.0	13.4
ANS	11.6	11.1	14.6
Corporate and other	2.1	4.7	5.3
Consolidated additions to property, plant and equipment	$ 50.9	$ 97.3	$ 125.8

The corporate and other line item above reflects general corporate costs that were previously allocated to the Home segment. These indirect expenses have been classified as continuing operations for all periods presented since the costs were not directly attributable to the discontinued operations of the Home segment. These costs will be reallocated to the Company's remaining segments in future years and will be at least partially offset by income from a transition services agreement with Vantiva or other restructuring actions.

Customer and Supplier Information

No direct customer accounted for 10% or more of the Company's total net sales during the years ended December 31, 2023, 2022 or 2021. Accounts receivable from Comcast Corporation (Comcast) represented approximately 11% of accounts receivable as of December 31, 2023. Accounts receivable from Charter Communications, Inc. (Charter) represented approximately 14% of accounts receivable as of December 31, 2022. Other than Comcast and Charter, no direct customer accounted for 10% or more of the Company's accounts receivable as of December 31, 2023 or 2022.

The Company relies on sole suppliers or a limited group of suppliers for certain key components, subassemblies and modules and a limited group of contract manufacturers to manufacture a significant portion of its products. Any disruption or termination of these arrangements could have a material adverse impact on the Company's results of operations.

Related Party Transactions

See Note 14 for a discussion of the Convertible Preferred Stock issued to Carlyle to finance the ARRIS acquisition. Other than transactions related to the Convertible Preferred Stock, there were no material related party transactions for the years ended December 31, 2023, 2022 or 2021.

Geographic Information

Sales to customers located outside of the U.S. comprised 35%, 33% and 40% of total net sales during the years ended December 31, 2023, 2022 and 2021, respectively. Sales by geographic region, based on the destination of product shipments or service provided, were as follows:

	Year Ended December 31, 2023				
	CCS	OWN	NICS	ANS	Total
Geographic Region:					
United States	$ 1,730.4	$ 549.6	$ 663.6	$ 806.4	$ 3,750.0
Europe, Middle East and Africa	402.0	206.9	286.2	85.3	980.4
Asia Pacific	381.4	97.1	127.9	25.1	631.5
Caribbean and Latin America	146.4	17.2	20.4	105.8	289.8
Canada	50.0	9.2	19.6	58.7	137.5
Consolidated net sales	$ 2,710.2	$ 880.0	$ 1,117.7	$ 1,081.3	$ 5,789.2

	Year Ended December 31, 2022				
	CCS	OWN	NICS	ANS	Total
Geographic Region:					
United States	$ 2,513.6	$ 1,062.6	$ 539.5	$ 902.4	$ 5,018.1
Europe, Middle East and Africa	571.9	218.0	250.7	124.2	1,164.8
Asia Pacific	431.4	120.8	114.1	79.2	745.5
Caribbean and Latin America	179.3	32.8	20.3	164.3	396.7
Canada	93.4	33.7	15.1	57.4	199.6
Consolidated net sales	$ 3,789.6	$ 1,467.9	$ 939.7	$ 1,327.5	$ 7,524.7

Long-lived assets, excluding intangible assets, consist substantially of property, plant and equipment and right of use assets. The Company's long-lived assets, excluding intangible assets, located in the U.S., EMEA, APAC and CALA regions represented the following percentages of such long-lived assets: 61%, 14%, 16% and 9%, respectively, as of December 31, 2023 and 61%, 14%, 17% and 8%, respectively, as of December 31, 2022.

18. QUARTERLY FINANCIAL DATA (UNAUDITED)

The unaudited interim financial information below has been recast to reflect the discontinuation of the Home business. See Note 3 for further information on discontinued operations.

	First Quarter 2023	Second Quarter 2023	Third Quarter 2023	Fourth Quarter 2023
Net sales	$ 1,664.4	$ 1,588.8	$ 1,350.1	$ 1,185.9
Gross profit	630.2	568.8	501.0	448.3
Operating income (loss)	183.1	98.1	(331.8)	(62.3)
Loss from continuing operations	29.9	(63.2)	(479.0)	(339.0)
Net income (loss)	3.4	(100.4)	(828.7)	(525.2)
Net loss attributable to common stockholders	(11.7)	(115.7)	(844.2)	(541.1)
Basic earnings (loss) from continuing operations per share	$ 0.07	$ (0.37)	$ (2.33)	$ (1.67)
Diluted earnings (loss) from continuing operations per share	$ 0.07	$ (0.37)	$ (2.33)	$ (1.67)

	First Quarter 2022	Second Quarter 2022	Third Quarter 2022	Fourth Quarter 2022
Net sales	$ 1,732.9	$ 1,876.3	$ 1,990.0	$ 1,925.5
Gross profit	557.6	623.3	688.0	725.2
Operating income (loss)	30.7	77.3	182.0	(903.1)
Loss from continuing operations	(105.6)	(50.2)	31.6	(1,060.5)
Net income (loss)	(139.9)	(61.0)	22.9	(1,108.9)
Net income (loss) attributable to common stockholders	(154.4)	(75.7)	8.0	(1,123.8)
Basic earnings (loss) from continuing operations per share	$ (0.58)	$ (0.31)	$ 0.08	$ (5.16)
Diluted earnings (loss) from continuing operations per share	$ (0.58)	$ (0.31)	$ 0.08	$ (5.16)

19. SUBSEQUENT EVENTS

On January 9, 2024, the Company completed the previously announced sale of the Home Business to Vantiva pursuant to the Purchase Agreement, dated as of December 7, 2023. Vantiva acquired the Home Business in exchange for (i) 134,704,669 shares of Vantiva common stock, representing a 24.73% equity stake in Vantiva (determined on a fully diluted basis), (ii) $250,465 in cash (in addition to cash paid in exchange for the cash on the Home Business companies' balance sheets) and (iii) an earn-out of up to $100 million, subject to the satisfaction of certain conditions. The $250,465 in cash paid in connection with the closing is expected to be used to acquire additional shares of Vantiva common stock, following which the Company is expected to own a 25% equity stake in Vantiva (on a fully diluted basis). In conjunction with the closing of the transaction, the Company entered into a transition services agreement (TSA) with Vantiva, whereby the Company will provide and receive certain post-closing support on a transitional basis. The TSA has varying terms for duration up to sixteen months, depending on the service, and provides for options to extend services for up to two renewal terms of three months each. The loss recognized upon disposal at closing of the transaction is currently estimated to be less than $50 million.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

Not applicable.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of the Chief Executive Officer (CEO) and Chief Financial Officer (CFO), evaluated the effectiveness of our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) under the Securities Exchange Act of 1934, as amended (the Exchange Act)) as of the end of the period covered by this report.

Based on this evaluation, our CEO and CFO have concluded that, as of the end of the period covered by this report, these disclosure controls and procedures were effective and operating to provide reasonable assurance that information that we are required to disclose in the reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the rules and forms of the Securities and Exchange Commission, and that such information is accumulated and communicated to our management, including our CEO and CFO, as appropriate, to allow timely decisions regarding required disclosure.

Management's Report on Internal Control over Financial Reporting

The management of CommScope is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is defined in Rule 13a-15(f) and 15d-15(f) under the Exchange Act, as a process designed by, or under the supervision of, the company's principal executive and principal financial officers and effected by the company's Board of Directors, management and other personnel, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of consolidated financial statements for external purposes in accordance with generally accepted accounting principles and includes those policies and procedures that:

- Pertain to the maintenance of records that in reasonable detail accurately and fairly reflect the transactions and dispositions of the assets of the company;
- Provide reasonable assurance that transactions are recorded as necessary to permit preparation of consolidated financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and
- Provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of the company's assets that could have a material effect on the consolidated financial statements.

CommScope's management assessed the effectiveness of CommScope's internal control over financial reporting as of December 31, 2023. In making this assessment, CommScope's management used the criteria set forth by the Committee of Sponsoring Organizations of the Treadway Commission (COSO) in Internal Control – Integrated Framework (2013). Based on this assessment, management concluded that, as of December 31, 2023, CommScope's internal control over financial reporting is effective based on the COSO internal control criteria.

CommScope's independent registered public accounting firm, Ernst & Young LLP, has issued an attestation report on the effectiveness of CommScope's internal control over financial reporting, which is included in Item 8 of this Annual Report on Form 10-K.

Changes in Internal Control over Financial Reporting

There have been no changes in the Company's internal controls over financial reporting during the quarter ended December 31, 2023 that have materially affected, or are reasonably likely to materially affect, the Company's internal control over financial reporting.

Inherent Limitations of Disclosure Controls and Internal Control over Financial Reporting

Because of their inherent limitations, our disclosure controls and procedures and our internal control over financial reporting may not prevent all material errors or fraud. A control system, no matter how well conceived and operated, can provide only reasonable, not absolute, assurance that the objectives of the control system are met. The effectiveness of our disclosure controls and procedures and our internal control over financial reporting is subject to risks, including that the controls may become inadequate because of changes in conditions or that the degree of compliance with our policies or procedures may deteriorate.

ITEM 9B. OTHER INFORMATION

Insider Trading Arrangements

Our officers and directors did not enter into, modify or terminate any Rule 10b5-1 trading arrangements or non-Rule 10b5-1 trading arrangements (each as defined in Item 408(c) of Regulation S-K) during the quarter ended December 31, 2023.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

The information responsive to this item is incorporated herein by reference to our Proxy Statement for our 2024 annual meeting of stockholders to be filed with the Securities and Exchange Commission pursuant to Regulation 14A not later than 120 days after the end of the fiscal year covered by this Annual Report on Form 10-K.

Code of Ethics for Principal Executive and Senior Financial and Accounting Officers

We have adopted the CommScope Holding Company, Inc. Code of Ethics for Principal Executive and Senior Financial and Accounting Officers (the Senior Officer Code of Ethics), a code of ethics that applies to our Chief Executive Officer, Chief Financial Officer and Chief Accounting Officer. The Senior Officer Code of Ethics is publicly available on our website at *www.commscope.com*. If we make an amendment to, or grant a waiver from, a provision of the Senior Officer Code of Ethics, we will disclose the nature of such waiver or amendment on our website.

ITEM 11. EXECUTIVE COMPENSATION

The information responsive to this item is incorporated herein by reference to our Proxy Statement for our 2024 annual meeting of stockholders to be filed with the Securities and Exchange Commission pursuant to Regulation 14A not later than 120 days after the end of the fiscal year covered by this Annual Report on Form 10-K.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information responsive to this item is incorporated herein by reference to our Proxy Statement for our 2024 annual meeting of stockholders to be filed with the Securities and Exchange Commission pursuant to Regulation 14A not later than 120 days after the end of the fiscal year covered by this Annual Report on Form 10-K.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

The information responsive to this item is incorporated herein by reference to our Proxy Statement for our 2024 annual meeting of stockholders to be filed with the Securities and Exchange Commission pursuant to Regulation 14A not later than 120 days after the end of the fiscal year covered by this Annual Report on Form 10-K.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information responsive to this item is incorporated herein by reference to our Proxy Statement for our 2024 annual meeting of stockholders to be filed with the Securities and Exchange Commission pursuant to Regulation 14A not later than 120 days after the end of the fiscal year covered by this Annual Report on Form 10-K.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

(a)Documents Filed as Part of this Report:

1.*Audited Consolidated Financial Statements*

The following consolidated financial statements of CommScope Holding Company, Inc. are included under Part II, Item 8:

Reports of Independent Registered Public Accounting Firm
Consolidated Statements of Operations for the Years Ended December 31, 2023, 2022 and 2021
Consolidated Statements of Comprehensive Loss for the Years Ended December 31, 2023, 2022 and 2021
Consolidated Balance Sheets as of December 31, 2023 and 2022
Consolidated Statements of Cash Flows for the Years Ended December 31, 2023, 2022 and 2021
Consolidated Statements of Stockholders' Deficit for the Years Ended December 31, 2023, 2022 and 2021
Notes to Consolidated Financial Statements

2.*Financial Statement Schedules*

All schedules are omitted because they are not applicable or the required information is shown in the financial statements or notes thereto.

3.*List of Exhibits.* See Index of Exhibits included herein.

Index of Exhibits

	Exhibit No.	Description
*	2.1	Call Option Agreement, dated October 2, 2023, by and between CommScope Holding Company, Inc. and Vantiva S.A. (Incorporated by reference to Exhibit 2.1 to the Registrant's Current Report on Form 8-K filed with the SEC on October 5, 2023).
*	2.2	Purchase Agreement by and between CommScope Holding Company, Inc. and Vantiva S.A. (Incorporated by reference to Exhibit 2.2 to the Registrant's Current Report on Form 8-K filed with the SEC on October 5, 2023).
*	3.1	Amended and Restated Certificate of Incorporation of CommScope Holding Company, Inc. (Incorporated by reference to Exhibit 3.1 of the Registrant's Form 10-Q (File No. 001-36146), filed with the SEC on November 7, 2013).
*	3.2	Certificate of Designations Designating Series A Preferred Stock (Incorporated by reference to Exhibit 3.1 to the Registrant's Current Report on Form 8-K filed with the SEC on April 4, 2019).
**	3.3	Certificate of Increase of Shares Designated as Series A Convertible Preferred Stock, par value $0.01, of CommScope Holding Company, Inc.
*	3.4	Certificate of Amendment of Amended and Restated Certificate of Incorporation of CommScope Holding Company, Inc. (Incorporated by reference to Exhibit 3.2 of the Registrant's Registration Statement on Form S-8 (File No. 333-256539), filed with the SEC on May 27, 2021).
*	3.5	Sixth Amended and Restated By-Laws of CommScope Holding Company, Inc. (as adopted November 30, 2023) (Incorporated by reference to Exhibit 3.1 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on December 4, 2023).
*	4.1	Indenture governing the 6.000% Senior Notes due 2025 by and between CommScope Technologies Finance LLC and Wilmington Trust, National Association, as trustee, dated as of June 11, 2015 (including form of 6.000% Senior Note due 2025) (Incorporated by reference to Exhibit 4.2 to the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on June 12, 2015).
*	4.2	First Supplemental Indenture, dated August 28, 2015, by and among CommScope Technologies LLC, the Guarantors party thereto and Wilmington Trust, National Association, as trustee (Incorporated by reference to Exhibit 4.1 to the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on August 28, 2015).
*	4.3	Indenture governing the 5.000% Senior Notes due 2027, by and among CommScope Technologies LLC, the guarantors named therein and Wilmington Trust, National Association, as trustee and as collateral agent, dated as of March 13, 2017, (including form of 5.000% Senior Note due 2027) (Incorporated by reference to Exhibit 4.1 to the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on March 13, 2017).
*	4.4	Indenture, dated as of February 19, 2019, by and between CommScope Finance LLC and Wilmington Trust, National Association, as trustee, including the form of 8.25% Senior Note due 2027 (Incorporated by reference to Exhibit 4.1 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on February 19, 2019).
*	4.5	First Supplemental Indenture, dated as of April 4, 2019, by and among CommScope, LLC (f/k/a CommScope, Inc.), the guarantors party thereto and Wilmington Trust, National Association, as trustee (Incorporated by reference to Exhibit 4.2 to the Registrant's Current Report on Form 8-K filed with the SEC on April 4, 2019).
*	4.6	Indenture, dated as of February 19, 2019, by and between CommScope Finance LLC and Wilmington Trust, National Association, as trustee and collateral agent, including the form of 5.50% Senior Secured Note due 2024 and form of 6.00% Senior Secured Note due 2026 (Incorporated by reference to Exhibit 4.3 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on February 19, 2019).
*	4.7	First Supplemental Indenture, dated as of April 4, 2019, by and among CommScope, LLC (f/k/a CommScope, Inc.), CommScope Holding Company, Inc., the other guarantors party thereto, Wilmington Trust, National

		Association, as trustee, and Wilmington Trust, National Association, as collateral agent (Incorporated by reference to Exhibit 4.4 to the Registrant's Current Report on Form 8-K filed with the SEC on April 4, 2019).
*	4.8	Indenture, dated as of July 1, 2020, by and among CommScope, LLC (f/k/a CommScope, Inc.), the guarantors party thereto and Wilmington Trust, National Association, as trustee, including the form of 7.125% Senior Note due 2028 (Incorporated by reference to Exhibit 4.1 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on July 2, 2020).
*	4.9	Indenture, dated as of August 23, 2021, by and among CommScope, LLC (f/k/a CommScope, Inc.), the guarantors party thereto and Wilmington Trust, National Association, as trustee and collateral agent, including the form of 4.750% Senior Secured Note due 2029 (Incorporated by reference to Exhibit 4.1 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on August 23, 2021).
*	4.10	Description of Securities Registered Pursuant to Section 12 of the Exchange Act (Incorporated by reference to Exhibit 4.10 to the Registrant's Annual Report on Form 10-K filed with the SEC on February 20, 2020).
*	10.1	Success Bonus Agreement, dated September 15, 2023, between CommScope, LLC (f/k/a CommScope, Inc.) and Gonzaga Chow (Incorporated by reference to Exhibit 10.1 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36046), filed with the SEC on November 9, 2023).***
*	10.2	Employment Agreement between Charles L. Treadway and CommScope, LLC (f/k/a CommScope, Inc.), dated October 1, 2020 (Incorporated by reference to Exhibit 10.1 to the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on October 1, 2020).***
*	10.3	Amendment to Employment Agreement between Charles L. Treadway and CommScope, LLC (f/k/a CommScope, Inc.), dated October 4, 2022 (Incorporated by reference to Exhibit 10.2 of the Registrant's Annual Report on Form 10-K (File No. 001-36046), filed with the SEC on February 17, 2023).***
*	10.4	Employment Agreement between Claudius E. Watts IV and CommScope, LLC (f/k/a CommScope, Inc.), dated October 1, 2020 (Incorporated by reference to Exhibit 10.4 to the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on October 1, 2020).***
*	10.5	Amendment to Employment Agreement between Claudius E. Watts, IV and CommScope, LLC (f/k/a CommScope, Inc.), dated October 4, 2022 (Incorporated by reference to Exhibit 10.4 of the Registrant's Annual Report on Form 10-K (File No. 001-36046), filed with the SEC on February 17, 2023).***
*	10.6	Form of Severance Protection Agreement among CommScope, LLC (f/k/a CommScope, Inc.) and Kyle D. Lorentzen, Justin C. Choi and Robyn T. Mingle (Incorporated by reference to Exhibit 10.6 of the Registrant's Annual Report on Form 10-K (File No. 001-36046), filed with the SEC on February 17, 2023).***
*	10.7	Severance Protection Agreement between Charles L. Treadway and CommScope, LLC (f/k/a CommScope, Inc.), dated October 4, 2022 (Incorporated by reference to Exhibit 10.7 of the Registrant's Annual Report on Form 10-K (File No. 001-36046), filed with the SEC on February 17, 2023).***
*	10.8	Severance Protection Agreement between Claudius E. Watts and CommScope, Inc., dated October 4, 2022 (Incorporated by reference to Exhibit 10.8 of the Registrant's Annual Report on Form 10-K (File No. 001-36046), filed with the SEC on February 17, 2023).***
*	10.9	Form of Indemnification Agreement (Incorporated by reference to Exhibit 10.22 of Amendment No. 2 to the Registrant's Registration Statement on Form S-1 (File No. 333-190354), filed with the SEC on September 12, 2013).***
*	10.10	CommScope Holding Company, Inc. Amended and Restated 2013 Long-Term Incentive Plan (as amended and restated effective February 21, 2017) (Incorporated by reference to Exhibit 10.28 of the Registrant's Annual Report on Form 10-K (File No. 001-36146), filed with the SEC on February 23, 2017).***
*	10.11	Form of Non-Qualified Stock Option Certificate under the CommScope Holding Company, Inc. 2013 Long-Term Incentive Plan (Incorporated by reference to Exhibit 10.4 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on April 30, 2015).***
*	10.12	CommScope Holding Company, Inc. Amendment to Outstanding Options, effective March 7, 2016 (Incorporated by reference to Exhibit 10.5 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on April 28, 2016).***

*	10.13	Form of Restricted Stock Unit Award Certificate under the CommScope Holding Company, Inc. 2013 Long-Term Incentive Plan (for grants in 2016 and later) (Incorporated by reference to Exhibit 10.2 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on April 28, 2016).***
*	10.14	Form of Performance Share Unit Award Certificate under the CommScope Holding Company, Inc. 2013 Long-Term Incentive Plan (for grants in 2016 and later) (Incorporated by reference to Exhibit 10.3 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on April 28, 2016).***
*	10.15	Form of Non-Qualified Stock Option Certificate under the CommScope Holding Company, Inc. 2013 Long-Term Incentive Plan (for grants in 2016 and later) (Incorporated by reference to Exhibit 10.4 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on April 28, 2016).***
*	10.16	CommScope Holding Company, Inc. Annual Incentive Plan, as amended and restated February 21, 2023. (Incorporated by reference to Exhibit 10.1 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36046), filed with the SEC on May 4, 2023).***
*	10.17	CommScope Holding Company, Inc. Non-Employee Director Compensation Plan, as amended on February 19, 2019 (Incorporated by reference to Exhibit 10.19 of the Registrant's Annual Report on Form 10-K (File No. 001-36046), filed with the SEC on February 17, 2023).
*	10.18	Form of Restricted Stock Unit Award Certificate under the CommScope Holding Company, Inc. Non-Employee Director Compensation Plan, which is operated as a subplan of the CommScope Holding Company, Inc. 2019 Long-Term Incentive Plan (Incorporated by reference to Exhibit 10.20 of the Registrant's Annual Report on Form 10-K (File No. 001-36046), filed with the SEC on February 17, 2023).
*	10.19	Form of Non-Qualified Stock Option Certificate under the CommScope Holding Company, Inc. Amended and Restated 2013 Long-Term Incentive Plan (for grants to senior executive officers in 2019) (Incorporated by reference to Exhibit 10.5 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on August 8, 2019).***
*	10.20	Form of Restricted Stock Unit Award Certificate under the CommScope Holding Company, Inc. Amended and Restated 2019 Long-Term Incentive Plan (Incorporated by reference to Exhibit 10.2 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on October 1, 2020).***
*	10.21	Form of Performance Share Unit Award Certificate under the CommScope Holding Company, Inc. Amended and Restated 2019 Long-Term Incentive Plan (service and average stock price vesting) (Incorporated by reference to Exhibit 10.3 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on October 1, 2020). ***
*	10.22	Form of Performance Share Unit Award Certificate under the CommScope Holding Company, Inc. Amended and Restated 2019 Long-Term Incentive Plan (Incorporated by reference to Exhibit 10.1 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on May 5, 2022).***
*	10.23	CommScope Holding Company, Inc. Deferred Compensation Plan (as amended and restated effective January 1, 2017) ((Incorporated by reference to Exhibit 10.41 of the Registrant's Annual Report on Form 10-K (File No. 001-36146), filed with the SEC on February 23, 2017).***
*	10.24	CommScope Holding Company, Inc. Amended and Restated 2019 Long-Term Incentive Plan (Incorporated by reference to Exhibit 99.1 to the Registrant's Registration Statement on Form S-8 (File No. 333-265198), filed with the Commission on May 24, 2023).***
*	10.25	Investment Agreement, dated November 8, 2018, by and between CommScope Holding Company, Inc. and Carlyle Partners VII S1 Holdings, L.P. (Incorporated by reference to Exhibit 10.1 of the Registrant's Current Report on Form 8-K (File No. 001-36146), filed with the SEC on November 8, 2018).
*	10.26	Registration Rights Agreement, dated as of April 4, 2019, by and between CommScope Holding Company, Inc. and Carlyle Partners VII S1 Holdings, L.P. (Incorporated by reference to Exhibit 10.1 to the Registrant's Current Report on Form 8-K filed with the SEC on April 4, 2019).
*	10.27	Revolving Credit Agreement, dated as of April 4, 2019, and as amended by that certain Amendment Agreement, dated August 11, 2021, among CommScope Holding Company, Inc., CommScope, LLC (f/k/a CommScope, Inc.), the co-borrowers named therein, JPMorgan Chase Bank, N.A., as administrative agent and collateral agent,

		and the other agents and lenders party thereto (Incorporated by reference to Exhibit 10.2 to the Registrant's Current Report on Form 8-K filed with the SEC on April 4, 2019).
*	10.28	Amendment No. 2, dated October 19, 2022, to the Revolving Credit Agreement, dated as of April 4, 2019, among CommScope Holding Company, Inc., CommScope, LLC (f/k/a CommScope, Inc.), the co-borrowers named therein, J.P. Morgan Chase Bank, N.A., as administrative agent and collateral agent, and the other agents and lenders party thereto (incorporated by reference to Exhibit 10.1 to the Registrant's Current Report on Form 8-K filed with the SEC on October 20, 2022).
*	10.29	Term Loan Credit Agreement, dated as of April 4, 2019, and as amended by that certain Amendment Agreement, dated August 11, 2021, among CommScope, LLC (f/k/a CommScope, Inc.), as the borrower, CommScope Holding Company, Inc., as holdings, JPMorgan Chase Bank, N.A., as administrative agent and collateral agent, and the other agents and lenders party thereto (Incorporated by reference to Exhibit 10.3 to the Registrant's Current Report on Form 8-K filed with the SEC on April 4, 2019).
*	10.30	Amendment No. 2 to Credit Agreement, dated as of June 8, 2023, by and among CommScope, LLC (f/k/a CommScope, Inc.), CommScope Holding Company, Inc., the subsidiary guarantors and JPMorgan Chase Bank, N.A., as administrative agent and collateral agent (Incorporated by reference to Exhibit 10.1 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36046), filed with the SEC on August 3, 2023).
*	18.1	Preferability Letter from Ernst & Young LLP, Independent Registered Public Accounting Firm (Incorporated by Reference to Exhibit 18.1 of the Registrant's Quarterly Report on Form 10-Q (File No. 001-36146), filed with the SEC on August 8, 2019).
**	19.1	CommScope Holding Company, Inc. Inside Information and Insider Trading Policy, effective May 18, 2023 (including Guidelines for a Rule 10b5-1 Plan).
**	21.1	List of Subsidiaries
**	23.1	Consent of Ernst & Young LLP, Independent Registered Public Accounting Firm
**	31.1	Certification of Principal Executive Officer pursuant to Rule 13a-14(a).
**	31.2	Certification of Principal Financial Officer pursuant to Rule 13a-14(a).
±	32.1	Certification of Principal Executive Officer and Principal Financial Officer pursuant to 18 U.S.C. Section 1350 as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002 (furnished pursuant to Item 601(b)(32)(ii) of Regulation S-K).
**	97.1	CommScope Holding Company, Inc. Compensation Recovery Policy.
†	101.INS	Inline XBRL Instance Document – The instance document does not appear in the interactive data file because its XBRL tags are embedded within the inline XBRL document.
†	101.SCH	Inline XBRL Schema Document, furnished herewith.
†	101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.
†	101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document.
†	101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document,
†	101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document.
	104	Cover Page Interactive Data File (embedded within the Inline XBRL document).

* Previously filed

** Filed herewith

*** Management contract or compensatory plan or arrangement.

† In accordance with Rule 406T of Regulation S-T, the information in these exhibits is furnished and deemed not filed or part of a registration statement or prospectus for purposes of Sections 11 or 12 of the Securities Act of 1933, is deemed not filed for purposes of Section 18 of the Exchange Act of 1934, and otherwise is not subject to liability under these sections.

± In accordance with Item 601(b)(32)(ii) of Regulation S-K and SEC Release No. 33-8238 and 34-47986, Final Rule: Management's Reports on Internal Control Over Financial Reporting and Certification of Disclosure in Exchange Act Periodic Reports, the certification furnished in Exhibit 32.1 hereto is deemed to accompany this Form 10-K and will not be deemed "filed" for purposes of Section 18 of the Exchange Act. Such certification will not be deemed to be incorporated by reference into any filings under the Securities Act or the Exchange Act, except to the extent that the registrant specifically incorporates it by reference.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, as amended, the Registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

COMMSCOPE HOLDING COMPANY, INC

DATE: February 28, 2024

BY: /s/ Charles L. Treadway
Charles L. Treadway
President and Chief Executive Officer

Pursuant to the requirements of the Securities Exchange Act of 1934, as amended, this Annual Report on Form 10-K has been signed below by the following persons on behalf of the Registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ CHARLES L. TREADWAY Charles L. Treadway	President, Chief Executive Officer and Director (Principal Executive Officer)	February 28, 2024
/s/ KYLE D. LORENTZEN Kyle D. Lorentzen	Executive Vice President and Chief Financial Officer (Principal Financial Officer)	February 28, 2024
/s/ LAURIE S. ORACION Laurie S. Oracion	Senior Vice President and Chief Accounting Officer (Principal Accounting Officer)	February 28, 2024
/s/ CLAUDIUS E. WATTS IV Claudius E. Watts IV	Director and Chairman of the Board	February 28, 2024
/s/ STEPHEN C. GRAY Stephen C. Gray	Director	February 28, 2024
/s/ SCOTT H. HUGHES Scott H. Hughes	Director	February 28, 2024
/s/ L. WILLIAM KRAUSE L. William Krause	Director	February 28, 2024
/s/ JOANNE M. MAGUIRE Joanne M. Maguire	Director	February 28, 2024
/s/ THOMAS J. MANNING Thomas J. Manning	Director	February 28, 2024
/s/ PATRICK R. MCCARTER Patrick R. McCarter	Director	February 28, 2024
/s/ DERRICK A. ROMAN Derrick A. Roman	Director	February 28, 2024
/s/ TIMOTHY T. YATES Timothy T. Yates	Director	February 28, 2024